

MSc Mathematics
Track: Data Science and Optimization

Master Thesis

Overcoming Video Compression for Remote Heart Rate Estimation: H.264 Parameter Analysis and Robust Self-Supervised Learning Solutions

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October 2025



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Abstract

Remote photoplethysmography (rPPG) enables non-contact heart rate monitoring by analyzing video recordings of a subject's face. In practical settings, however, videos are often compressed, commonly using the H.264 codec, which can reduce physiological signal quality and affect the accuracy of heart rate estimation.

This thesis starts by exploring the relationships among various H.264 compression parameters, focusing on those that have noticeable effects on compression efficiency, such as Constant Rate Factor (CRF), Group-of-Pictures (GOP) structure, and B-frames. We study how these parameters influence the quality of rPPG signals extracted from video data using several quantitative metrics. A detailed literature review of the H.264 encoder and its impact on video quality serves as the basis for this analysis.

Next, we review deep learning methods for rPPG estimation, which leads to selecting the RS-rPPG framework because of its promising results on uncompressed videos. Since RS-rPPG was originally developed for uncompressed data, we adapt it to work on compressed videos with different CRF levels, evaluating its performance on both the PURE dataset and a proprietary VicarVision rPPG dataset. An important aspect of our approach is the inclusion of uncompressed temporal maps during both pretraining and training, which helps the model better recover physiological signals from compressed inputs. This results in improved performance compared to commercial baselines such as FaceReader, especially when tested on combined compressed videos with higher CRF values.

Experiments conducted across various compression levels and datasets show the method's ability to provide reliable heart rate estimation even under compression. The thesis ends with a discussion on possible future work, including the use of signal magnification techniques, and considers the trade-offs between compression rate, signal distortion, and model generalization for practical applications.

Keywords: H.264, video compression, remote photoplethysmography, rPPG, heart rate estimation, RS-rPPG, temporal maps, contrastive learning.

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Chapter 1

Introduction

1.1 Research Questions and Approach

Remote photoplethysmography (rPPG) is a promising technique for measuring heart rate without physical contact, using only a video camera. It works by detecting subtle changes in skin color caused by blood flow, which are invisible to the naked eye but can be captured by sensitive image processing algorithms. With the widespread availability of cameras in smartphones, laptops, and other consumer devices, rPPG has strong potential for accessible health monitoring, telemedicine, and fitness tracking.

However, a major practical challenge is that videos captured in real-world settings are almost always compressed to save storage and bandwidth. The dominant standard, H.264/AVC, uses advanced techniques such as block prediction, quantization, and in-loop filtering, which can degrade the subtle color variations that rPPG relies on. This raises the question:

How do different H.264 compression parameters affect the accuracy of heart rate estimation via rPPG?

To address this, the thesis focuses on two main research questions:

- **RQ1:** Which H.264 encoding parameters most affect the quality of physiological signals extracted by rPPG?
- **RQ2:** How can deep learning methods be adapted to improve robustness against compression artifacts in rPPG estimation?

Our approach is twofold. First, we systematically study key H.264 encoding parameters and their relationships, focusing on those with noticeable effects on compression efficiency and signal quality:

- **CRF (Constant Rate Factor):** Controls the overall compression level by adjusting the quantization parameter uniformly across frames. Lower CRF values yield higher video quality at the cost of larger file sizes, while higher values increase compression and may degrade subtle skin color changes essential for rPPG.
- **B-frames:** Bidirectionally predicted frames referencing both past and future frames to improve compression efficiency. Extensive use may introduce temporal smoothing or artifacts affecting physiological signal clarity.
- **GOP (Group of Pictures) length:** Defines the interval between intra-coded keyframes. Shorter GOPs improve error resilience but reduce compression efficiency, while longer GOPs increase efficiency at the risk of error propagation that may distort rPPG signals.
- **Reference frames (REF):** Number of past frames used for motion compensation. More ref-

erence frames allow better motion prediction but increase encoder complexity and may affect compression artifacts.

- **Sub-pixel motion estimation (SUBQ):** Enables motion vector calculation at fractional pixel accuracy, improving prediction precision but potentially impacting temporal signal consistency.

By varying these parameters within practical ranges, we generated a dataset of compressed videos exhibiting diverse signal distortions for detailed analysis.

Second, we develop a novel compression-aware training method that pairs compressed and uncompressed temporal maps during pretraining and training. This strategy guides the network to focus on relevant physiological features while reducing the impact of compression noise.

Overall, this thesis investigates how common video compression settings affect rPPG signals and introduces methods to improve deep learning model robustness on compressed video data. By bridging the gap between ideal uncompressed data and realistic compressed video, this work aims to make remote heart rate monitoring more accurate and practical for everyday use.

1.2 Motivation and Context

VicarVision is dedicated to developing innovative computer vision solutions that enable non-invasive, real-time physiological monitoring to improve health and well-being. Their technology aims to make vital sign measurement more accessible by leveraging everyday devices such as smartphones and webcams, facilitating applications ranging from healthcare and fitness to workplace safety.

The increasing demand for contactless and continuous health monitoring has never been more relevant, as global populations face growing healthcare challenges including aging demographics and chronic disease management. Remote photoplethysmography (rPPG) aligns closely with VicarVision’s vision by offering contactless heart rate measurement through video analysis, enabling scalable and user-friendly monitoring outside clinical environments.

However, real-world deployment faces practical challenges: videos captured in everyday settings are often compressed for efficient storage and transmission, which can degrade the subtle physiological signals that rPPG methods rely on. Most existing algorithms are developed and validated on high-quality, uncompressed data, limiting their performance on such compressed videos.

This thesis addresses this critical gap by investigating the influence of video compression parameters on rPPG signal quality and by developing robust deep learning methods that enhance model performance on compressed data. These advancements support VicarVision’s goal of bringing reliable, non-contact physiological sensing to broad, real-world applications, ultimately contributing to improved health outcomes and greater accessibility of monitoring technologies.

1.3 Contributions

This thesis advances remote physiological measurement through the following key contributions:

- **Comprehensive analysis of H.264 parameters:** In Chapter 2, we systematically examine how critical compression settings, including CRF, B-frames, GOP length, reference frames, and

sub-pixel motion estimation, affect physiological data quality, with a focus on correlation and signal-to-noise ratio (SNR) as primary evaluation metrics.

- **Compression-aware extension of RS-rPPG:** In Chapter 3, building on the original RS-rPPG framework tested on uncompressed videos, we introduce a novel pretraining strategy that incorporates uncompressed temporal maps (Tmap0) as a supervisory signal during both pretraining and training. This guides the model to emphasize physiological features while reducing sensitivity to compression artifacts.
- **Empirical validation on benchmark datasets:** Later in Chapter 3, our enhanced RS-rPPG model is evaluated on the PURE dataset compressed at multiple CRF levels, as well as on a proprietary VicarVision dataset, demonstrating improved robustness and signal fidelity. The approach outperforms both the baseline RS-rPPG and the commercial FaceReader system, particularly at higher compression rates.
- **Insightful discussion of trade-offs:** In Chapter 4, we analyze the balance between compression rate, signal distortion, and model generalization in resource-constrained environments, and outline future directions such as combining the proposed framework with the existing FaceReader model and incorporating signal magnification techniques to further improve performance.

1.4 Thesis Roadmap

The thesis is organized as follows:

- Chapter 1 introduces the research questions, motivation, contributions, and thesis roadmap.
- Chapter 2 covers video compression fundamentals, including an overview of H.264, key compression parameters, experimental methods for varying these parameters, and evaluation metrics.
- Chapter 3 details the RS-rPPG framework and remote heart rate estimation, including the methodology, contrastive learning approach, novelty, and results on Pure and VicarVision datasets.
- Chapter 4 concludes the thesis with a summary of contributions and outlines future research directions.
- Appendix A develops the Mathematical Modeling of Signal Stability under Compression, providing a formal analysis of how compression affects rPPG signal estimation.
- Appendix B contains additional investigations and experiments on the VicarrPPG model.
- Appendix C presents the Theoretical Explanation of RS-rPPG Robustness under Compression, offering a mathematical justification for the model's stability.

Through systematic analysis of compression effects and model improvements, this thesis aims to enhance the accuracy and usability of remote heart rate estimation in real-world scenarios.

Chapter 2

Video Compression Fundamentals

2.1 What is H.264?

H.264, also known as Advanced Video Coding (AVC), is a widely adopted video compression standard designed to effectively reduce the size of digital video data while preserving high visual quality. Due to its efficiency, H.264 is extensively used in various applications including video streaming, conferencing, and digital recording. Its effectiveness comes from its ability to minimize redundant information both within individual frames (spatial redundancy) and across consecutive frames (temporal redundancy), significantly decreasing the data needed for storage or transmission [1,2].

A clear understanding of the H.264 encoding architecture is important, particularly in applications like remote photoplethysmography (rPPG), where subtle signal details must be preserved during compression. Figure 2.1 provides an overview of the main components of an H.264 encoder and their interactions.

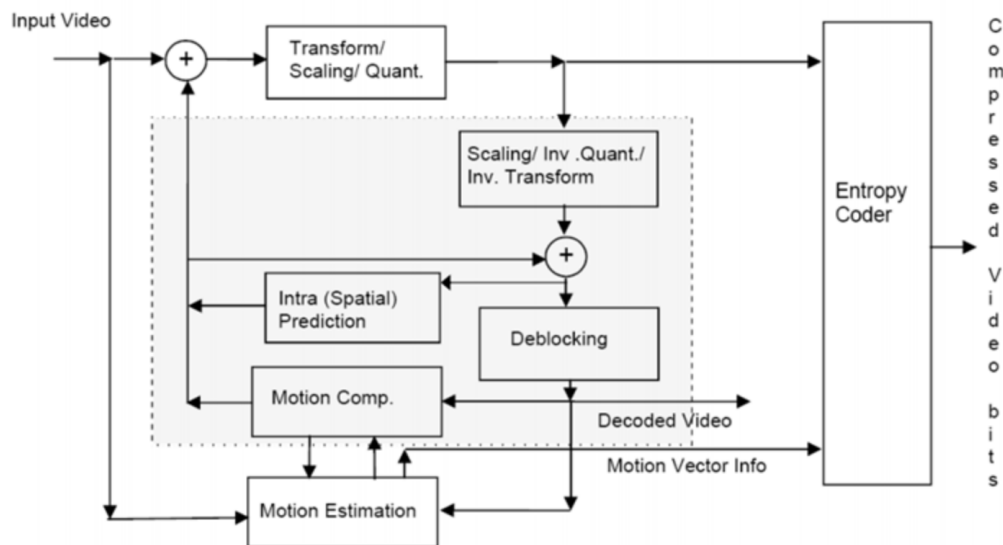


Figure 2.1: Simplified block diagram of the H.264 encoder showing key components such as motion estimation, intra/inter prediction, transform and quantization, entropy coding, and deblocking filter. [2].

The H.264 video encoding process begins with a stream of raw, uncompressed video frames, where each frame is a collection of pixel values representing color and brightness. To achieve efficient compression, the encoder avoids storing every frame independently. Instead, it predicts the content of each new frame using information from previous frames, as well as from the current frame itself. This prediction

occurs in two primary ways: intra (spatial) prediction, which estimates pixel values using neighboring pixels within the same frame, and inter (temporal or motion) prediction, which estimates blocks in the current frame based on similar blocks from past (reference) frames. For motion prediction, the encoder calculates motion vectors that describe how parts of the image have shifted between frames, enabling it to capture temporal redundancy.

Even with accurate prediction, some differences, called residuals, remain between the predicted and actual frames. These residuals represent the new or changed information that cannot be derived from prediction alone. Instead of directly encoding these residuals, the encoder transforms them into the frequency domain, using a mathematical operation called the Discrete Cosine Transform (DCT). This concentrates most of the important information into just a few frequency coefficients, making it easier to compress. The coefficients are then scaled and quantized, which means their values are rounded to reduce precision and further lower the data size. The degree of quantization is controlled by a parameter (the Quantization Parameter, QP, or Constant Rate Factor, CRF), which determines the balance between compression and visual quality: stronger quantization (higher QP/CRF) saves space but discards more fine details [1].

Once quantized, the residual coefficients, along with motion information and other prediction data, are compressed using entropy coding methods such as Context-Adaptive Variable Length Coding (CAVLC) or Context-Adaptive Binary Arithmetic Coding (CABAC). These techniques assign shorter codes to more common patterns in the data, further shrinking the file size without losing any additional information [1]. However, to perform future predictions, the encoder needs to “reconstruct” each frame from the compressed data as it goes. It does this by inversely scaling and transforming the quantized coefficients back to the spatial domain, creating an approximation of the original residual, which is then added to the predicted frame to produce a reconstructed version of the current frame. Because compression operates on blocks of pixels, this reconstruction can introduce visible artifacts at block boundaries. To smooth out these edges, the encoder applies a deblocking filter, which helps improve both visual quality and the accuracy of subsequent predictions.

The reconstructed and filtered frames are continuously used as references for predicting new frames, closing the feedback loop and maintaining temporal coherence. Meanwhile, the fully compressed video bitstream, containing the encoded residuals, motion vectors, and prediction information, is generated for storage or transmission.

In summary, H.264 encoding works by repeatedly predicting frames based on past information, encoding only what cannot be predicted, and applying mathematical transformations and lossless compression to reduce data size. This cyclical process allows the encoder to achieve high levels of compression while preserving acceptable video quality [1]. For remote photoplethysmography (rPPG) applications, understanding each step is essential, as choices like quantization strength and motion compensation can directly affect the subtle pixel changes needed to estimate heart rate from video. Thus, a clear grasp of the H.264 workflow is critical for designing robust algorithms capable of extracting physiological signals from compressed video sources [3].

2.2 Parameters of H.264

Video compression using the H.264 standard depends on many encoder settings that greatly influence both how much the video is compressed and how good it looks after compression. These settings control important processes like frame prediction, how image data is simplified (quantization), motion estimation, and more. For remote photoplethysmography (rPPG), which picks up very small color and movement changes, it is vital to understand how these settings affect the quality of the physiological signals [1, 4].

In our study, we examined more than thirty H.264 parameters to understand how they interact with each other and how they influence both the degree of video compression and the preservation of subtle physiological details. Figure 2.2 presents an overview of these parameters and their interconnections. The dark circles highlight the key parameters selected for our experiments, as they have the greatest impact on physiological signal quality [5]. The connecting lines in the figure indicate the dependencies between parameters, modifying one parameter can directly or indirectly affect others, underscoring the importance of considering their combined effects when optimizing compression for rPPG signal fidelity.

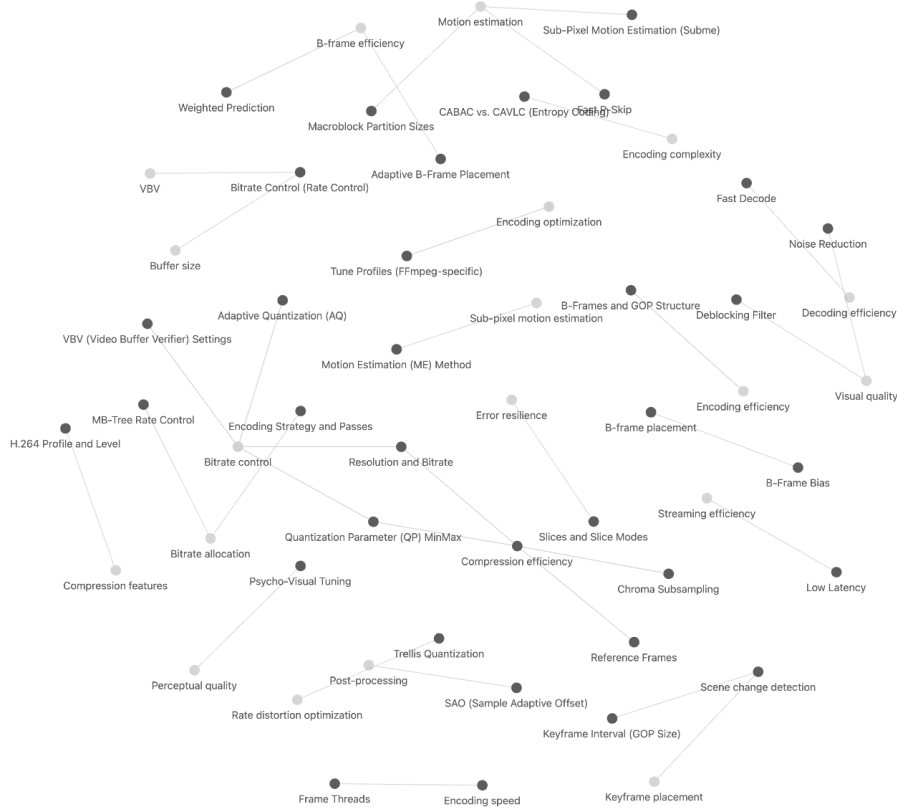


Figure 2.2: Dependency and correlation map of H.264 encoding parameters developed from our analysis. Dark nodes show the primary parameters varied in experiments.

From this analysis, we summarized ten important parameters that balance good compression with keeping the video details necessary for accurate rPPG (see Table 2.1).

These parameters were chosen because they significantly impact both how well the video compresses and how well it keeps the tiny details needed for physiological analysis. Some parameters depend on others; for example, GOP length limits how many B-frames and reference frames can be used, and bitrate control modes affect how quantization is applied [4,5].

- **Constant Rate Factor (CRF):** This parameter globally controls the degree of quantization applied across all frames, thus directly affecting the trade-off between compression level and video quality. Lower CRF values preserve more of the subtle color and texture details crucial for rPPG, at the cost of larger file sizes. Higher CRF values increase compression by discarding fine details, which can degrade the physiological signal quality. *Typical range:* 0 (lossless) to 51 (lowest quality), with common practical values between 18 and 28. *Common default:* 23 [4,5].
- **Group of Pictures (GOP) Length:** GOP length determines the frequency of intra-coded keyframes, I-frames, in the video. Longer GOPs reduce the number of keyframes, enhancing compression efficiency by relying more on predictive frames, but can cause error propagation and temporal distortion, which negatively impacts the fast, subtle changes important for rPPG. *Typical range:* 12 to 300 frames, depending on application. *Common default:* 60 frames [5,6].
- **B-Frames:** B-frames are frames predicted from both previous and future frames, improving compression efficiency through bidirectional prediction. However, excessive B-frame usage can cause temporal smoothing, potentially blurring the minute motion and color changes that rPPG algorithms depend on. *Typical range:* 0 to 16 frames, depending on GOP length and encoder capabilities. *Common default:* 3 B-frames [6].
- **Reference Frames:** This sets how many past frames are used for motion estimation. More reference frames typically improve prediction accuracy and compression by allowing the encoder to find better temporal matches, but they also increase encoder complexity and memory usage. Additionally, using many reference frames can sometimes propagate compression artifacts or noise from previous frames, which may affect the quality of physiological signals extracted from the video. *Typical range:* 1 to 16 frames. *Common default:* 3 frames [4,7].
- **Sub-Pixel Motion Estimation (SUBQ):** This parameter refines motion vectors to fractional pixel accuracy, most often quarter-pixel, improving motion prediction precision and compression efficiency. While this enhances overall quality, it can also smooth out subtle motion details, potentially diminishing rPPG signal fidelity. *Typical range:* 1 (integer-pixel only) to 12 (maximum quarter-pixel refinement with rate-distortion optimization, RDO) [1,4]. *Common default:* 7 or 8 in x264-based encoders. Higher values yield better compression but increase computational cost and may mask very fine physiological changes.
- **Bitrate Control Mode:** This parameter governs how the encoder allocates bitrate across the video, including modes like CRF (quality-based), Constant Bitrate (CBR), and Variable Bitrate (VBR). Proper bitrate control helps maintain consistent video quality, which is important for preserving subtle physiological signals critical for rPPG analysis. Choosing CRF ensures that each frame is compressed to a visually similar quality regardless of scene complexity, while CBR enforces a fixed data rate, which can result in variable quality, especially in challenging scenes. VBR allows the bitrate to adjust dynamically, offering a balance between quality and file size.

Common modes: CRF (quality-based), CBR (fixed bitrate), VBR (variable bitrate). *Common default:* CRF mode [4, 5].

- **Quantization Parameter (QP) Min/Max:** QP settings limit the minimum and maximum quantization levels allowed, constraining quality variation across frames. Stabilizing quantization helps maintain consistent detail, avoiding sudden quality drops that could impair rPPG. *Typical range:* 0 (lossless) to 51 (lowest quality). *Common default:* Limits typically set within 10–40 to avoid extreme quality swings [4].
- **Adaptive Quantization (AQ):** AQ dynamically adjusts quantization based on video content complexity, allocating more bits to complex areas to enhance perceptual quality, which helps retain fine details necessary for physiological analysis. *Typical settings:* Enabled or disabled; several modes exist for spatial or temporal AQ. *Common default:* Enabled with spatial AQ mode [4].
- **Deblocking Filter Strength:** This controls the intensity of the in-loop filter that smooths block edges to reduce visible compression artifacts. While stronger filtering improves visual quality, it may also suppress subtle texture variations critical to rPPG. *Typical range:* Adjustable strength levels from off (0) to strong (2 or 3). *Common default:* Moderate strength enabled [1].
- **Motion Search Range:** This parameter sets how far the encoder looks, in pixels, when searching for matching blocks between frames during motion estimation. A larger search range enables the encoder to track larger or faster movements, improving motion compensation accuracy—especially for scenes with rapid or large-scale motion. However, increasing the search range also significantly raises computational complexity and encoding time, while having only a marginal effect on overall video quality if set too high. *Typical range:* 16 to 128 pixels. *Common default:* 32 or 64 pixels [1].

After testing different settings and reviewing practical guidelines such as those provided by YouTube [5], we selected the following defaults for our experiments. These settings aim to strike a balance between efficient video compression and the preservation of subtle color and motion variations needed for accurate rPPG estimation (see Table 2.2). In addition, we identified the most influential compression parameters, which are further examined experimentally in the next section.

These default settings achieve a practical balance between keeping the video files compact and preserving the subtle color and motion variations needed for accurate rPPG signal extraction. In this way, the compressed videos still contain the key information for reliable heart-rate estimation while taking full advantage of the efficiency offered by the H.264 standard.

Parameter	Description	Dependencies	Effect on Video Quality / Compression
CRF	Global quantization control	Independent	Major impact on quality and file size
GOP Length	Frequency of I-frames	Affects B-frames, Ref Frames	Long GOP = better compression, higher risk of error
B-Frames	Bidirectional frames	Depends on GOP	May blur motion if excessive
Reference Frames	Used for prediction	Linked with GOP/B	Improves compression, more complex
SUBQ	Motion vector precision	Independent	Higher precision, better motion, possibly smoother
Bitrate Mode	CRF / CBR / VBR	Related to CRF	Controls quality consistency
QP Min/Max	Limits on quantization	Related to CRF	Constrains quality/compression bounds
AQ	Dynamic quantization	Related to QP/CRF	Improves perceptual quality
Deblocking	Edge smoothing	Independent	Improves visuals, may lose detail
Motion Range	Search area for vectors	Affects complexity	Improves motion match, adds cost

Table 2.1: Summary of key H.264 parameters and their influence on video compression and quality relevant to rPPG.

Parameter	Default Setting	Description / Effect
vcodec	libx264	H.264 codec
preset	slow	Better compression efficiency
crf	18	Balanced quality and compression
profile	high	Supports high-quality settings
level	4.2	Enables 1080p resolution
maxrate	8000k	Peak bitrate cap
bufsize	16000k	Buffer for rate control
g	60	GOP interval
refs	3	Reference frames
me_method	umh	Balanced motion estimation
subq	7	Quarter-pixel precision
deblock	1:1	Moderate filtering
trellis	1	Improves quality via entropy coding
pix_fmt	yuv420p	Standard for video encoding
threads	0	Auto CPU usage
acodec	aac	Audio codec
audio_bitrate	192k	Audio quality
ar	48000	Sampling rate for audio
x264opts	bframes=2	Bidirectional frames count

Table 2.2: Default H.264 encoder settings used in our rPPG compression experiments.

2.3 Experimenting by Changing the Parameters

To systematically evaluate the impact of H.264 encoder parameters on the fidelity of physiological signals in compressed video, we performed a series of experiments. For each experiment, a specific parameter was varied while others were held constant at their recommended defaults, allowing us to observe the resulting changes in rPPG signal estimation metrics across different video types and scenarios.

The primary evaluation metrics were:

- **SNR (Signal-to-Noise Ratio):** Indicates the clarity of the extracted physiological signal.
- **Correlation:** Measures the similarity between estimated and ground-truth heart rate signals.
- **DTW (Dynamic Time Warping):** Quantifies the alignment error between two signals time series.
- **BPM Error:** The absolute error between estimated and ground-truth heart rate in beats per minute.

We explore these metrics more in the upcoming section.

It is important to note that all reported metrics (SNR, correlation, DTW, and BPM error) are calculated as averages over groups of participants and video trials for each parameter setting. Specifically, the data represent the mean values across 16 subjects, each of whom was recorded under seven different video conditions. For clarity and consistency, we report results for three representative video types: type 2 (where participants are holding their breath relatively still), type 4 (featuring deliberate side-to-side head movements), and type 7 (captured after physical activity, when heart rate is elevated). These types were chosen because they cover a broad range of realistic motion and physiological scenarios encountered in remote photoplethysmography (rPPG) studies, allowing us to generalize the observed trends and parameter effects. This design ensures that the results are not biased by any single recording but instead reflect the typical performance across diverse conditions.

Effect of CRF Level on rPPG Signal Quality

We first examined the influence of the Constant Rate Factor (CRF), which directly controls the quantization and thus the trade-off between compression and quality. For this analysis, we compared the rPPG performance across a range of CRF levels for three representative video types. All other encoding settings were held constant, matching the default profile described previously.

Table 2.3 presents the results for the ground-truth reference, while Table 2.4 shows the results for uncompressed video. For each video type, four metrics are reported: SNR, Correlation, DTW, and BPM error.

CRF	Video Type 2				Video Type 4				Video Type 7			
	SNR	Corr.	DTW	BPM	SNR	Corr.	DTW	BPM	SNR	Corr.	DTW	BPM
uncompressed	1.949	0.616	3.39	0.50	1.361	0.523	3.63	1.24	1.477	0.587	4.06	1.76
0	1.977	0.562	3.40	0.50	1.386	0.493	3.60	1.24	1.477	0.530	4.07	1.77
5	1.660	0.451	3.69	1.10	1.189	0.414	3.70	0.52	1.213	0.427	3.96	1.85
10	1.074	0.377	3.73	1.53	0.840	0.358	3.93	0.54	0.911	0.361	4.01	1.16
15	0.698	0.270	4.05	0.92	0.651	0.309	4.04	3.18	0.709	0.279	4.17	10.39
20	0.296	0.096	4.56	12.11	0.433	0.196	4.24	4.30	0.367	0.156	4.77	14.40
25	0.201	-0.004	5.13	11.32	0.328	0.049	4.20	11.59	0.134	0.066	5.13	17.87
FacerPPG	0.875	0.385	4.07	1.85	0.575	0.357	3.85	10.71	0.766	0.384	4.33	12.29

Table 2.3: rPPG performance metrics across different CRF levels (using ground-truth signal as reference).

CRF	Video Type 2				Video Type 4				Video Type 7			
	SNR	Corr.	DTW	BPM	SNR	Corr.	DTW	BPM	SNR	Corr.	DTW	BPM
uncompressed	1.952	1.000	0.000	0.00	1.373	1.000	0.000	0.00	1.493	1.000	0.000	0.00
0	1.980	0.962	0.353	0.01	1.397	0.965	0.389	0.01	1.493	0.959	0.393	0.01
5	1.683	0.799	1.252	1.44	1.207	0.820	1.087	0.86	1.234	0.788	1.271	0.19
10	1.112	0.680	1.711	1.12	0.860	0.727	1.624	0.86	0.936	0.681	1.765	1.84
15	0.725	0.551	2.097	0.50	0.692	0.626	1.845	1.99	0.750	0.555	2.045	11.65
20	0.306	0.264	3.086	11.90	0.498	0.460	2.436	3.72	0.478	0.380	2.678	16.16
25	0.191	0.068	3.675	11.14	0.365	0.194	3.076	10.75	0.183	0.179	3.360	19.43
FacerPPG	0.875	0.385	4.07	1.85	0.575	0.357	3.85	10.71	0.766	0.384	4.33	12.29

Table 2.4: rPPG performance metrics across different CRF levels (using uncompressed video as reference).

The above tables present a comprehensive overview of how rPPG signal quality metrics respond to varying levels of CRF (Constant Rate Factor), reported for both the ground-truth physiological signal and the uncompressed video as reference. These two reference choices allow us to separate the effects of video compression from other sources of signal distortion, providing a detailed understanding of how compression alone influences signal quality. Using the ground-truth signal as a reference evaluates the physiological accuracy of the recovered rPPG, whereas comparison with the uncompressed video isolates the degradation caused specifically by compression artifacts.

A clear pattern is visible in both tables: as CRF increases, the Signal-to-Noise Ratio (SNR) and correlation coefficient consistently decrease, while the Dynamic Time Warping (DTW) distance and BPM error rise sharply. This trend is remarkably consistent across all three video types, though the absolute values and the steepness of the changes differ between types. For example, in video type 2, SNR drops from approximately 1.95 (uncompressed) to just 0.20 at CRF 25 when using ground-truth as reference, while

the corresponding DTW increases from 3.39 to 5.13 and BPM error surges from 0.5 to 11.3. Similar trends are observed in video types 4 and 7, although the rate of signal degradation may vary based on scene content and motion.

The underlying cause of these changes is the progressive loss of fine temporal and color information as compression becomes more aggressive. At low CRF values (i.e., higher video quality), both SNR and correlation remain relatively high, indicating that most physiologically relevant information is preserved and the extracted rPPG waveform closely matches the reference. In these settings, DTW distances and BPM errors are minimal, reflecting both accurate signal shape and frequency estimation. However, as CRF increases beyond 10, there is a rapid degradation: SNR and correlation drop off quickly, indicating a loss of pulse-related features and temporal coherence, while DTW and especially BPM error rise substantially. Notably, the BPM error remains low and stable for $\text{CRF} \leq 10$, but then increases dramatically at CRF 15 and above, reaching extreme values (e.g., 10–18 BPM error) for the highest compression levels. This indicates that the signal may still exhibit superficial periodicity (allowing DTW-based alignment), but its frequency content becomes too distorted for reliable heart rate estimation.

When comparing the use of ground-truth versus uncompressed video as a reference, we see that both reveal the same qualitative patterns: all metrics degrade with stronger compression, but the absolute values differ. The uncompressed video reference yields perfect or near-perfect metrics at the lowest CRF, reflecting that this comparison isolates only the effect of compression itself (removing any algorithmic bias). In contrast, the ground-truth reference reflects the full signal recovery challenge, incorporating not only compression losses but also the inherent difficulty of recovering physiological signals from video. The fact that both reference choices show parallel degradation as CRF increases supports the conclusion that compression by changing CRF level, is the dominant factor controlling rPPG signal fidelity in this experiment.

Overall, these findings underscore the importance of careful compression parameter selection for rPPG-based physiological monitoring. While moderate compression (e.g., CRF 10 or below) preserves the majority of physiologically relevant information, higher CRF values introduce pronounced signal loss, affecting not just waveform shape and correlation, but especially the accuracy of frequency-based metrics such as BPM. It is thus crucial to avoid aggressive compression when reliable heart rate extraction is required. These patterns are visually summarized in Figure 2.3, which illustrates the effect of CRF on all four metrics and three representative video types.

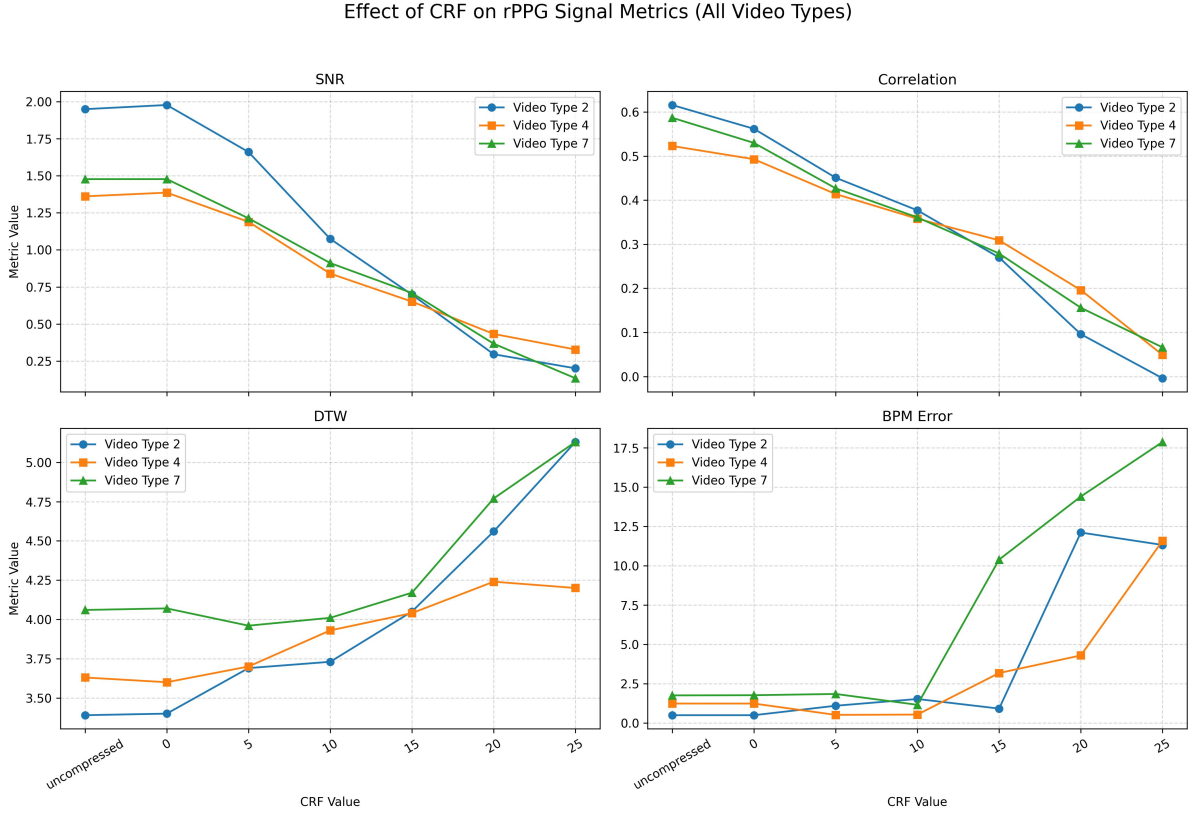


Figure 2.3: Effect of CRF levels on rPPG signal quality across video types 2, 4, and 7, measured by SNR, correlation, DTW, and BPM. Each plot shows the trend for one metric as compression increases.

Effect of GOP Size, B-Frames, Reference Frames, and Sub-Pixel Motion Estimation

Next, we fixed the CRF at 10 and systematically varied four key encoder parameters: GOP size, number of reference frames, sub-pixel motion estimation setting (SUBQ), and the number of B-frames. The results of these experiments, using the ground-truth physiological signal as reference, are summarized in Table 2.5. This table presents the effects of each parameter variation across all evaluation metrics and video types side by side, enabling a direct and comprehensive comparison of how each setting influences rPPG signal preservation and estimation quality under controlled compression conditions.

Parameter	Setting	Video Type 2				Video Type 4				Video Type 7			
		SNR	Corr.	DTW	BPM	SNR	Corr.	DTW	BPM	SNR	Corr.	DTW	BPM
B-Frames	uncompressed	1.949	0.616	3.39	0.50	1.361	0.523	3.63	1.24	1.477	0.587	4.06	1.76
	0	1.662	0.506	3.56	0.61	1.103	0.459	3.69	1.29	1.290	0.495	4.09	1.86
	1	1.333	0.485	3.59	0.17	0.904	0.423	3.76	2.62	1.006	0.459	4.09	2.13
	2 (default)	1.274	0.405	3.80	1.46	0.885	0.375	3.71	3.84	1.047	0.384	4.20	1.44
	3	0.983	0.388	3.77	1.84	0.796	0.383	3.71	1.21	0.888	0.364	4.17	2.48
	4	1.015	0.388	4.14	2.00	0.831	0.360	3.75	0.50	1.022	0.364	4.39	2.17
	5	0.796	0.381	3.79	2.07	0.763	0.381	3.80	1.31	0.885	0.373	4.28	1.64
	6	1.148	0.409	3.76	0.54	0.830	0.380	3.77	0.88	0.970	0.386	4.19	2.59
	facerPPG	0.875	0.385	4.07	1.85	0.575	0.357	3.85	10.71	0.766	0.384	4.33	12.29
GOP Size	uncompressed	1.949	0.616	3.39	0.50	1.361	0.523	3.63	1.24	1.477	0.587	4.06	1.76
	1	1.721	0.550	3.40	0.16	1.284	0.474	3.63	1.25	1.325	0.512	4.03	2.12
	60 (default)	1.074	0.377	3.73	1.53	0.840	0.358	3.93	0.54	0.911	0.361	4.01	1.16
	120	1.126	0.383	3.75	1.54	0.866	0.367	3.93	0.54	0.946	0.361	4.06	1.36
	240	1.104	0.382	3.79	1.55	0.912	0.369	3.98	1.30	1.005	0.367	4.04	0.98
	facerPPG(30)	0.875	0.385	4.07	1.85	0.575	0.357	3.85	10.71	0.766	0.384	4.33	12.29
	facerPPG(30)	0.875	0.385	4.07	1.85	0.575	0.357	3.85	10.71	0.766	0.384	4.33	12.29
Ref. Frames	uncompressed	1.949	0.616	3.39	0.50	1.361	0.523	3.63	1.24	1.477	0.587	4.06	1.76
	0	1.208	0.390	3.66	1.50	0.934	0.376	3.87	0.59	0.997	0.369	3.98	1.45
	1	1.208	0.390	3.66	1.50	0.934	0.376	3.87	0.59	0.997	0.369	3.98	1.45
	3 (default)	1.074	0.377	3.73	1.53	0.840	0.358	3.93	0.54	0.911	0.361	4.01	1.16
	6	0.967	0.365	3.83	1.54	0.890	0.363	3.92	0.53	0.871	0.353	4.11	2.87
	9	0.894	0.365	4.01	1.46	0.857	0.359	3.92	0.56	0.837	0.355	4.04	1.15
	12	0.904	0.358	3.85	1.51	0.904	0.359	3.93	0.49	0.826	0.357	4.13	8.51
	15	0.877	0.363	3.83	1.52	0.852	0.361	3.97	0.51	0.792	0.363	4.17	2.23
	facerPPG(2)	0.875	0.385	4.07	1.85	0.575	0.357	3.85	10.71	0.766	0.384	4.33	12.29
SUBQ	uncompressed	1.949	0.616	3.39	0.50	1.361	0.523	3.63	1.24	1.477	0.587	4.06	1.76
	0	1.180	0.357	3.84	1.50	0.922	0.361	4.06	2.06	1.061	0.342	4.18	1.52
	1	1.048	0.355	3.89	1.52	0.764	0.332	3.89	1.33	0.855	0.343	4.05	1.46
	3	0.934	0.301	4.08	1.17	0.773	0.319	3.93	0.58	0.821	0.306	4.22	1.58
	5	0.897	0.308	4.08	1.53	0.847	0.340	3.97	0.61	0.843	0.320	4.22	0.61
	7 (default)	1.074	0.377	3.73	1.53	0.840	0.358	3.93	0.54	0.911	0.361	4.01	1.16
	9	1.099	0.384	3.79	1.52	0.919	0.367	3.86	0.59	0.891	0.364	4.02	1.43
	12	1.099	0.384	3.79	1.52	0.919	0.367	3.86	0.59	0.891	0.364	4.02	1.43
	facerPPG(2)	0.875	0.385	4.07	1.85	0.575	0.357	3.85	10.71	0.766	0.384	4.33	12.29

Table 2.5: Effect of GOP size, reference frames, and sub-pixel motion estimation (SUBQ) on rPPG signal quality at CRF=10. Each parameter is varied independently; metrics shown are SNR, correlation, DTW, and BPM error for each video type.

This table provides an overview of how four central H.264 encoder parameters, GOP size, reference frames, sub-pixel motion estimation (SUBQ), and the number of B-frames, affect rPPG signal quality when the CRF is held constant at 10. For each parameter, results are shown across all video types and reported for standard rPPG evaluation metrics. In each block, a single parameter is varied while the

others are kept fixed, making it possible to isolate and compare the individual effect of each setting on signal fidelity. In the following of this section, we examine the behavior of each parameter in more detail, interpreting the results for each metric to highlight their practical impact on physiological signal preservation.

B-Frames. Changing the number of B-frames has a clear effect on rPPG signal quality, as seen in both Table 2.5 and Figure 2.4. Both SNR and correlation drop quickly as we add more B-frames. For example, in video type 2, the SNR starts at about 1.9 for uncompressed video and falls to around 0.8 when the number of B-frames is set to 5. Correlation follows a similar trend, dropping from 0.62 to below 0.4. This shows that increasing B-frames generally makes it harder for the model to preserve a strong and accurate physiological signal.

DTW, which measures the alignment between estimated and reference signals, also gets worse (higher values) as we increase B-frames, but the change isn't as sharp as with SNR or correlation.

When it comes to BPM error, things get a bit chaotic. Instead of a smooth increase or decrease, the BPM error jumps up and down at different B-frame settings. For instance, in video type 4, BPM error goes from 1.3 at zero B-frames to nearly 3.8 at two B-frames, then back down again. In other video types, the same irregular pattern appears. This shows that BPM error is much less reliable as a sole indicator of signal quality when B-frames are varied, since it doesn't follow a predictable trend and can be influenced by small distortions in unpredictable ways. So, while more B-frames usually make the signal noisier and less correlated, BPM error can be quite erratic, which is why it's important not to rely only on this metric, but to consider the full set of signal quality measures.

GOP Size. The effect of GOP size on rPPG signal quality is clear when looking at both Table 2.5 and Figure 2.5. When the GOP size is very small, meaning there are frequent keyframes, such as GOP size 1, the SNR and correlation values stay relatively high. For example, with GOP size 1, video type 2 shows an SNR of 1.72 and a correlation of 0.55, while video type 4 has an SNR of 1.28 and correlation of 0.47. As the GOP size increases to the default value of 60, SNR and correlation drop to around 1.07 and 0.38 for video type 2, and to 0.84 and 0.36 for video type 4. When the GOP size is extended further to 120 or 240, SNR and correlation remain low and quite stable, showing only minor changes e.g., SNR 1.10 and correlation 0.38 for video type 2 at 240.

At the same time, DTW values gradually increase as the GOP size gets larger. In video type 2, DTW rises from 3.40 at GOP size 1 to 3.73 at 60, and up to 3.79 at 240. This indicates a growing misalignment as fewer keyframes are used.

BPM error, on the other hand, does not follow any clear or consistent pattern. For video type 2, the BPM error is very low at GOP size 1 0.16, but then jumps to 1.53 at 60, 1.54 at 120, and 1.55 at 240. For other video types, BPM error fluctuates in a similarly unpredictable way, with no obvious relationship to GOP size. These irregular jumps again suggest that BPM error is not a reliable indicator of overall signal quality in this context.

The patterns in both the table and Figure 2.5 make it clear that more frequent keyframes, which means shorter GOPs, help preserve physiological signal quality, while larger GOP sizes allow more temporal drift and degradation. However, to accurately interpret signal quality, it is important to rely on multiple metrics, as BPM error alone can be misleading.

Reference Frames. Changing the number of reference frames has a noticeable impact on rPPG signal quality, as seen in Table 2.5 and Figure 2.6. When no reference frames are used like setting 0 or 1, SNR and correlation drop sharply compared to the uncompressed case: for video type 2, SNR drops from 1.95 (uncompressed) to about 1.21, and correlation falls from 0.62 to around 0.39. As the number of reference frames increases to the default of 3, SNR and correlation both reach their moderate compressed values, SNR is 1.07 and correlation is 0.38 for video type 2, and similar changes appear for the other video types. Beyond this point, further increasing the reference frame count leads to diminishing returns. For instance, SNR gradually decreases from 0.97 at 6 frames down to 0.88 at 15 frames for video type 2, and correlation shows only minor changes, mostly flattening out around 0.36. DTW error also rises slightly as reference frames increase, indicating a growing misalignment. Most notably, BPM error can spike at higher settings, such as a jump to 8.5 for video type 7 with 12 reference frames, which is a clear outlier in an otherwise relatively stable trend.

Overall, using a moderate number of reference frames such as the default 3 provides the reasonable balance, yielding high SNR and correlation with low error rates. Increasing reference frames further does not improve results and can even degrade signal quality, likely due to excessive temporal blending that starts to obscure the subtle, frame-to-frame color changes needed for accurate rPPG estimation. This pattern holds across all three video types, and most compressed settings outperform the FacerPPG baseline, especially when reference frame counts are set moderately.

Sub-Pixel Motion Estimation (SUBQ). Adjusting the SUBQ parameter has a clear influence on rPPG signal quality, as illustrated in both Table 2.5 and Figure 2.7. When SUBQ is set to very low values such as 0 or 1, both SNR and correlation drop sharply, SNR for video type 2, for example, falls from 1.95 (uncompressed) to 1.18 at SUBQ 0, with correlation dropping from 0.62 to 0.36. Video type 4 shows a similar decline, with SNR dropping from 1.36 (uncompressed) to 0.92 at SUBQ 0. Increasing SUBQ to intermediate values, particularly around the default of 7, provides the best reasonable results across all metrics. For instance, at SUBQ 7, SNR for video type 2 rebounds to 1.07 and correlation improves to 0.38, and these improvements are echoed for the other video types.

Pushing SUBQ higher to values like 9 or 12, however, does not continue to boost performance. Instead, SNR and correlation tend to decline slightly, suggesting there's no advantage in maximizing SUBQ beyond a moderate setting. DTW also remains relatively flat across SUBQ values but tends to be lowest at the intermediate settings, which aligns with improved alignment between reference and estimated signals.

As seen elsewhere, BPM error does not follow a consistent pattern, jumping around between SUBQ values and video types, making it less reliable for drawing conclusions.

Overall, these results highlight that setting SUBQ too low or too high both negatively impact signal quality, but the default value of 7 offers a sweet spot. This intermediate level of motion estimation accuracy appears to be sufficient for capturing the subtle motion changes needed for rPPG, without introducing too much smoothing or losing fine detail.

Conclusion. These experiments show just how much the main H.264 encoder settings can impact rPPG signal quality. Increasing the number of B-frames made the SNR and correlation drop, and BPM error became more unpredictable. For GOP size, shorter intervals, especially the default of 60, helped

keep the signal strong, while very large intervals led to lower quality. With reference frames, 3 was the sweet spot; adding more did not help and sometimes made things worse. And for sub-pixel motion estimation (SUBQ), the default setting of 7 worked best, while going too low or too high reduced signal quality. Overall, using these moderate, literature-backed defaults, GOP=60, B-frames=2, references=3, and SUBQ=7, gives the most robust performance and even outperforms tools like FaceReader on most metrics.

The plots below summarize all the numbers and trends for each parameter.

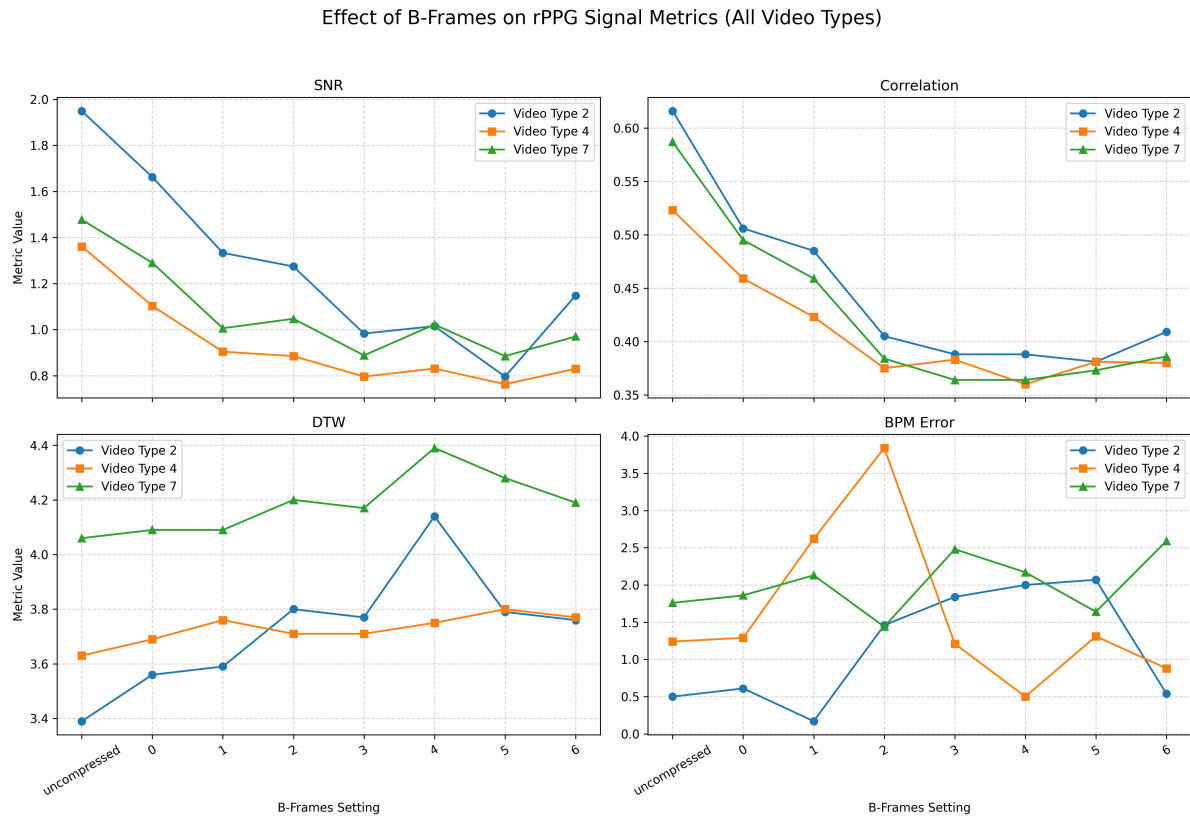


Figure 2.4: Effect of B-Frames on rPPG signal quality across video types 2, 4, and 7.

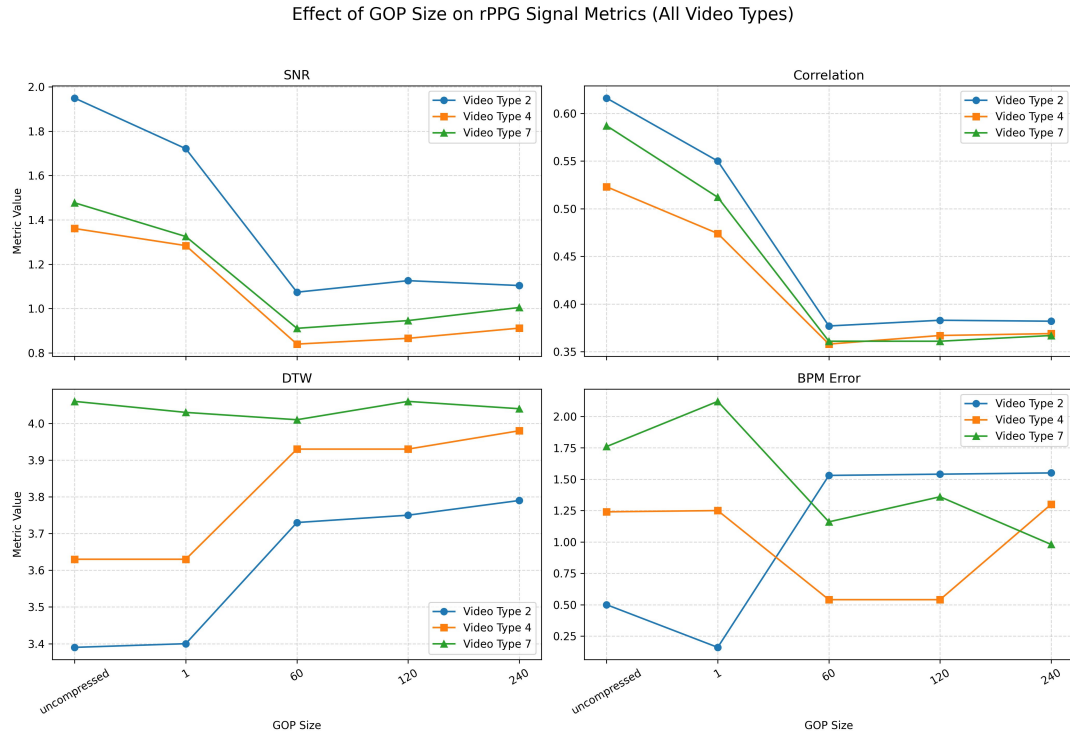


Figure 2.5: Effect of GOP size on rPPG signal quality across video types.

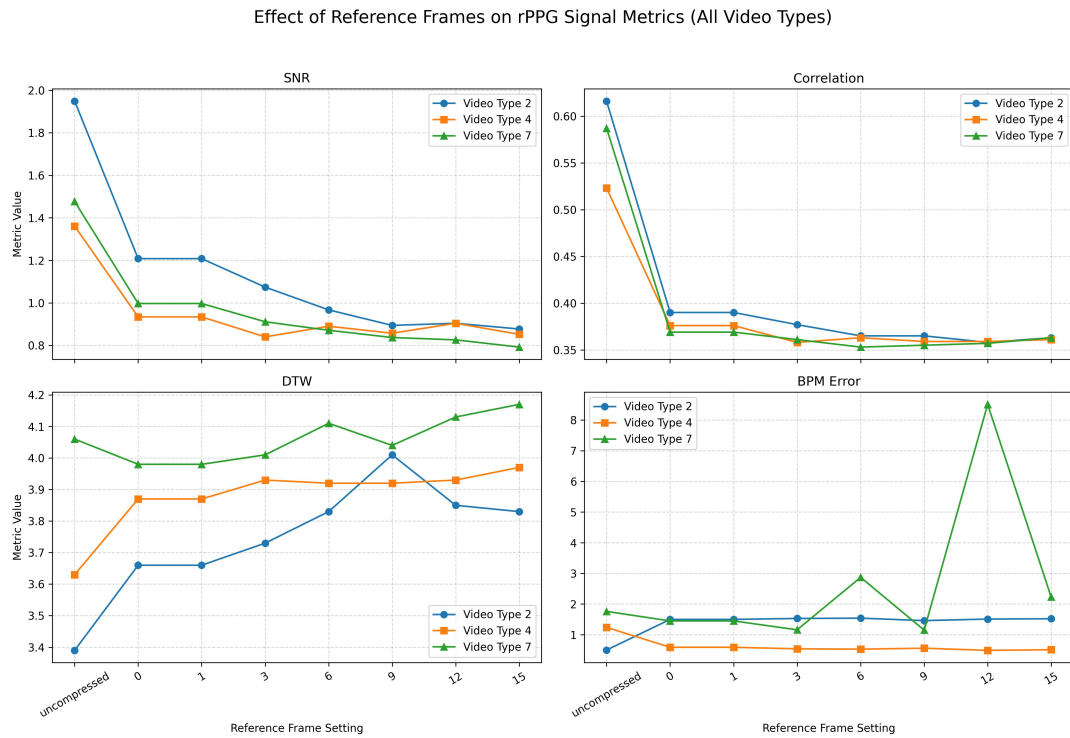


Figure 2.6: Effect of reference frame count on rPPG signal quality.

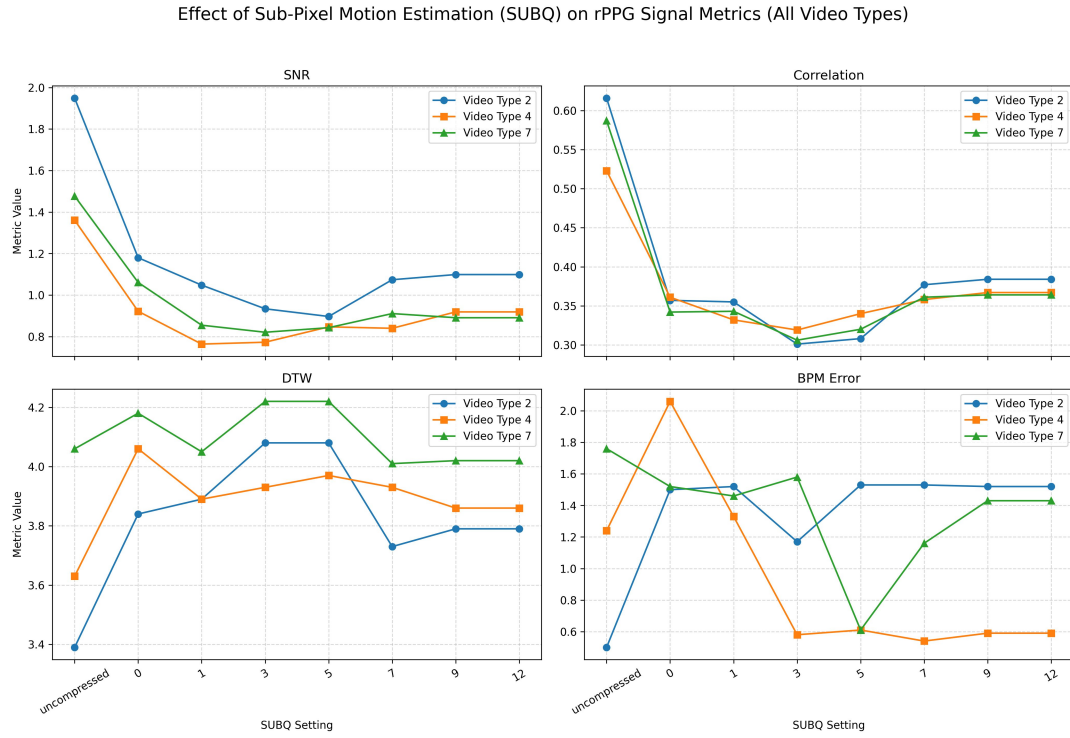


Figure 2.7: Effect of SUBQ setting on rPPG signal quality across all metrics and video types.

2.4 Metrics in Use

In the previous section, we reported the effects of different H.264 encoder parameters on several key metrics. Here, we go into mathematical depth for each metric, Signal-to-Noise Ratio (SNR), correlation coefficient, Dynamic Time Warping (DTW), and BPM error, explaining their definitions simply, how we apply them to the rPPG signal, and what they reveal about signal fidelity after compression.

Signal-to-Noise Ratio (SNR)

Signal-to-Noise Ratio (SNR) quantifies how much of the desired physiological signal remains in the presence of noise or error introduced by compression or algorithmic estimation. Mathematically, let $s = [s_1, s_2, \dots, s_N]$ denote the ground-truth rPPG signal and $x = [x_1, x_2, \dots, x_N]$ the estimated signal extracted from video, both of equal length N . The error at each sample n is defined as $e_n = s_n - x_n$. SNR is calculated in decibels as

$$\text{SNR} = 10 \log_{10} \left(\frac{\sum_{n=1}^N s_n^2}{\sum_{n=1}^N (s_n - x_n)^2} \right)$$

where the numerator represents the total power of the reference (signal), and the denominator is the power of the error signal (difference between ground-truth and estimated signals). In this thesis, SNR is computed for every trial and subject, and then averaged across groups of interest, such as all recordings for a given video type. All signals are normalized prior to calculation to eliminate amplitude scaling effects and ensure comparability across experiments. High SNR values indicate that the estimated rPPG signal retains most of the true physiological variations with minimal noise or distortion, while low SNR reflects increasing loss of signal fidelity, which can result from aggressive video compression or limitations in the extraction algorithm. Since rPPG relies on the detection of subtle changes in color, maintaining high SNR is essential for reliable heart rate extraction; significant reductions in SNR can make physiological measurement from video data unreliable.

Correlation Coefficient

Correlation is a statistical measure that quantifies the degree to which two time series co-vary with each other in terms of their shape and temporal fluctuations, regardless of their absolute amplitudes. In the context of rPPG evaluation, the Pearson correlation coefficient is used to capture the linear relationship between the estimated physiological signal and the ground-truth reference signal. For two sequences $x = [x_1, x_2, \dots, x_N]$, which is the estimated signal, and $s = [s_1, s_2, \dots, s_N]$ that represents the ground-truth, their respective means are computed as $\bar{x} = \frac{1}{N} \sum_{n=1}^N x_n$ and $\bar{s} = \frac{1}{N} \sum_{n=1}^N s_n$. The Pearson correlation coefficient is then defined as

$$\text{Corr}(x, s) = \frac{\sum_{n=1}^N (x_n - \bar{x})(s_n - \bar{s})}{\sqrt{\sum_{n=1}^N (x_n - \bar{x})^2} \sqrt{\sum_{n=1}^N (s_n - \bar{s})^2}}$$

This expression yields a value between -1 and 1 , where values near 1 indicate that the estimated and reference signals fluctuate together in a highly similar, temporally aligned manner, while values near 0

suggest a lack of linear relationship, and negative values indicate that the signals change in opposing directions. In practice, before calculating the correlation, both the rPPG and ground-truth signals are first adjusted by removing their mean. If needed, they are also aligned in time by shifting or resampling them, so that small delays caused by processing or recording do not affect the comparison. In this thesis, correlation is computed for every subject and trial, and then averaged over the relevant groups. High correlation coefficients in the context of rPPG analysis show that the waveform structure of the estimated pulse signal closely matches the true physiological dynamics, confirming the preservation of the important temporal features even after video compression.

Dynamic Time Warping (DTW)

Dynamic Time Warping (DTW) is a metric that measures how similar two signals are, even if their features do not line up perfectly in time. This is very helpful for rPPG analysis, since heart rate signals from video and the ground truth are often not exactly synchronized due to natural variations or processing delays.

To put it simply, DTW works by stretching and shifting one signal so that its important features match up as closely as possible with the other signal. Mathematically, suppose we have an estimated signal $x = [x_1, \dots, x_N]$ and a ground-truth signal $s = [s_1, \dots, s_M]$, which can be of different lengths. DTW constructs a cost matrix, where each entry measures how different the points x_i and s_j are:

$$D(i, j) = (x_i - s_j)^2$$

Then, DTW finds the best path through this matrix, called a warping path, that adds up the differences along the way. The DTW distance is simply the total cost along this optimal path:

$$\text{DTW}(x, s) = \min_W \sum_{(i,j) \in W} D(i, j)$$

where W is a sequence of matching points from both signals, always moving forward in time. This approach allows DTW to align similar shapes in the two signals, even if they occur at slightly different times. In practice, both signals are usually normalized for example, with z-score normalization, and resampled to the same length before computing DTW, to make sure that differences in scale or sampling do not affect the result.

In this thesis, DTW is used to compare the estimated and ground-truth pulse signals for every trial and subject. A lower DTW value means that the estimated signal closely follows the timing and shape of the ground-truth signal, which is what we want for reliable rPPG.

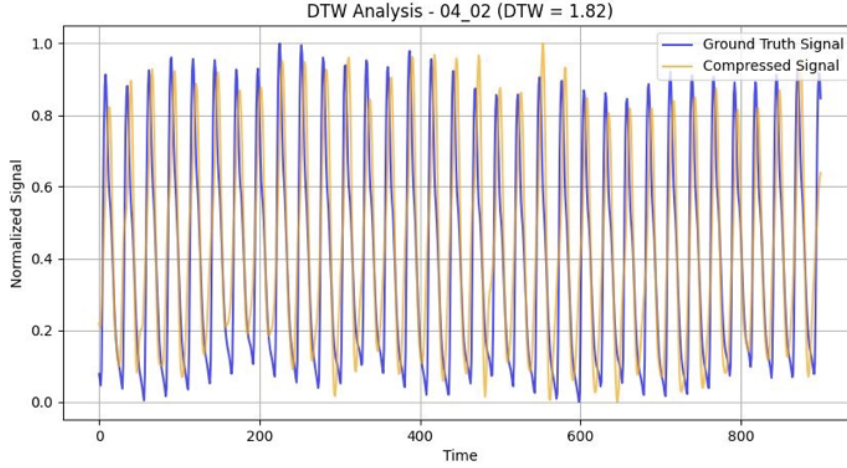


Figure 2.8: Example of DTW analysis comparing a ground-truth and compressed rPPG signal from the same video. The DTW value quantifies how well the two signals match in shape and timing, even when there are local shifts.

BPM Error (Heart Rate Estimation Error)

BPM error is a straightforward metric used to assess the accuracy of heart rate estimation from video-based rPPG signals, relative to a ground-truth physiological reference. The calculation is based on comparing the heart rate, expressed in beats per minute (BPM), derived from the estimated rPPG signal with that obtained from a reference measurement.

To compute BPM error, both the estimated and reference signals are typically segmented into fixed analysis windows e.g., 30 seconds, detrended, and normalized. Frequency analysis, such as the Fast Fourier Transform (FFT), is then applied to each segment to identify the dominant frequency component, which corresponds to the heart rate for that window. This frequency is converted to BPM by multiplying by 60. Denoting the heart rate estimated from the rPPG signal as BPM_{est} and the heart rate from the reference as BPM_{gt} , the BPM error for each window is defined as:

$$\text{BPM Error} = |\text{BPM}_{\text{est}} - \text{BPM}_{\text{gt}}|$$

Depending on the analysis, this error can be computed for each window or over the entire signal duration. In this work, the average BPM error across all subjects, video types, and encoder settings is reported to summarize the impact of compression on heart rate estimation performance.

BPM error is widely used due to its intuitive interpretation and relevance in practical and clinical contexts, where accurate heart rate monitoring is essential. However, as demonstrated in previous results, BPM error does not always align with other signal quality metrics such as SNR or correlation. The metric may vary unpredictably, particularly when compression artifacts shift the dominant frequency or introduce false spectral peaks. Consequently, while BPM error provides a useful summary of heart rate estimation accuracy, it may not always reliably capture the underlying fidelity of the rPPG signal in all conditions.

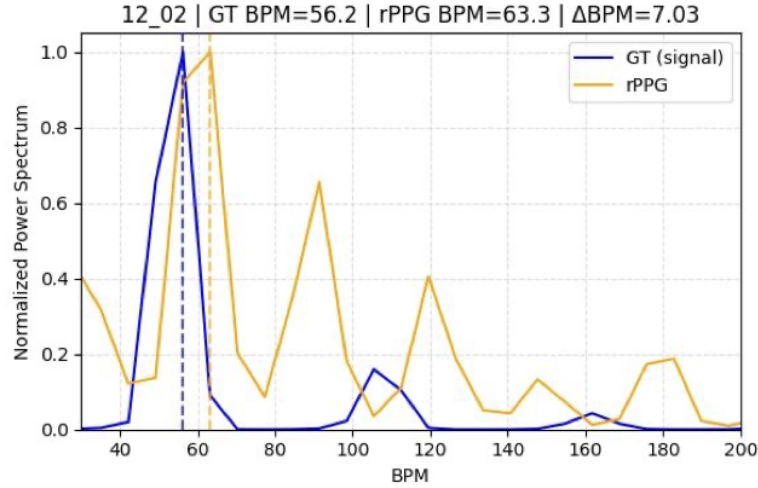


Figure 2.9: Power spectrum analysis of the ground-truth and estimated rPPG signals from the same video. The peak frequencies correspond to the true and estimated heart rates. The distance between peaks indicates the BPM error for this example.

For all the metrics discussed above, the results were then averaged across relevant groups, typically all 16 subjects within each video type and experimental condition. The analysis focuses on three key video types: type 2 (rest), type 4 (head movement), and type 7 (post-exercise), which together represent a range of realistic situations and motion levels encountered in remote heart rate monitoring. By combining SNR, correlation, DTW, and BPM error, the impact of H.264 compression on rPPG signal quality can be examined from different perspectives: how well the signal is preserved, how similar its shape is to the original, how robust it is to timing differences, and how accurately heart rate can be estimated. This thorough and multi-angle evaluation makes it possible to clearly compare the effects of different encoding settings and understand what they mean for developing reliable, compression-aware rPPG methods in practice.

2.5 Frequency-Domain Analysis of rPPG Signals

In the previous section, H.264 compression settings and their effects on rPPG signal quality were carefully described using different metrics such as SNR, correlation, DTW, and BPM error. The meaning and calculation of these metrics were also explained. While these numbers help to show how compression changes the rPPG signal, they do not fully explain why these changes happen.

In this section, the focus is on the frequency content of rPPG signals to better understand the results. The goal is to see how compression settings affect the important heart rate frequencies, that is usually between 0.7 and 3.5 Hz). By looking at how well the heart rate peak is kept in the frequency spectrum for each setting, the changes seen in SNR, correlation, DTW, and BPM error can be better understood. This frequency-based view helps explain how compression changes the true physiological information in the rPPG signals.

2.5.1 Mathematical Framework for Frequency Analysis and Power Spectral Density

Understanding the frequency content of physiological signals is fundamental to remote photoplethysmography (rPPG) and to evaluating the effects of video compression. This section presents the mathematical tools needed to analyze and quantify these frequency characteristics: the Discrete Fourier Transform (DFT), Parseval's theorem, Power Spectral Density (PSD), the Wiener–Khinchin theorem, and robust estimation via Welch's method.

Discrete Fourier Transform (DFT)

Frequency analysis begins with the Discrete Fourier Transform, which decomposes a discrete-time signal into its frequency components. For a signal $x[n]$, $n = 0, 1, \dots, N - 1$, sampled at frequency f_s , the DFT and its inverse are defined as

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}, \quad k = 0, 1, \dots, N - 1 \quad (2.1)$$

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j2\pi kn/N}, \quad n = 0, 1, \dots, N - 1 \quad (2.2)$$

Each k corresponds to the frequency $f_k = \frac{k f_s}{N}$.

It is important to ensure that energy is preserved when moving between time and frequency domains. Parseval's theorem formalizes this conservation, which is essential for meaningful spectral analysis and signal comparison.

Theorem 1 (Parseval's Theorem). *Let $x[n]$ be a discrete-time signal of length N with discrete Fourier transform (DFT) $X[k]$. Then the total energy of the signal in the time domain equals the total energy in the frequency domain, scaled by $1/N$:*

$$\sum_{n=0}^{N-1} |x[n]|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X[k]|^2. \quad (2.3)$$

Proof. We begin by writing the energy of the sequence $x[n]$ as a sum of squared magnitudes:

$$\sum_{n=0}^{N-1} |x[n]|^2 = \sum_{n=0}^{N-1} x[n] x^*[n] \quad (2.4)$$

Next, substitute the inverse DFT for both $x[n]$ and $x^*[n]$:

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j2\pi kn/N}, \quad x^*[n] = \frac{1}{N} \sum_{l=0}^{N-1} X^*[l] e^{-j2\pi ln/N} \quad (2.5)$$

Plugging these into the sum, we get:

$$\sum_{n=0}^{N-1} x[n] x^*[n] = \sum_{n=0}^{N-1} \left(\frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j2\pi kn/N} \right) \left(\frac{1}{N} \sum_{l=0}^{N-1} X^*[l] e^{-j2\pi ln/N} \right) \quad (2.6)$$

Expand both sums and collect all exponents:

$$= \frac{1}{N^2} \sum_{n=0}^{N-1} \sum_{k=0}^{N-1} \sum_{l=0}^{N-1} X[k] X^*[l] e^{j2\pi n(k-l)/N} \quad (2.7)$$

Now, move the sum over n to the innermost position :

$$= \frac{1}{N^2} \sum_{k=0}^{N-1} \sum_{l=0}^{N-1} X[k] X^*[l] \left(\sum_{n=0}^{N-1} e^{j2\pi n(k-l)/N} \right) \quad (2.8)$$

The inner sum is a sum of complex exponentials. By the orthogonality property of the DFT:

$$\sum_{n=0}^{N-1} e^{j2\pi n(k-l)/N} = \begin{cases} N & \text{if } k = l \\ 0 & \text{if } k \neq l \end{cases} \quad (2.9)$$

Therefore, only terms with $k = l$ remain:

$$= \frac{1}{N^2} \sum_{k=0}^{N-1} X[k] X^*[k] N = \frac{1}{N} \sum_{k=0}^{N-1} |X[k]|^2 \quad (2.10)$$

This completes the proof. [8]

□

Power Spectral Density (PSD) and the Periodogram

To study how the energy of a signal is distributed across different frequencies, we use the *Power Spectral Density* (PSD). The PSD characterizes the power of a signal per unit frequency and is a fundamental tool for analyzing physiological signals such as rPPG, where the dominant frequency corresponds to the heart-rate component.

For a deterministic discrete-time signal $x[n]$ of length N , with sampling frequency f_s and discrete Fourier transform $X[k]$, the *periodogram* provides a simple estimate of the PSD at frequency f_k :

$$P_{\text{periodogram}}(f_k) = \frac{1}{N f_s} |X[k]|^2. \quad (2.11)$$

Intuitively, the periodogram measures how much of the signal's total energy is concentrated around each frequency component. As the observation length $N \rightarrow \infty$, this estimate converges to the true PSD and becomes asymptotically unbiased. However, for finite N , the periodogram exhibits high variance due to the limited data length, making it an inconsistent estimator of the true spectrum. This is why, in practice, averaging or windowing techniques e.g., Welch's method are often applied to produce smoother and more reliable PSD estimates.

For random signals, particularly wide-sense stationary (WSS) processes, the PSD has a direct theoretical relationship with the autocorrelation function $R_x[\tau]$. This relationship is established by the *Wiener-Khinchin theorem*, which states that the PSD is the discrete-time Fourier transform of the

autocorrelation sequence:

$$S_x(f) = \sum_{\tau=-\infty}^{\infty} R_x[\tau] e^{-j2\pi f\tau}. \quad (2.12)$$

Hence, the PSD captures how the signal's statistical dependence over time (as represented by its autocorrelation) translates into energy distribution over frequency. This connection provides the theoretical foundation for estimating spectral properties of stochastic physiological signals directly from data, and therefore motivates the detailed discussion of the Wiener–Khinchin theorem.

Theorem 2 (Wiener–Khinchin Theorem). *Let $x[n]$ be a zero-mean, wide-sense stationary (WSS) discrete-time process with autocorrelation sequence*

$$R_{xx}[m] = \mathbb{E}[x[n] x^*[n - m]].$$

Then the power spectral density (PSD) $S_{xx}(f)$ is the discrete-time Fourier transform of the autocorrelation function:

$$S_{xx}(f) = \sum_{m=-\infty}^{\infty} R_{xx}[m] e^{-j2\pi fm/f_s}. \quad (2.13)$$

Conversely, the autocorrelation sequence $R_{xx}[m]$ can be recovered by the inverse Fourier transform of the PSD:

$$R_{xx}[m] = \int_{-f_s/2}^{f_s/2} S_{xx}(f) e^{j2\pi fm/f_s} df. \quad (2.14)$$

Proof. Let $x[n]$ be a realization of a zero-mean, wide-sense stationary (WSS) discrete-time process, and let $X_N(f)$ denote its discrete-time Fourier transform (DTFT) over N samples:

$$X_N(f) = \sum_{n=0}^{N-1} x[n] e^{-j2\pi fn/f_s}.$$

To relate the power spectral density (PSD) to the autocorrelation function, we consider the expected value of the periodogram:

$$\mathbb{E}\left[\frac{1}{N}|X_N(f)|^2\right] = \mathbb{E}\left[\frac{1}{N}X_N(f)X_N^*(f)\right].$$

Expanding the product and bringing the expectation inside yields:

$$\frac{1}{N} \sum_{n=0}^{N-1} \sum_{m=0}^{N-1} \mathbb{E}[x[n] x^*[m]] e^{-j2\pi f(n-m)/f_s}.$$

Since $x[n]$ is WSS, its autocorrelation depends only on the lag $k = n - m$, i.e. $\mathbb{E}[x[n] x^*[m]] = R_{xx}[k]$. Substituting this into the expression gives:

$$\frac{1}{N} \sum_{n=0}^{N-1} \sum_{m=0}^{N-1} R_{xx}[n - m] e^{-j2\pi f(n-m)/f_s}.$$

By introducing the change of variables $k = n - m$ and observing that, as $N \rightarrow \infty$, the double sum

covers all integer values of $k \in \mathbb{Z}$, the average converges to

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} \sum_{m=0}^{N-1} R_{xx}[n-m] e^{-j2\pi f(n-m)/f_s} = \sum_{k=-\infty}^{\infty} R_{xx}[k] e^{-j2\pi f k/f_s}.$$

Hence, the expected value of the periodogram converges to the discrete-time Fourier transform of the autocorrelation sequence as $N \rightarrow \infty$. This demonstrates that the PSD and the autocorrelation form a Fourier transform pair, completing the proof. [8]

□

The Wiener–Khinchin theorem establishes the fundamental relationship between the autocorrelation function and the power spectral density, providing the theoretical foundation for spectral analysis of stochastic signals. However, directly estimating the PSD from finite, noisy data using this definition or the basic periodogram often results in high variance and unstable spectral estimates. To address these limitations, practical methods such as the *Welch method* are employed, which reduce variance by segmenting the signal, applying windowing, and averaging multiple periodograms.

Welch’s Method for Robust PSD Estimation

Welch’s method is a practical and statistically robust approach for estimating the power spectral density (PSD) of a signal, particularly when the available data are noisy or of limited length. Unlike the plain periodogram, which often yields high-variance estimates due to reliance on a single realization of the data, Welch’s method reduces variance by segmenting the signal into overlapping parts, applying windowing to each segment, and averaging the resulting periodograms.

Let $x[n]$ be a discrete-time signal of length N , sampled at rate f_s . The signal is divided into M segments, each of length L . A typical choice is a 50% overlap, which increases statistical reliability without excessive redundancy. Each segment $x_m[n]$, for $m = 1, 2, \dots, M$, is multiplied by a window function $w[n]$ (e.g., Hann or Hamming) to minimize spectral leakage at the segment boundaries:

$$x_m^{(w)}[n] = x_m[n] w[n], \quad n = 0, 1, \dots, L-1. \quad (2.15)$$

The discrete Fourier transform (DFT) of the windowed segment is then computed as:

$$X_m^{(w)}[k] = \sum_{n=0}^{L-1} x_m^{(w)}[n] e^{-j2\pi kn/L}, \quad k = 0, 1, \dots, L-1. \quad (2.16)$$

To ensure proper normalization, the average power of the window is defined as:

$$U = \frac{1}{L} \sum_{n=0}^{L-1} w^2[n]. \quad (2.17)$$

Using this normalization, the periodogram (PSD estimate) for the m -th segment at frequency bin $f_k = \frac{k f_s}{L}$ is given by:

$$P_m(f_k) = \frac{1}{L f_s U} \left| X_m^{(w)}[k] \right|^2. \quad (2.18)$$

Finally, Welch’s PSD estimate is obtained by averaging the individual segment periodograms:

$$\hat{P}(f_k) = \frac{1}{M} \sum_{m=1}^M P_m(f_k). \quad (2.19)$$

Averaging across M partially overlapping, windowed segments effectively reduces the variance of the PSD estimate by approximately a factor of M . The windowing step mitigates spectral leakage—artificial spreading of power across adjacent frequency bins—by smoothly tapering the signal at each segment boundary. Consequently, Welch’s method produces smoother and more reliable PSD estimates compared to the basic periodogram, making it particularly suitable for analyzing real-world physiological signals, such as rPPG, where the data may be corrupted by noise, motion, or video compression artifacts [9].

The mathematical results and theorems discussed above provide the essential foundation for the frequency-domain analyses carried out in this thesis. A clear understanding of the Discrete Fourier Transform, power spectral density, and the advantages of robust estimation methods such as Welch’s approach enables meaningful investigation of rPPG signals and their behavior under compression. In the implementation, the Welch method was used as the primary technique for frequency analysis, providing stable and reliable PSD estimates from the video-derived signals. In the following section, these theoretical tools are put into practice for the visualization and interpretation of frequency content. This mathematical groundwork directly supports the creation of all spectral plots and the comparisons presented next.

2.5.2 Practical Implementation and Frequency-Domain Analysis

For each rPPG extraction parameter and subject group, the Welch power spectral density (PSD) was computed on a per-trial basis and then averaged across all available subjects within each group. This approach produces a robust estimate of the typical spectral content for each setting, reducing the influence of outliers and inter-individual variability. The procedure was repeated for all parameter values e.g., all CRF levels, all B-frame counts, all GOP sizes, reference frames, and SUBQ settings.

To provide an intuitive overview of how compression and encoder parameters affect the preservation of physiological frequencies, results are visualized in two formats:

- **Overlay plots:** For each group and parameter, all parameter settings are overlaid in a single figure. This facilitates direct comparison of the main spectral peak, corresponding to the heart rate band, and shows how its magnitude and shape evolve with parameter changes.
- **Side-by-side plots:** For each group, all parameter values are arranged in a grid, allowing a detailed view of how the frequency spectrum changes across the parameter sweep. This format makes it easier to track the loss, broadening, or shifting of the heart rate peak and harmonics as compression increases or encoder settings are varied.

Interpretation of the Side-by-Side Frequency Plots:

The side-by-side grids (see Figures 2.13, 2.14, 2.15) reveal clear and interpretable trends in the preservation or degradation of physiological signal content for each subject group as encoder parameters are varied.

Spectral Analysis for Group 2. The average power spectral density (PSD) plots for group 2 show how different H.264 compression settings affect the preservation of heart rate information in the signal.

For the **CRF** parameter, the main peak in the spectrum around 1.0 to 1.2 Hz, which corresponds to the heart rate, stays strong and clear for uncompressed and low CRF settings (0, 5, 10). When the CRF is increased to 15 or 20, this peak gets weaker and more noise appears in the background. At CRF 25, the signal looks very noisy: the main peak is no longer easy to spot and a lot of smaller peaks appear, making it difficult to recognize the heart rate.

Looking at the **B-frame** parameter, using a low number of B-frames (0, 1, or 2) keeps the heart rate peak mostly intact, but as the number goes higher, the peak becomes less clear and some unwanted extra peaks start to show up. This means that using too many B-frames can add artifacts to the signal that might interfere with heart rate detection.

When it comes to **GOP size**, having keyframes very often (GOP 1) gives a nice, sharp peak. If the GOP is set to a moderate size, like 60 or 120, the main peak is still clear, although it gets a little broader. With very large GOP sizes (like 240), the main peak gets flatter and there are more fluctuations and bumps in the background, which suggests that errors build up when keyframes are too far apart.

For the **Reference Frames** parameter, keeping the setting at 0 to 3 helps to maintain the original peak and overall spectral shape. If you increase the number of reference frames above this, some small extra peaks and a bit more noise appear, which shows a slight loss in signal quality.

Finally, for **SUBQ** (sub-pixel motion estimation), using a very low setting (0 or 1) makes the main peak weaker and raises the noise. On the other hand, moderate or high SUBQ values (3 to 12) bring back a strong, clear peak, although the very highest setting (12) causes a little drop in peak strength.

All in all, these results show that the heart rate signal is best preserved with moderate, balanced encoder settings. Too much compression or extreme parameter values introduce noise and distortions that make it harder to accurately recover the heart rate from the video.

Spectral Analysis for Group 4. Looking at Group 4, which includes videos where subjects show more motion, we notice many of the same trends as in Group 2, but with a few notable differences. When compression is low or encoder settings are moderate, the main heart rate peak near 1.0–1.2 Hz is still visible, although it is often not as sharp or pronounced—likely because subject motion adds extra variability to the signal. As compression becomes stronger or parameter values move away from the recommended range, the heart rate peak gradually fades or spreads out. In some cases, especially at very high CRF or B-frame values, extra peaks and more background noise begin to appear, making it harder to distinguish the true physiological signal from artifacts. Compared to Group 2, these changes happen a bit more quickly in Group 4, showing that motion makes the signal more sensitive to compression and parameter choices. Overall, while the general pattern holds, the effects of both motion and compression

are more obvious here, reinforcing the importance of careful parameter selection for reliable heart rate estimation when subjects are moving.

Spectral Analysis for Group 7. In Group 7, which includes recordings after exercise, with higher heart rates, the overall trends are quite similar to those seen in Groups 2 and 4. The dominant heart rate peak, now around 1.5 Hz, is clear and sharp when compression is low or encoding parameters are moderate. As with the other groups, increasing CRF makes the main peak less distinct: at CRF 20 and especially CRF 25, the peak is no longer clearly visible and the spectrum becomes noisy, making it hard to identify the true heart rate.

The effect of B-frames also follows the same pattern, using a few B-frames (0 to 3) preserves the peak, but higher values blur the frequency content and introduce more side peaks. For GOP size, the peak remains well defined up to GOP 60, but with larger values like 120 or 240, it broadens and becomes less isolated, though the change is not as dramatic as with CRF.

For reference frames, lower values (0 to 3) keep the main peak strong, while higher settings add a little more noise and make the peak less clear. The SUBQ parameter has the same effect as before: low values soften the peak and raise the noise floor, while moderate to high values (especially around 7) maintain the heart rate signal.

Compared to Groups 2 and 4, the main difference here is that the higher physiological frequency in Group 7 makes the signal a bit more sensitive to distortion, so the main peak disappears or spreads out more quickly at extreme compression or parameter values. This suggests that aggressive encoding choices have an even greater impact on heart rate estimation when the underlying heart rate is elevated, highlighting the importance of careful parameter selection in motion-rich or post-exercise scenarios.

Conclusion. A clear pattern emerges across all three groups when comparing the spectral plots. We can observe that, moderate encoder settings consistently preserve the main heart rate peak, making it possible to reliably extract physiological information from the video. When compression becomes too strong, or when parameter values are set too high or low, the heart rate peak either weakens, broadens, or gets lost among extra noise and side peaks. This effect is seen in every group, but it is especially pronounced in Group 7, where higher heart rates are more easily distorted by aggressive encoding. These frequency-domain results provide a straightforward visual explanation for the trends reported in the quantitative metrics, directly linking encoder choices to the quality of rPPG signal recovery and heart rate estimation.

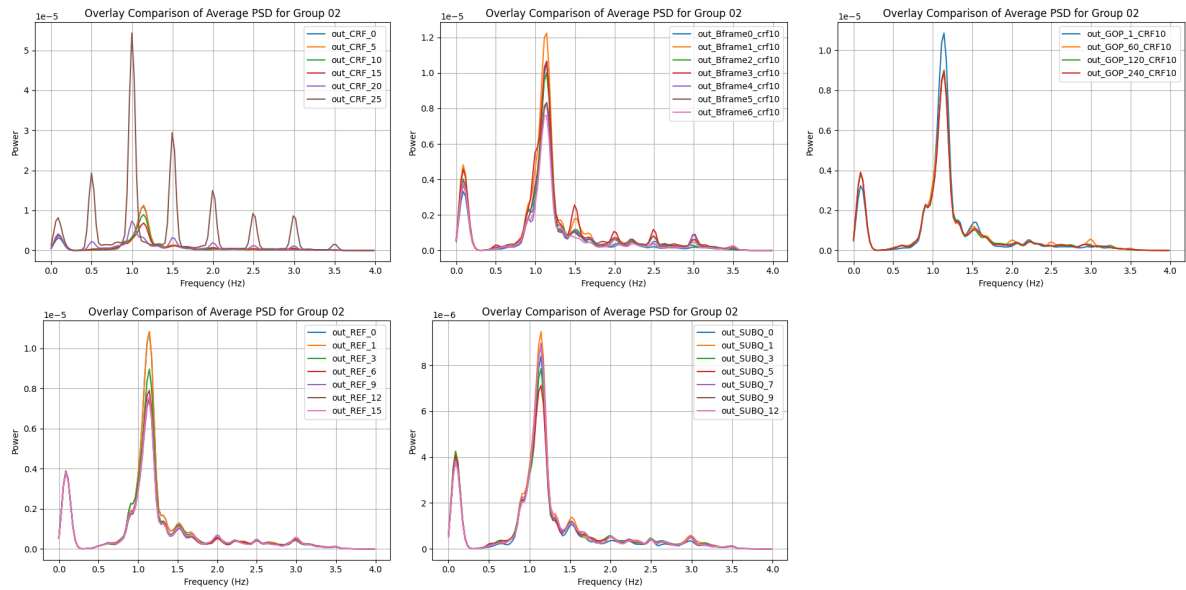


Figure 2.10: Overlay of average PSDs for Group 02 for all parameter values.

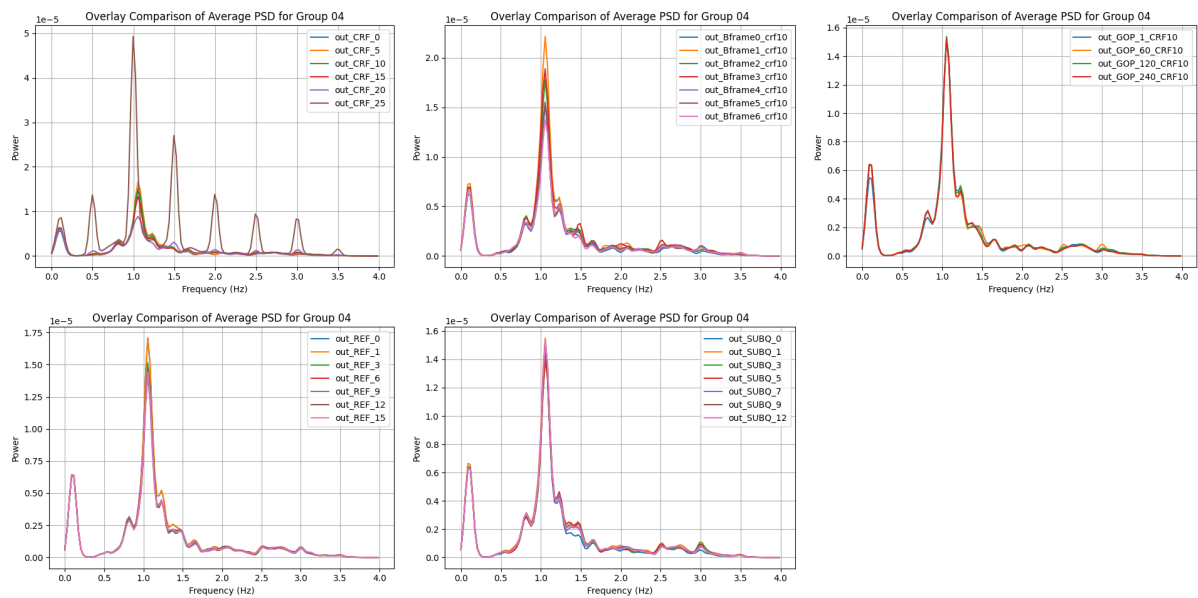


Figure 2.11: Overlay of average PSDs for Group 04 for all parameter values.

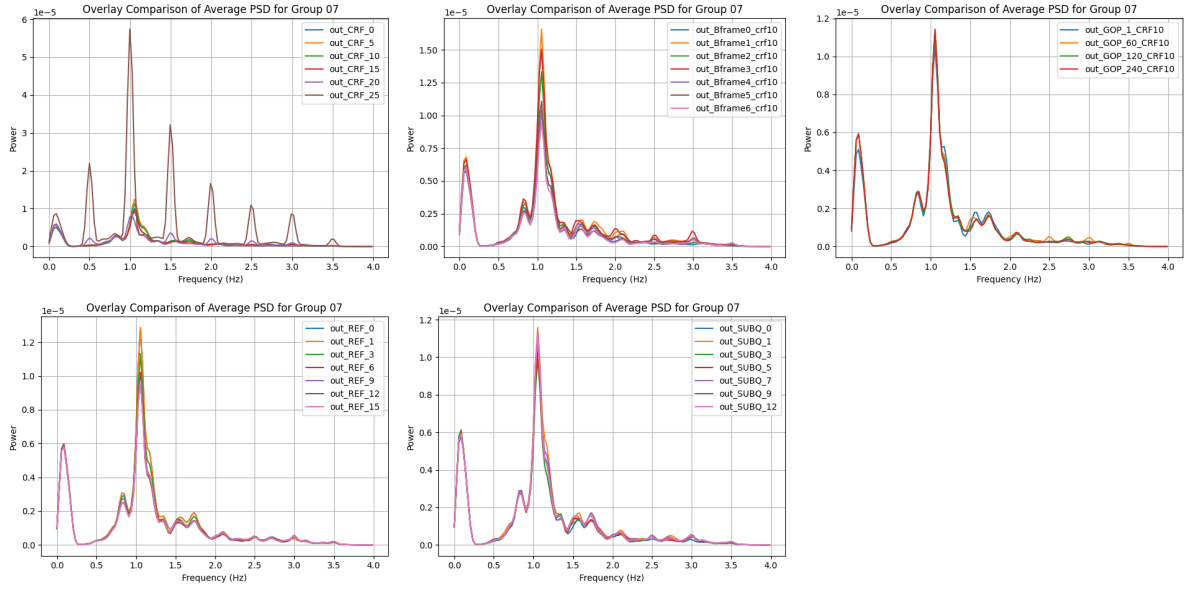


Figure 2.12: Overlay of average PSDs for Group 07 for all parameter values.

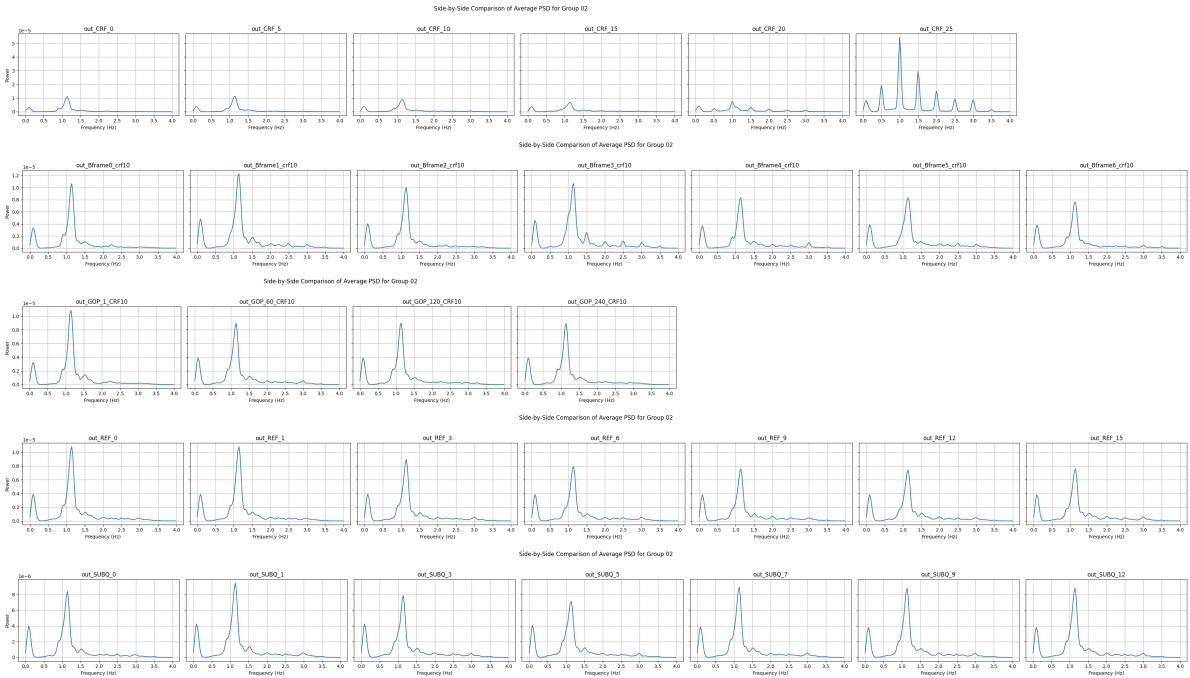


Figure 2.13: Side-by-side comparison of PSDs for Group 02, each subplot corresponds to one value of the tested parameter.

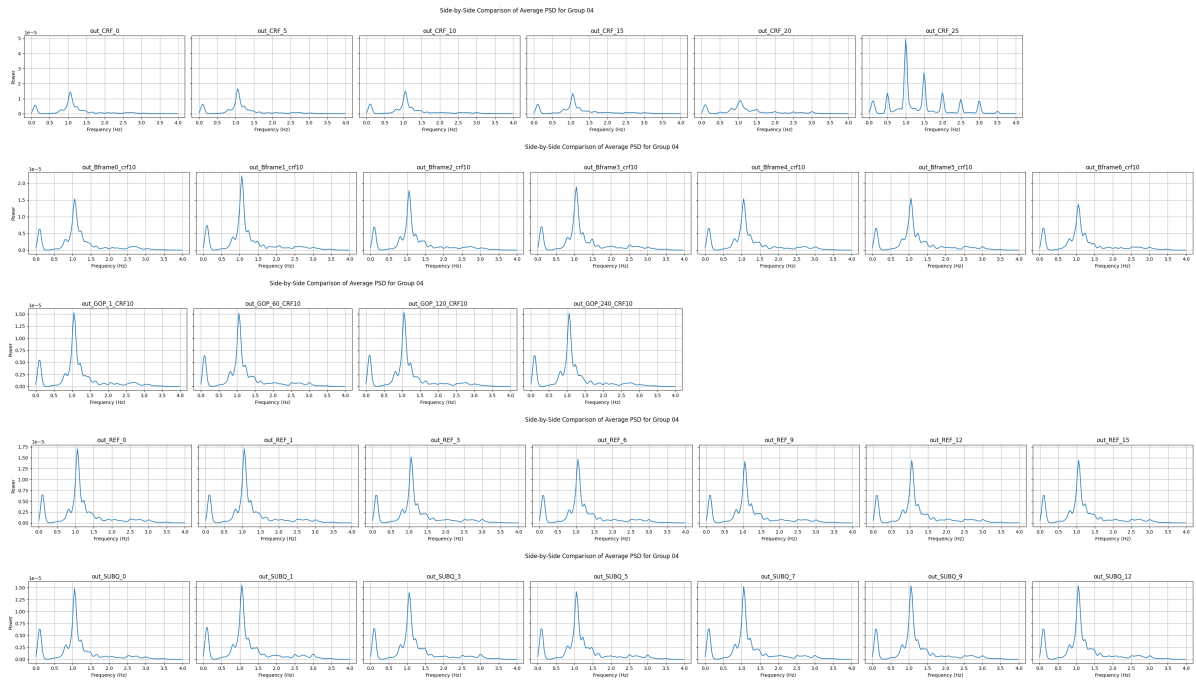


Figure 2.14: Side-by-side comparison of PSDs for Group 04, each subplot corresponds to one value of the tested parameter.

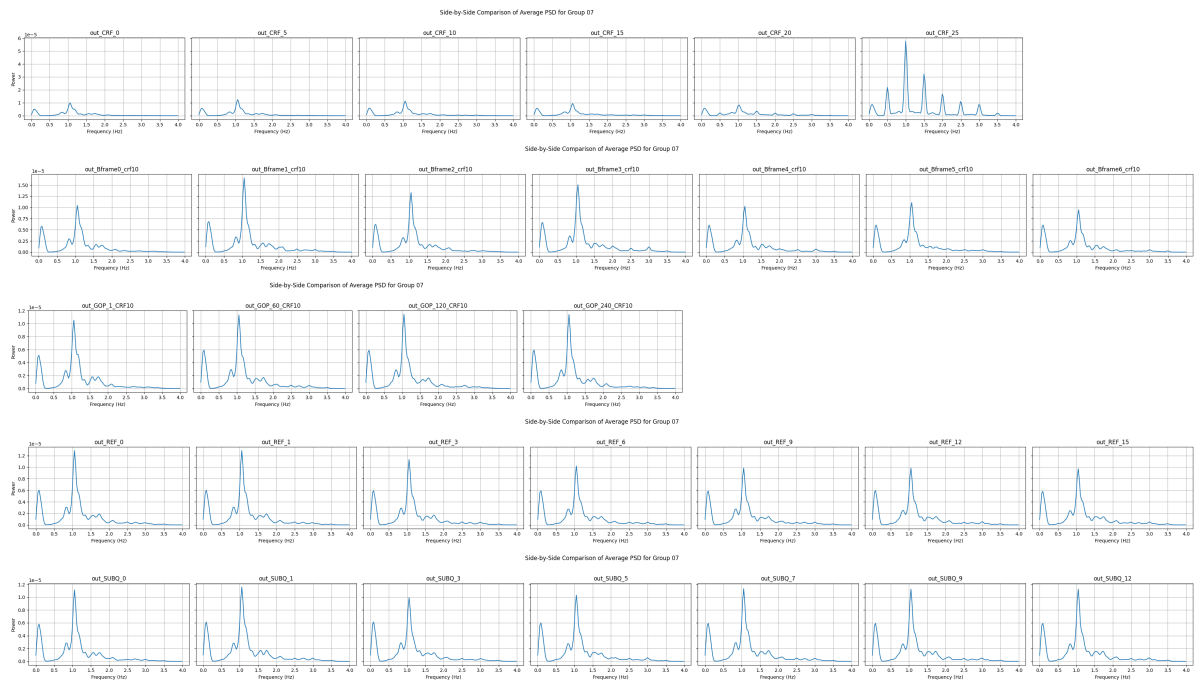


Figure 2.15: Side-by-side comparison of PSDs for Group 07, each subplot corresponds to one value of the tested parameter.

2.6 Rationale for the Selected Compression Parameter (CRF)

Across all the experiments and analyses in this chapter, the Constant Rate Factor (CRF) turned out to be the most influential compression setting for preserving rPPG signal quality. This was clear in Section 2.2 (see Table 2.3 and Figure 2.3), where CRF had a strong and steady effect on all metrics, SNR, correlation, DTW, and even BPM error. In contrast, other encoder settings, such as GOP size, reference frames, and sub-pixel motion estimation, tended to have weaker or less consistent impacts.

One of the main reasons for focusing on CRF is how consistent and predictable its effects are. As the CRF value goes up, which means more aggressive compression, the quality of the rPPG signal drops across the board. This trend is especially visible in the BPM error metric, while other parameters sometimes caused the BPM error to jump around with no clear pattern, changing the CRF produced a sharp and clear threshold. For values of CRF up to about 15, the BPM error stays low, but as soon as CRF increases beyond this point, the error rises dramatically. SNR and correlation also show a steady decrease as CRF gets higher, so it is much easier to interpret and trust the results.

The frequency-domain plots in Section 2.3 (Figures 2.10 to 2.15) reinforce this finding. At lower CRF values, the main frequency peak, which corresponds to heart rate, remains sharp and well defined. As CRF increases, this peak becomes blurred or even disappears, and the overall noise in the signal grows. This makes it clear that CRF has a direct effect on whether key physiological information is preserved or lost during compression.

To validate this choice, a comprehensive dataset was constructed by combining all available videos in which various encoder parameters, such as CRF, GOP length, and the number of reference frames, had been systematically varied. For each video, the corresponding CRF value was extracted, and key performance metrics including SNR, correlation, DTW distance, and BPM error were computed. Rather than analyzing each experiment independently, this aggregated approach enabled a unified assessment by grouping all results according to their CRF level, regardless of other encoding settings. The resulting distributions are visualized as boxplots in Figure 2.16. These plots reveal clear trends, as CRF increases, both SNR and correlation steadily decrease, DTW error rises, and BPM error remains relatively stable up to CRF 15 before increasing sharply.

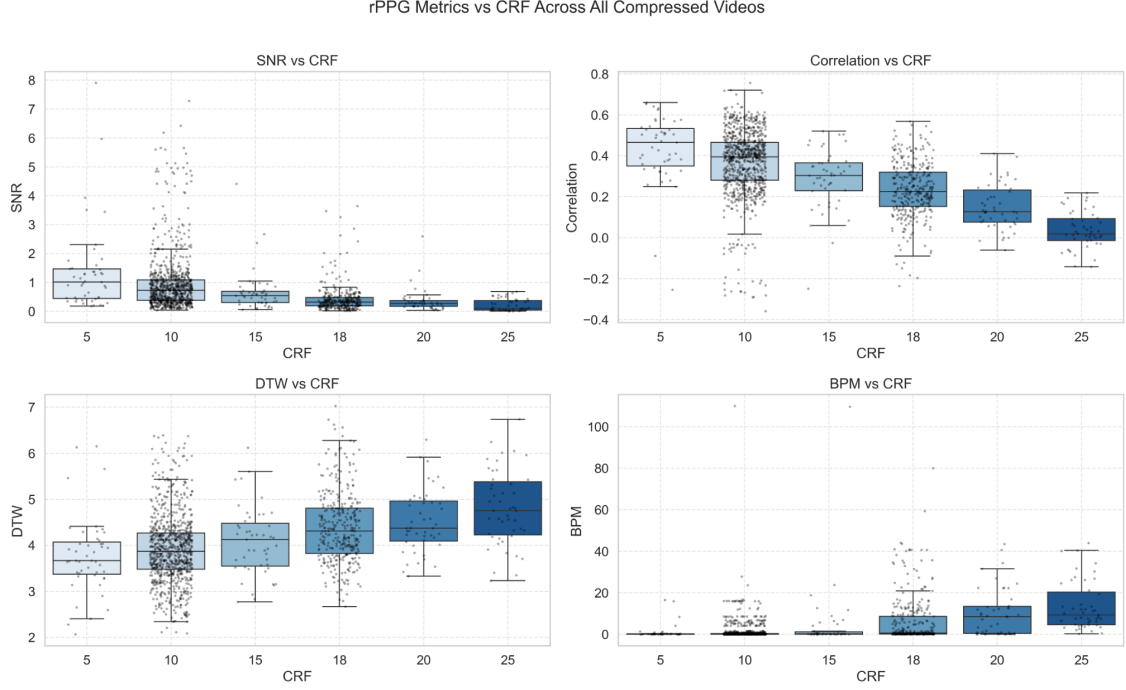


Figure 2.16: rPPG metrics (SNR, Correlation, DTW, BPM error) versus CRF level across all compressed videos in the dataset. Each boxplot shows the spread of metric values for each CRF, highlighting the clear effect of compression strength on signal quality.

Overall, the CRF parameter provides a clear and practical way to control the strength of video compression and evaluate its effect on rPPG signal quality. Among all encoding parameters studied, CRF consistently showed the most direct and interpretable relationship with the main performance metrics, making it a reliable indicator of how compression influences physiological signal recovery. As compression becomes stronger, the progressive degradation observed across SNR, correlation, and BPM accuracy demonstrates how sensitive the underlying color and motion cues are to information loss.

For these reasons, all deep learning experiments presented in Chapter 3 are conducted using videos compressed at different CRF levels, enabling a systematic exploration of how varying compression strength affects the accuracy and robustness of heart-rate estimation. The theoretical foundation behind this behavior, where compression is viewed as a bounded perturbation applied to the signal and its stability is analyzed within that framework, is developed in Appendix A: Mathematical Modeling of Signal Stability under Compression as a Perturbation Operator.

Chapter 3

RS-rPPG and Remote Heart-Rate Estimation

3.1 Literature Review of RS-rPPG

In this section, we review the RS-rPPG framework proposed by Savić and Zhao [10], which forms the foundation of this thesis. Their work focuses on one of the main challenges in remote photoplethysmography (rPPG), which is how to train deep models for heart-rate estimation when labeled data are limited and recordings are affected by noise, motion, or compression artifacts. To overcome these limitations, the authors introduce a self-supervised learning approach that leverages physiological priors and contrastive training in the frequency domain, enabling robust heart-rate estimation without the need for large annotated datasets. RS-rPPG combines structured input representations (MSTmaps and Tmaps), a transformer-based backbone, and a two-stage training procedure to produce models that generalize well across datasets and conditions.

We describe the key design principles of RS-rPPG, including its input representations, contrastive learning setup, model architecture, and training strategy in detail, providing the necessary background to understand both the original method and our extensions in later sections.

Physiological Priors in RS-rPPG

The RS-rPPG framework is guided by a set of physiological and data-driven priors, which together form the foundation of the self-supervised learning strategy. These priors are derived from well-established characteristics of cardiac signals and video-based pulse extraction, and they ensure that the model focuses on physiologically meaningful information rather than noise or spurious correlations. By explicitly encoding these assumptions into the training process, RS-rPPG is able to learn robust representations even without direct access to ground-truth labels. The priors are:

- **P1: Spatial-temporal averaging reduces noise.** Spatial-temporal representations such as MSTmaps are assumed to be less affected by noise than raw pixel intensities. By averaging the color variations over multiple regions of interest (ROIs), random fluctuations caused by motion or illumination changes are reduced, while the underlying pulse-related variations are preserved. This is why MSTmaps are used as input instead of raw video frames or single-ROI signals.
- **P2: Transformer architectures capture long-range temporal structure.** Transformer-based models can capture dependencies across longer time intervals compared to traditional convolutional networks. Since heart-rate signals evolve over time in a quasi-periodic manner,

this prior assumes that attention-based architectures are better suited for modeling the temporal consistency of the cardiac signal and suppressing short-term irregularities.

- **P3: Handcrafted signals contain useful physiological information.** Traditional rPPG algorithms such as CHROM, POS, and GREEN, although simple, still encode meaningful physiological information. This prior assumes that these handcrafted methods can serve as reference signals during training, helping the model align its predictions with the true physiological rhythm rather than arbitrary image features.
- **P4: Each facial video contains a distinct rPPG signal.** Different videos usually correspond to different subjects or recording conditions, and thus their rPPG waveforms are expected to differ. This assumption is used to define negative pairs in the contrastive learning framework, ensuring that embeddings from separate videos are pushed apart in the feature space.
- **P5: Signals from different ROIs and color channels share the same physiology.** Although signals from various ROIs (e.g., forehead, cheeks) or color channels may differ slightly in strength or noise, they originate from the same underlying heart-rate signal. This motivates the creation of positive pairs during training, where embeddings from different ROIs or channels of the same video are encouraged to align in the latent space.
- **P6: The rPPG signal is band-limited.** Human heart-rate signals lie within a specific frequency band, typically between 0.5 Hz and 3 Hz (30–180 BPM). Energy outside this range is unlikely to correspond to true cardiac activity. This prior supports the use of band-limited regularization terms that suppress spectral energy outside the expected physiological range.
- **P7: The rPPG spectrum is sparse.** A clean rPPG signal has a distinct dominant frequency peak corresponding to the heart rate, while noise tends to distribute energy across many frequencies. This assumption motivates the inclusion of spectral sparsity regularization, encouraging the model to produce sharp, physiologically realistic spectral peaks at the true heart-rate frequency.

Together, these priors form the foundation of the RS-rPPG framework. Priors P1–P3 motivate the model’s input representation and architectural design, favoring spatial–temporal maps and transformer-based encoders that emphasize physiologically meaningful signal components. Priors P4–P7, on the other hand, govern the construction of positive and negative sample pairs within the contrastive learning setup, as well as the inclusion of frequency-domain regularizations that encourage band-limited and sparse spectral representations. By embedding these physiological and structural assumptions into the training process, RS-rPPG learns robust cardiac representations directly from unlabeled video data, eliminating the need for explicit ground-truth supervision.

3.1.1 Input Map Generation

The first step in RS-rPPG is to transform raw facial video into structured spatial–temporal representations that are suitable for learning. To this end, two types of maps are constructed: the Mean Spatial–Temporal map (MSTmap) and the Physiological Temporal map (Tmap). Both representations are built from regions of interest (ROIs) defined on the face, typically covering the forehead, cheeks, and mouth region (see Fig. 3.1).

MSTmap. For each region of interest (ROI) and each color channel, the mean pixel intensity is

computed across all frames in a video, producing temporal sequences that reflect color variations over time. Concatenating these sequences across ROIs and channels yields the Multi-Spatio-Temporal Map (MSTmap), which provides a compact summary of the spatial and temporal dynamics within the facial video. Although the MSTmap contains both physiological information and non-physiological variations such as motion or illumination changes, it serves as the primary input representation during training and forms the basis for learning robust rPPG features.

Tmap. In parallel, a second representation is generated by applying traditional rPPG algorithms, GREEN [11], CHROM [12], and POS [13], to the same ROI-based temporal signals. These classical methods are designed to enhance pulse-related color variations and suppress unrelated noise, resulting in signals that better reflect the true physiological dynamics. The outputs of the three algorithms are then concatenated to form the Temporal Map (Tmap), which serves as a physiology-informed reference representation for guiding the learning process.

The key idea is that MSTmaps and Tmaps capture the same underlying cardiac activity from complementary perspectives. MSTmaps represent raw spatio-temporal color variations across facial regions, while Tmaps emphasize the physiologically meaningful components extracted by traditional rPPG algorithms. Within the contrastive learning framework, MSTmaps serve as anchor inputs and Tmaps as their corresponding positive references. This setup enables the model to learn how to associate noisy, unprocessed observations with their physiology-driven counterparts, while distinguishing them from unrelated signals. The design directly reflects Priors P1 and P3, which together motivate the use of spatial-temporal averaging to suppress noise and physiology-informed references to guide the learning process toward meaningful cardiac representations.

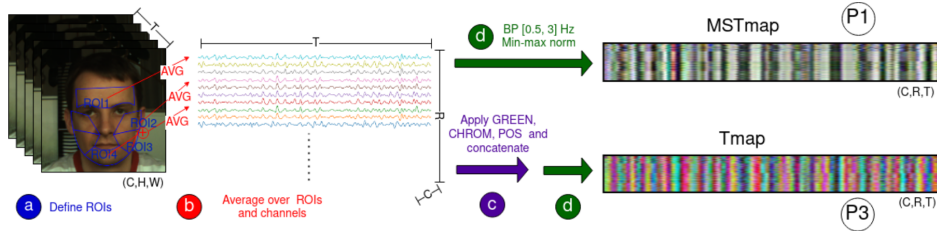


Fig. 1. **Input map generation:** a) ROIs are defined for each frame b) $C \times R$ temporal sequences are extracted by averaging pixels for each channel and ROI combination. c) GREEN, CHROM, POS are applied on the signals and results in $C = 3$ signal maps of $(1, R, T)$, that are concatenated to form the Tmap (C, R, T) d) The sequences are filtered with a pass band of $[0.5, 3] \text{ Hz}$ and min-max normalized to form the MSTmap or Tmap.

Figure 3.1: Generation of MSTmaps and Tmaps from facial ROIs [10]. MSTmaps are constructed by averaging normalized, band-pass-filtered ROI color signals, capturing raw spatio-temporal variations. Tmaps are generated by applying traditional rPPG methods (GREEN, CHROM, POS) to the same signals, providing physiology-guided references for training.

3.1.2 Pretraining: Spectrum Regression

The first stage of RS-rPPG is a pretraining phase that initializes the network with physiology-aware representations before contrastive learning. This phase consists of two main components, as illustrated in Fig. 3.2.

(a) Spectrum regression. The network receives MSTmaps as input and is trained to reconstruct the corresponding Tmaps. Since MSTmaps and Tmaps represent the same cardiac dynamics at different noise levels, this task encourages the model to learn mappings from noisy, raw inputs to cleaner,

physiology-driven signals. To capture the quasi-periodic nature of the heartbeat, the regression is performed in the frequency domain rather than the time domain. Both predicted and ground-truth Tmaps are transformed into power spectral densities (PSDs), and the model minimizes their L_2 difference within the physiological heart-rate range (0.5–3 Hz):

$$L_f = \sqrt{\sum_{r=0}^R \sum_{c=0}^C \sum_{f=f_{\min}}^{f_{\max}} (\bar{x}_{r,c}(f) - \bar{x}'_{r,c}(f))^2},$$

where $\bar{x}_{r,c}(f)$ and $\bar{x}'_{r,c}(f)$ denote the normalized ground-truth and predicted spectra for ROI r and channel c . By constraining the learning objective to the relevant frequency band, the network is explicitly guided to emphasize cardiac-related oscillations and to disregard irrelevant variations caused by illumination or motion.

(b) Dataset ordering and mini-batch construction. After pretraining, rough heart-rate estimates are obtained from each Tmap by identifying the dominant PSD peak. The dataset is then sorted based on these estimated heart rates to enable the construction of mini-batches with diverse physiological frequencies. This diversity is essential for the subsequent contrastive learning stage, since positive pairs should represent similar cardiac activity, while negative pairs should correspond to clearly different heart rates. By enforcing diversity during mini-batch sampling, the model encounters more informative comparisons, improving its ability to generalize across subjects and recording conditions.

Together, these steps provide an effective initialization for contrastive learning. Through spectrum regression, the model learns to recognize the frequency structure of cardiac signals, while dataset ordering ensures that mini-batches contain physiologically distinct examples that maximize the contrastive objective.

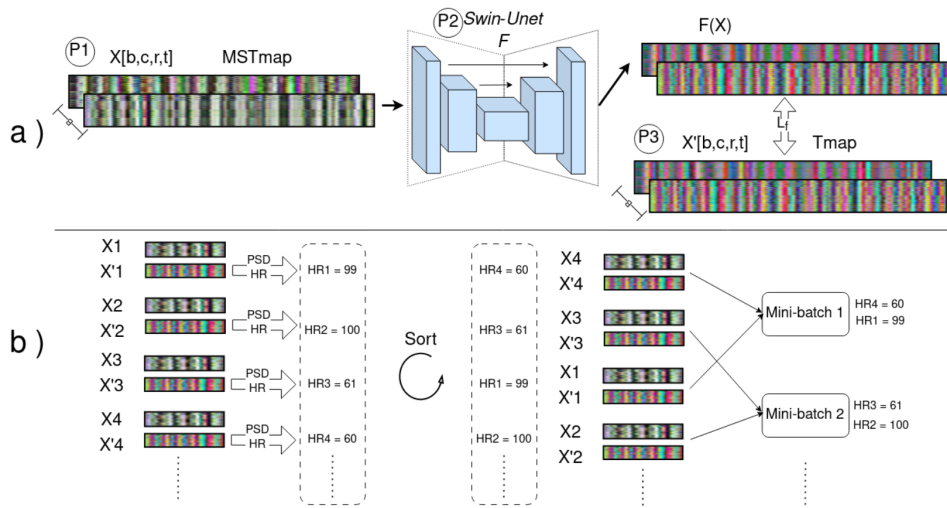


Fig. 3. a) The network F is initialised by pre-training it to predict Tmaps from MSTmaps via the frequency based loss L_f (P3). b) The dataset is ordered by using rough HR predictions calculated via PSD peak from the Tmaps (traditional method signals) resulting in more diverse mini-batch sampling (P4).

Figure 3.2: Pretraining pipeline of RS-rPPG [10]. (a) The network is initialized by predicting Tmaps from MSTmaps using a frequency-based regression loss L_f , encouraging attention to cardiac-related frequency components. (b) The dataset is then sorted by rough heart-rate estimates, enabling diverse and informative mini-batch construction for contrastive learning.

3.1.3 Training: Contrastive Learning

After pretraining, the RS-rPPG framework is fine-tuned through a self-supervised training stage based on contrastive learning. The aim of this phase is to learn meaningful and discriminative representations of the rPPG signal without relying on explicit heart-rate labels. Instead, the model learns through structured comparisons between different signal embeddings, guided by physiological priors. This approach helps the network focus on cardiac information that is consistent across views of the same video while ignoring unrelated variations caused by motion, lighting, or compression. The overall training pipeline is shown in Fig. 3.3.

Anchor–Positive–Negative Construction

Contrastive learning is based on comparing three related examples:

- **Anchor (A):** an embedding of the MSTmap from the current video, serving as the reference sample.
- **Positive (P):** an embedding of the corresponding Tmap, which encodes the same underlying cardiac dynamics but through a physiology-informed representation. The model is trained to consider the anchor and positive as similar.
- **Negative (N):** an embedding of a Tmap from a different video, typically representing another subject or heart rate. The model learns to separate the anchor and negative in the feature space.

Through this triplet setup, the network organizes its learned representations such that embeddings with the same physiological content (anchor and positive) are close together, while those with different cardiac signals (anchor and negative) are pushed apart. This forms the core idea of contrastive learning, which has been formalized in the frequency domain.

Frequency-Domain Training Objective

The key insight behind RS-rPPG is to embed cardiac priors directly into the contrastive learning objective by operating in the frequency domain. After pretraining, the temporal signals derived from the anchor, positive, and negative embeddings are transformed into their normalized power spectral densities (PSDs), denoted as $\bar{a}(f)$, $\bar{p}(f)$, $\bar{n}(f)$, where each function describes how the signal energy is distributed across frequency. All spectra are normalized so that their total energy equals one.

Triplet Loss in the Frequency Domain. The model is optimized to ensure that the anchor spectrum is more similar to the positive than to the negative within the physiological frequency band $[f_{\min}, f_{\max}] = [0.5, 3]$ Hz. This is enforced through a triplet loss defined as

$$f^T = \max \left(m + \frac{1}{f_{\max} - f_{\min}} \sum_{f=f_{\min}}^{f_{\max}} \left(\|\bar{a}(f) - \bar{p}(f)\|_1 - \|\bar{a}(f) - \bar{n}(f)\|_1 \right), 0 \right),$$

where $\|\cdot\|_1$ denotes the absolute difference between spectra and m is a small margin constant that enforces a minimum separation. Intuitively, this loss decreases when the anchor and positive spectra are close, which means high similarity, and increases when the anchor resembles the negative spectrum.

By averaging the differences across frequency bins, the model learns to group signals with similar heart-rate content.

Physiological Regularizations. While the triplet loss promotes discriminative features, it does not guarantee that the spectra themselves remain physiologically realistic. To address this, two additional regularization terms are introduced:

- **Band-limited power regularization.** The first regularization term penalizes spectral energy outside the expected cardiac range:

$$\text{Reg}_{bw} = \frac{\sum_{f \in \mathbb{F} \setminus [f_{\min}, f_{\max}]} \bar{x}(f)}{\sum_{f \in \mathbb{F}} \bar{x}(f)},$$

This term encourages the network to concentrate spectral power within the 0.5–3 Hz band, corresponding to typical human heart-rate frequencies, while suppressing irrelevant energy caused by motion or illumination changes.

- **Spectral sparsity regularization.** The second regularization enforces sparsity by promoting the presence of a single dominant spectral peak. It is computed as the normalized variance of the spectrum within the heart-rate band:

$$\text{Reg}_{sp} = \sqrt{\frac{1}{f_{\max} - f_{\min}} \sum_{f=f_{\min}}^{f_{\max}} \left(\frac{\bar{x}(f) - \min_f \bar{x}(f)}{\max_f \bar{x}(f) - \min_f \bar{x}(f)} \right)^2}.$$

This term penalizes flat or noisy spectra, encouraging the model to produce sharp, well-defined peaks corresponding to the heart rate.

Overall Training Objective. The total loss combines the frequency-domain triplet loss with the two regularization terms:

$$\mathcal{L}_{\text{total}} = f^T + \lambda_{bw} \text{Reg}_{bw} + \lambda_{sp} \text{Reg}_{sp},$$

where λ_{bw} and λ_{sp} are weighting factors that control the relative contribution of each regularization term. This objective ensures that the network not only distinguishes between different cardiac signals via the triplet loss, but also learns to produce power spectra that are physiologically plausible, band-limited to the expected heart-rate range and dominated by a clear spectral peak.

Architecture and Training Strategy. The RS-rPPG framework is built upon the Swin-UNet backbone [14], a hierarchical vision transformer originally developed for medical image segmentation. Its shifted-window self-attention mechanism enables the model to capture long-range temporal dependencies across spatial regions of the face, while the U-shaped encoder–decoder design preserves fine-grained local details through skip connections. This architecture is particularly well suited to the spatio-temporal structure of MSTmaps and Tmaps, where both localized fluctuations e.g., subtle color changes in specific ROIs, and overall periodicity of the cardiac signal are essential for accurate

representation.

The training process follows a two-stage strategy designed to progressively inject physiological knowledge into the network. In the first stage, **spectrum regression pretraining** teaches the model to map noisy MSTmaps to their physiology-informed Tmaps by minimizing the spectral difference between predicted and reference signals. This stage initializes the backbone with cardiac-aware representations that emphasize periodic and band-limited spectral features.

In the second stage, **contrastive learning fine-tuning** further refines these representations without requiring heart-rate labels. The model is trained using a frequency-domain triplet loss that enforces similarity between spectra from the same physiological source and dissimilarity across different subjects or heart rates. Additional spectral regularizations ensure that the learned features remain physiologically plausible, concentrated within the cardiac frequency band and dominated by a clear spectral peak.

Mini-batches are organized to maximize diversity in heart-rate content, which strengthens the contrastive comparisons and improves robustness. Together, this two-stage training procedure stabilizes convergence, enhances cross-dataset generalization, and enables the model to maintain performance even under strong video compression.

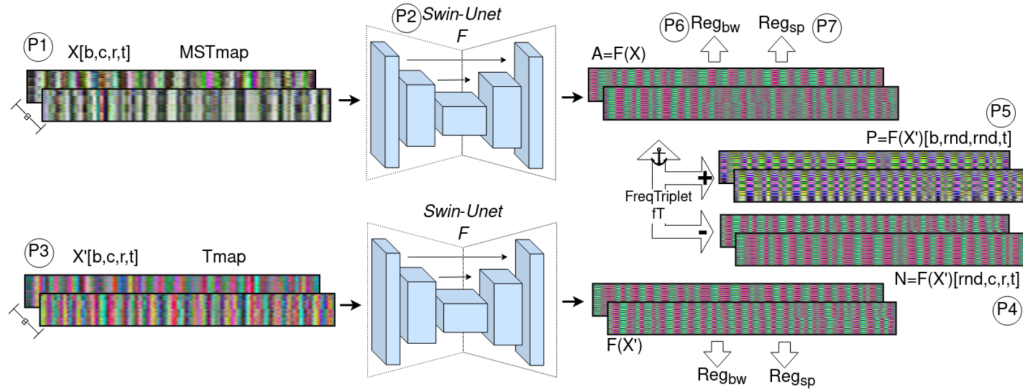


Fig. 2. **Overview:** A mini-batch of the input X (P1) and its traditional augmentation X' (P3) are fed into F (P2). For our frequency triplet loss learning $F(X)$ is the anchor, the positive (P5) is generated by shuffling $F(X')$ along the ROI and channel dimensions and the negative (P4) by shuffling $F(X')$ along the batch dimension. $F(X')$ and $F(X)$ are regularized to have limited bandwidth and low normalized spectral power with Reg_{bw} (P6) and Reg_{sp} (P7).

Figure 3.3: Training pipeline of RS-rPPG [10]. Overview of the contrastive learning framework [10]. MSTmaps and Tmaps are passed through a shared Swin-UNet encoder to generate embeddings. The anchor (A) comes from an MSTmap, the positive (P) from the corresponding Tmap (shuffled across ROIs and channels), and the negative (N) from a different Tmap. Training uses the frequency-domain triplet loss (f^T) combined with spectral regularization terms (Reg_{bw} and Reg_{sp}) to enforce physiological consistency.

All in all, The RS-rPPG framework introduces a self-supervised approach for remote heart-rate estimation that combines structured input representations (MSTmaps and Tmaps) with a two-stage training strategy. Spectrum regression pretraining aligns raw and physiology-informed signals, while frequency-domain contrastive learning, with spectral regularization, ensures physiologically meaningful and discriminative embeddings. Built on a transformer-based Swin-UNet backbone, RS-rPPG captures long-range spatio-temporal dependencies and achieves strong robustness to motion, noise, and compression, forming the basis for the extended analysis presented in this thesis.

3.2 Contrastive Learning Method

Contrastive learning seeks to learn an embedding space where samples that are meant to be similar are mapped close together, while those that differ are placed farther apart. In self-supervised learning, the concept of similarity is determined directly from the data, for example by comparing two augmented versions of the same signal. In the context of rPPG, similarity can be defined through shared physiological information, which means that, segments that contain the same heart-rate pattern, even if obtained from different regions or processing methods, should produce similar embeddings, whereas segments corresponding to different heart rates should be distinct. This chapter introduces the mathematical principles of contrastive learning as applied in this work and related studies, and explains how they extend naturally to the frequency-domain formulation used for rPPG.

Setup and Notation

Contrastive learning provides a framework for learning data representations by comparing examples based on their similarity. The central idea is to map inputs into a space where samples that represent the same underlying information are close together, and those that differ are farther apart.

Let $x \in \mathcal{X}$ be an input signal drawn from the input space $\mathcal{X}$. An *encoder network* $f_\theta : \mathcal{X} \rightarrow \mathbb{R}^d$ parameterized by θ extracts a d -dimensional feature vector $h = f_\theta(x)$ that captures meaningful structure in the data. Because these representations are not yet optimized for direct similarity comparisons, a smaller *projection head*

$$g_\phi : \mathbb{R}^d \rightarrow \mathbb{R}^{d'}$$

parameterized by ϕ is used to transform h into a new embedding space suited for contrastive learning. This head is typically a shallow multi-layer perceptron (MLP).

The final embedding is given by

$$z = \frac{g_\phi(f_\theta(x))}{\|g_\phi(f_\theta(x))\|_2},$$

which projects the sample onto the unit hypersphere $\mathbb{S}^{d'-1}$. This normalization ensures that all embeddings have the same length, so similarity comparisons depend only on the angle between them rather than their scale. In this geometry, the cosine similarity between two embeddings z_i and z_j is defined as

$$s(z_i, z_j) = z_i^\top z_j,$$

ranging from -1 (opposite directions) to 1 (perfect alignment).

To train the model, each batch consists of three related elements. The *anchor* x_i serves as the reference input, the *positive* x_i^+ is another input that shares the same underlying information, and the *negatives* $\{x_k^-\}$ are samples that differ in content. After passing through the encoder and projection head, these become embeddings z_i , z_i^+ , and $\{z_k^-\}$, respectively. The training objective encourages the anchor and positive embeddings to be close in cosine similarity, while pushing the anchor away from the negatives:

$$s(z_i, z_i^+) \rightarrow 1, \quad s(z_i, z_k^-) \rightarrow -1.$$

In other words, the model learns to cluster related samples near each other on the unit hypersphere

and to separate unrelated ones.

This setup forms the basis for modern contrastive objectives such as InfoNCE and triplet loss [15--17], which formalize this geometric intuition into differentiable optimization functions used throughout this thesis.

The NT-Xent (InfoNCE) Objective

A key step in contrastive learning is to translate the intuitive idea of bringing similar samples closer and pushing dissimilar ones apart into a well-defined optimization objective. The most widely used formulation is the *Normalized Temperature-scaled Cross-Entropy* loss, known as NT-Xent or InfoNCE [15--18]. This objective provides both a probabilistic interpretation and a computationally efficient mechanism for training large-scale self-supervised models.

The InfoNCE loss can be viewed as a classification problem over similarity scores. For a given batch of B samples, each input is augmented twice, yielding $2B$ embeddings. For an anchor embedding z_i , its corresponding positive z_j (derived from the same source) should be identified among $2B - 1$ possible candidates that include all other embeddings as negatives. The similarity between any two embeddings is measured by the cosine similarity.

$$s(u, v) = u^\top v,$$

and these similarities are converted into a probability distribution using a temperature-scaled softmax:

$$p(j | i) = \frac{\exp(s(z_i, z_j)/\tau)}{\sum_{\substack{k=1 \\ k \neq i}}^{2B} \exp(s(z_i, z_k)/\tau)},$$

where $\tau > 0$ is the temperature parameter controlling the sharpness of the distribution. The loss for anchor i is the negative log-likelihood of selecting the correct positive:

$$\mathcal{L}_{\text{NCE}}^{(i)} = -\log p(j | i) = -\log \frac{\exp(s(z_i, z_j)/\tau)}{\sum_{\substack{k=1 \\ k \neq i}}^{2B} \exp(s(z_i, z_k)/\tau)}. \quad (3.1)$$

Averaging over all $2B$ embeddings yields the batch objective:

$$\mathcal{L}_{\text{NCE}} = \frac{1}{2B} \sum_{i=1}^{2B} \mathcal{L}_{\text{NCE}}^{(i)}.$$

This formulation effectively performs a softmax classification over all possible pairings, where the model learns to assign high probability to the true positive and low probability to the negatives. The temperature parameter τ scales similarity values before the softmax operation: smaller values of τ produce a sharper distribution that emphasizes hard negatives, those closest to the anchor, while larger values yield smoother gradients and more stable training. Choosing τ appropriately is therefore crucial

to balancing discriminative power and training stability.

From an information-theoretic perspective [15], minimizing $\mathcal{L}_{\text{NCE}}$ maximizes a lower bound on the mutual information between the paired embeddings z_i and z_j , encouraging the model to preserve informative dependencies while discarding irrelevant variations. In this way, the NT-Xent objective unifies similarity maximization, negative separation, and information preservation within a single probabilistic framework, forming the mathematical foundation of modern contrastive learning methods, including the one used in this thesis.

Mutual-information view. Beyond its classification interpretation, the InfoNCE objective can be viewed as maximizing a lower bound on the *mutual information* (MI) between two random variables. This perspective provides the theoretical foundation for contrastive learning, minimizing the InfoNCE loss drives the model to preserve information that is shared between paired samples while discarding irrelevant variations. In the self-supervised rPPG setting, this means retaining physiological information that is common to an MSTmap and its corresponding Tmap, while becoming invariant to factors such as illumination, motion, or video compression.

Let X and Y be random variables with joint distribution $p(x, y)$ and marginals $p(x)$ and $p(y)$. The *mutual information* between X and Y is defined as:

$$I(X; Y) = \mathbb{E}_{p(x, y)} \left[\log \frac{p(x, y)}{p(x)p(y)} \right] = D_{\text{KL}}(p(x, y) \parallel p(x)p(y)),$$

where $D_{\text{KL}}(P \parallel Q)$ denotes the Kullback-Leibler divergence

$$D_{\text{KL}}(P \parallel Q) = \mathbb{E}_P \left[\log \frac{P(x)}{Q(x)} \right].$$

The KL divergence measures how different one distribution P is from another Q ; it is non-negative and equals zero only when $P = Q$ almost everywhere. Intuitively, $I(X; Y)$ quantifies how much knowing X reduces uncertainty about Y ; it is large when Y can be reliably predicted from X and zero when they are statistically independent.

Using the chain rule of probability, the mutual information can equivalently be written as

$$I(X; Y) = \mathbb{E}_{p(x, y)} \left[\log \frac{p(y|x)}{p(y)} \right].$$

The term $\frac{p(y|x)}{p(y)}$ represents how much more likely a sample y is under the conditional distribution $p(y|x)$ than under its marginal $p(y)$. However, this ratio is difficult to compute directly since both $p(y|x)$ and $p(y)$ are unknown in practice. The InfoNCE loss circumvents this difficulty by constructing a discriminative objective that implicitly estimates this ratio.

Theorem 3 (InfoNCE as a Lower Bound on Mutual Information). *Let x be an anchor sample and y its corresponding positive drawn from $p(y|x)$. Let $\{y_k\}_{k=1}^{K-1}$ be a set of negatives drawn independently from a*

proposal distribution $q(y)$, which approximates the marginal $p(y)$. Define the density ratio

$$r(x, y) = \frac{p(y|x)}{q(y)}.$$

Then the expected InfoNCE loss

$$\mathbb{E}[\mathcal{L}_{\text{NCE}}] = -\mathbb{E}_{\substack{(x,y) \sim p \\ \{y_k\} \sim q}} \left[\log \frac{r(x, y)}{\sum_{k=0}^{K-1} r(x, \tilde{y}_k)} \right], \quad \tilde{y}_0 = y, \tilde{y}_{1:K-1} \sim q,$$

satisfies the inequality

$$I(X; Y) \geq \log K - \mathbb{E}[\mathcal{L}_{\text{NCE}}],$$

that is, minimizing $\mathcal{L}_{\text{NCE}}$ maximizes a variational lower bound on $I(X; Y)$.

Proof. We begin by expressing the mutual information as

$$I(X; Y) = \mathbb{E}_{p(x,y)}[\log r(x, y)].$$

Consider a classification problem where, for each anchor x , the learner must identify the true positive y among $K - 1$ distractors $\{y_k\}$ drawn from $q(y)$. The normalized probability of correctly identifying y is

$$p(j|i) = \frac{r(x, y)}{\sum_{k=0}^{K-1} r(x, \tilde{y}_k)}.$$

Taking the negative log-likelihood and expectation over the sampling process yields the expected InfoNCE loss:

$$\mathbb{E}[\mathcal{L}_{\text{NCE}}] = -\mathbb{E}_{\substack{(x,y) \sim p \\ \{y_k\} \sim q}} \left[\log \frac{r(x, y)}{\sum_{k=0}^{K-1} r(x, \tilde{y}_k)} \right].$$

We now derive the lower bound. Multiply both sides by -1 and rearrange the logarithm:

$$\mathbb{E}_{p(x,y)}[\log r(x, y)] = \log K - \mathbb{E}[\mathcal{L}_{\text{NCE}}] + \mathbb{E}_{p(x,y)} \left[\log \left(\frac{r(x, y)}{K r(x, y)} + \frac{K-1}{K} \right) \right].$$

Since $\log(\cdot)$ is a concave function, applying Jensen's inequality yields

$$\mathbb{E}_{p(x,y)}[\log r(x, y)] \geq \log K - \mathbb{E}[\mathcal{L}_{\text{NCE}}],$$

because for $r > 0$ and $K \geq 1$,

$$\frac{K r(x, y)}{r(x, y) + K - 1} \leq r(x, y).$$

Hence,

$$\boxed{I(X; Y) \geq \log K - \mathbb{E}[\mathcal{L}_{\text{NCE}}].}$$

This proves that the InfoNCE objective maximizes a variational lower bound on mutual information. $\square$

The tightness of this lower bound depends on the number and diversity of negative samples [19]. As K increases, the denominator in the loss better approximates the marginal distribution $p(y)$, and the

bound becomes tighter. This explains the empirical success of large-batch contrastive methods such as SimCLR [18] and MoCo [20], which use large memory banks or queues to increase the number of effective negatives. Conversely, biased or redundant negatives; for instance, two rPPG clips from different subjects with similar heart rates, can loosen the bound and impair training stability.

From this theoretical perspective, contrastive learning in rPPG can be understood as maximizing the shared information between an MSTmap and its corresponding Tmap, while reducing dependence on nuisance factors such as illumination, motion, and compression artifacts. This mutual-information interpretation thus provides both a rigorous justification and practical guidance for designing positive and negative pairs in self-supervised physiological signal learning.

Temperature and Gradient Dynamics in RS-rPPG. The temperature parameter τ plays a crucial role in the behavior of the InfoNCE objective. It determines how strongly the model distinguishes between similar and dissimilar samples by scaling the similarity scores before the softmax. Understanding its effect is particularly important for rPPG data, where signals from different subjects may share overlapping heart-rate frequencies, creating many near-duplicate *hard negative* examples.

Recall Eq. 3.1 that for an anchor embedding z_i and its positive z_j , the per-sample InfoNCE loss is:

$$\mathcal{L}_{\text{NCE}}^{(i)} = -\log \frac{\exp(s(z_i, z_j)/\tau)}{\sum_{\substack{k=1 \\ k \neq i}}^{2B} \exp(s(z_i, z_k)/\tau)}.$$

where $s(z_i, z_k) = z_i^\top z_k$ denotes cosine similarity between unit-normalized embeddings.

Defining:

$$p_k = \frac{\exp(s(z_i, z_k)/\tau)}{\sum_{\substack{\ell=1 \\ \ell \neq i}}^{2B} \exp(s(z_i, z_\ell)/\tau)}.$$

we can interpret p_k as the probability that candidate k is the positive for anchor i .

Differentiating the loss with respect to z_i gives:

$$\frac{\partial \mathcal{L}_{\text{NCE}}^{(i)}}{\partial z_i} = \frac{1}{\tau} \left(\sum_k p_k z_k - z_j \right),$$

which decomposes into two terms:

- a *pulling term* $-\frac{1}{\tau} z_j$ that attracts the anchor towards its true positive, and
- a *repulsive term* $\frac{1}{\tau} \sum_k p_k z_k$ that pushes it away from all negatives.

The temperature τ controls the balance between these two effects. When τ is small, the exponential terms become sharply peaked: only the most similar negatives dominate the denominator, making p_k highly concentrated. This causes strong repulsive gradients focused on the hardest negatives, yielding sharper class boundaries but less stable training. When τ is large, p_k spreads more uniformly, averaging the repulsion over many samples, stabilizing learning but reducing discriminative power.

In the RS-rPPG context, this balance is physiologically meaningful. Different subjects can have heart rates within a narrow frequency band, producing embeddings that are spectrally similar even if they come from distinct videos. If τ is too small, these physiological *hard negatives* are repelled excessively, leading the model to separate signals that represent the same underlying cardiac pattern. If τ is too large, the model becomes insensitive to subtle physiological differences, failing to discriminate between genuinely distinct rhythms. Choosing an appropriate temperature is therefore essential for achieving stable and physiologically consistent learning, ensuring that the model focuses on meaningful differences in heart-rate content rather than superficial noise or random variation.

Although the InfoNCE loss provides a strong mathematical foundation for contrastive learning, it operates only in the embedding space and does not directly consider the physiological nature of rPPG signals. In rPPG, the most important information, the heart-rate rhythm, is best represented in the frequency domain rather than in raw temporal features. Also, since different subjects can have similar heart-rate frequencies, a standard contrastive loss may confuse these cases and treat physiologically similar signals as negatives. To handle this, the RS-rPPG framework moves the contrastive learning process to the frequency domain and introduces a triplet-based loss that measures similarity directly between the spectra of the signals. This allows the model to focus on true physiological differences in heart-rate patterns rather than visual or spatial variations.

From InfoNCE to the Frequency-Domain Triplet Loss

While the InfoNCE objective effectively learns representations by bringing similar samples closer and pushing dissimilar ones apart, it works only in the latent embedding space. For physiological signals such as rPPG, this creates two main problems. First, the cosine similarity in Eq. 3.1 measures how close embeddings are in a high-dimensional space, but it does not necessarily reflect the actual physiological properties of the signal, such as its heart-rate frequency. Second, InfoNCE treats all negative pairs as different, even when two recordings have nearly the same heart rate. This leads to *false negatives*—samples that are physiologically similar but incorrectly treated as unrelated—which can harm model stability and reduce generalization.

To overcome these issues, RS-rPPG extends the contrastive objective into the *frequency domain*. Instead of comparing embeddings directly, the model compares their normalized *power spectral densities (PSDs)*, which describe how signal energy is distributed across frequencies. This transformation aligns the learning objective with the physiological nature of the data, encouraging the model to focus on frequency components that reflect true cardiac dynamics rather than irrelevant spatial or visual differences.

Let $\bar{a}(f)$, $\bar{p}(f)$, and $\bar{n}(f)$ denote the normalized PSDs of the anchor, positive, and negative samples, respectively. Here, the anchor corresponds to an MSTmap segment, the positive to the corresponding Tmap (shuffled across ROIs and channels to ensure invariance), and the negative to a Tmap from a different subject. The *frequency-domain triplet loss* is defined as:

$$\mathcal{L}_{\text{freq-trip}} = \max(0, m + d(\bar{a}, \bar{p}) - d(\bar{a}, \bar{n})), \quad (3.2)$$

where $m > 0$ is a margin parameter, and $d(\cdot, \cdot)$ measures the mean absolute spectral difference restricted

to the physiological heart-rate range $[f_{\min}, f_{\max}] = [0.5, 3]$ Hz:

$$d(\bar{u}, \bar{v}) = \frac{1}{f_{\max} - f_{\min}} \sum_{f=f_{\min}}^{f_{\max}} |\bar{u}(f) - \bar{v}(f)|. \quad (3.3)$$

This margin-based formulation enforces that the spectral distance between anchor and positive samples remains smaller than that between anchor and negative samples by at least m . By defining the comparison directly on the PSDs, the loss measures similarity according to the heart-rate structure of the signal rather than latent-space geometry. As a result, the model learns to align samples that share similar cardiac frequency patterns and to separate those with distinct physiological signatures.

Conclusion. This section outlined the mathematical foundations of contrastive learning and how they are applied to rPPG signal analysis. Starting from the InfoNCE objective, contrastive learning was shown to learn meaningful representations by maximizing similarity between related samples and separating unrelated ones. Its information-theoretic view connected this process to maximizing mutual information, enabling self-supervised learning without labeled data. To better capture physiological structure, the RS-rPPG framework extended this idea to the frequency domain, comparing normalized power spectra instead of abstract embeddings. This frequency-domain triplet loss embeds cardiac priors directly into the learning process, producing representations that are both discriminative and physiologically consistent with heart-rate dynamics.

3.3 Key Innovations of This Work

This thesis builds on the RS-rPPG framework of Savić and Zhao [10] by introducing two main extensions that make the method more relevant for real-world use. The first is that, while the original work focused only on uncompressed video data, here the models are trained and evaluated on compressed videos (H.264 with different CRF levels), similar to those typically produced by webcams, teleconferencing systems, or mobile devices. The second extension is a change in the training setup: during both pretraining and training, we use uncompressed physiological maps (Tmap0) as a form of supervision. This provides the model with a clean reference of the underlying signal while it learns from compressed inputs. In effect, the model benefits from extra guidance at training time, making the process closer to a semi-supervised setting where uncompressed data act as a strong proxy for ground truth.

Training and evaluating for compressed video.

The original RS-rPPG framework was designed and validated on uncompressed facial videos, where the subtle color changes caused by blood flow are preserved and the signal-to-noise ratio is relatively high [10]. In real-world applications, however, videos are almost always stored or transmitted in compressed form. Compression can blur or distort the very small, periodic color variations that rPPG relies on, making the task more challenging. To bridge this gap, we extended the training and evaluation of RS-rPPG to include controlled H.264 compression, focusing on different CRF levels and, where relevant, other encoder parameters discussed in Chapter 2. In practice, this means that the model inputs (MSTmaps) are generated from compressed rather than raw video clips, while the architecture and loss functions remain unchanged.

By testing the framework under these more realistic conditions, we aim to assess whether the model can still recover reliable pulse signals when video quality is degraded. Our results show that the models maintain good accuracy at moderate compression levels, degrade more gradually at higher compression, and in some cases even approach the performance of a commercial baseline (see the Results section). In this way, the method is shifted from a proof-of-concept validated only under ideal laboratory conditions to one that is explicitly stress-tested for bandwidth-limited scenarios such as teleconferencing or mobile health monitoring.

Using Tmap0 as supervision (semi-supervised)

The second contribution of this work is the introduction of a semi-supervised strategy that incorporates a clean, physiology-based signal during training. For each video clip, we compute Tmaps from the uncompressed version (Tmap0) and use them as supervision in both the pretraining and training stages. In this setup, the model receives compressed inputs but is guided by uncompressed reference targets. This resembles a teacher–student framework, where the student (trained on compressed data) learns from the teacher (uncompressed Tmap0), which is only available during training. By doing so, we combine the efficiency of self-supervised learning with the stability provided by a strong proxy target, making the approach a form of semi-supervised learning [21, 22].

In the pretraining stage, the model is presented with MSTmaps extracted from compressed videos and is trained to predict the corresponding uncompressed Tmap0. The loss is computed as the L_2 distance between the predicted and reference power spectra within the physiological band (0.5–3 Hz). This encourages the backbone to focus on cardiac-related frequencies and to learn features that remain consistent despite compression. As a result, pretraining aligns the model with physiologically meaningful representations rather than compression-induced noise.

During contrastive training, compressed MSTmaps are used as anchors, while positives come from the uncompressed Tmap0 of the same segment, and negatives from Tmap0 belonging to other subjects or recordings. This setup ensures that anchors are always pulled toward clean, physiology-aware positives instead of potentially noisy compressed signals. The triplet loss in the frequency domain then enforces this alignment, while band-limiting and sparsity regularizers preserve physiological plausibility. In effect, Tmap0 acts as a teacher signal, it does not provide explicit BPM labels, but it supplies a structured spectral reference that guides the student toward robust and physiologically accurate embeddings.

This semi-supervised design stabilizes training in two key ways. First, it reduces label noise by ensuring that positives always carry a clean spectral signature. Second, it bridges the gap between compressed inputs and physiological priors, helping the model to generalize more effectively across CRF levels and datasets. As our experiments show, this improves performance particularly at moderate compression and leads to more reliable results in cross-dataset evaluations.

The two contributions of this work, training directly on compressed video data and using uncompressed Tmap0 as a supervisory signal, build on the original RS-rPPG framework [10] while making it better suited to real-world conditions. These changes provide the model with both realistic inputs and cleaner guidance during training, combining the strengths of self-supervised learning with teacher-like supervision. The resulting pipeline is still label-efficient, but now it aligns more closely with the challenges of practical deployment, where compression is unavoidable, and delivers representations

that remain consistent with the physiological nature of the rPPG signal.

3.4 Metrics of the paper

To evaluate the performance of our heart rate estimation models, we use a set of well-established metrics that measure both accuracy and consistency. Following the RS-rPPG paper [10], we report mean absolute error (MAE), root mean square error (RMSE), Pearson’s correlation coefficient (R), and the standard deviation (STD) of the errors. Each of these metrics provides a distinct perspective on model performance across different datasets and compression settings.

Two of these metrics, correlation and MAE (which, as noted in Chapter 2, closely relate to the BPM error), have already been explained in detail. Briefly, the correlation coefficient quantifies how closely the estimated and reference heart rate signals align in shape and timing, while the mean absolute error (MAE) reflects the average difference between predicted and ground-truth heart rates, with lower values indicating higher accuracy.

In the remainder of this section, we provide the mathematical definitions and interpretation of the two additional metrics used in this study: root mean square error (RMSE) and standard deviation (STD) of the estimation errors. These metrics further contribute to a comprehensive assessment of model accuracy and reliability.

Root Mean Square Error (RMSE)

Root Mean Square Error (RMSE) is a standard metric for evaluating the accuracy of continuous predictions, and is particularly valued for its sensitivity to larger errors. For a sequence of N predicted values $x = [x_1, \dots, x_N]$ and corresponding ground-truth values $s = [s_1, \dots, s_N]$, RMSE is defined as:

$$\text{RMSE}(x, s) = \sqrt{\frac{1}{N} \sum_{n=1}^N (x_n - s_n)^2}$$

This formula computes the square root of the mean of squared differences between each prediction and its reference value. The squaring process means that larger errors have a disproportionately greater influence on the final value, making RMSE an appropriate choice for tasks where occasional large deviations are particularly undesirable. In this study, RMSE is used to assess not just the average accuracy of heart rate estimation, but also the model’s ability to avoid outliers or significant mistakes in its predictions. Lower RMSE values indicate that the model’s predictions are both close to the ground truth and consistently reliable, while higher RMSE values may reveal rare but substantial errors that could impact practical use. This metric thus complements others, such as MAE and correlation, providing a more complete view of model performance in real-world scenarios.

Standard Deviation (STD) of Error

Standard deviation (STD) of the prediction errors is a metric that captures the variability or spread of errors produced by the model. For a set of N error values $e_n = x_n - s_n$, where x_n is the model

prediction and s_n is the ground-truth value, the standard deviation is defined as:

$$\text{STD}(e) = \sqrt{\frac{1}{N} \sum_{n=1}^N (e_n - \bar{e})^2}$$

where $\bar{e}$ is the mean error across all samples. A lower STD indicates that the errors are consistently close to their mean value, reflecting stable and predictable model performance. Conversely, a higher STD means that the errors are more dispersed, pointing to fluctuations in the accuracy of the model’s predictions. In the context of rPPG estimation, a low standard deviation is desirable because it suggests that the model performs reliably not just on average, but also across individual samples and varying conditions. The RS-rPPG paper includes this metric to provide additional insight into the consistency and robustness of the model, ensuring that performance is not only accurate on average but also dependable from trial to trial.

By evaluating model performance with MAE, RMSE, correlation, and STD, we gain a comprehensive view of both the accuracy and the consistency of heart rate estimation under different experimental conditions. These metrics help to highlight the strengths and limitations of the RS-rPPG model, guiding future improvements and practical deployment.

3.5 PURE Experiments and Results

In this section, we present the results of our experiments on the PURE dataset. Since PURE is an uncompressed and relatively clean benchmark, it serves as an ideal baseline for controlled analysis. To study how video compression influences remote heart-rate estimation, we re-encoded all videos using the H.264 codec at different Constant Rate Factor (CRF) levels, thereby generating multiple versions of the dataset with varying compression strengths. Each experiment follows a standardized protocol: for every data fold, the model is first pretrained, then trained, and finally evaluated across all CRF levels to assess both performance and robustness under increasing compression.

Data Splits and Training Protocol

To ensure robust and reproducible evaluation, the PURE dataset was divided into a set of fixed, non-overlapping folds, with each fold corresponding to a contiguous segment of the video data. These folds were not created using random sampling, but rather by sequentially partitioning the dataset, allowing for consistent segmentation and evaluation across all experiments.

For each fold, our experimental pipeline consisted of two main stages:

- **Pretraining:** In the first stage, the model was pretrained to reconstruct physiological maps (tmaps) from compressed input representations (mstmaps). We compared two pretraining strategies: one where only compressed tmaps were used as targets, and another where the uncompressed tmap0 was always provided as the regression target, regardless of the compression level of the inputs. This comparison allowed us to assess whether exposing the model to a clean reference during pretraining encourages better feature learning and generalization.
- **Main Training:** After pretraining, the model was further trained for heart rate estimation, again

using two variants: either supervising with compressed tmaps only, or consistently using the uncompressed tmap0 as the target. The training framework combines frequency-domain loss functions, triplet-based contrastive objectives, and regularization techniques, all designed to help the model learn to extract accurate physiological information, even from highly compressed videos.

Evaluation Methodology

After each training run, the resulting models were evaluated on the full range of CRF levels to analyze their robustness to compression. The evaluation metrics reported include mean absolute error (MAE), root mean squared error (RMSE), standard deviation (STD), and correlation (R) with ground-truth heart rates. For every experiment, results were recorded for each CRF level and compared with the FacerPPG baseline.

Pretrained and Trained using paper implementation

As a reference, we first trained and evaluated models separately for each CRF level using the conventional approach established in the literature. Specifically, we followed the data split and evaluation protocol described for the PURE dataset as discussed in the paper. The results for this standard, single-CRF approach, where the uncompressed tmap0 was not included in either pretraining or training, are evaluated using four key metrics, and for each CRF level, the deep learning model's performance is compared to the FacerPPG baseline.

The results are also visualized in Figures 3.4, 3.5, and 3.8.

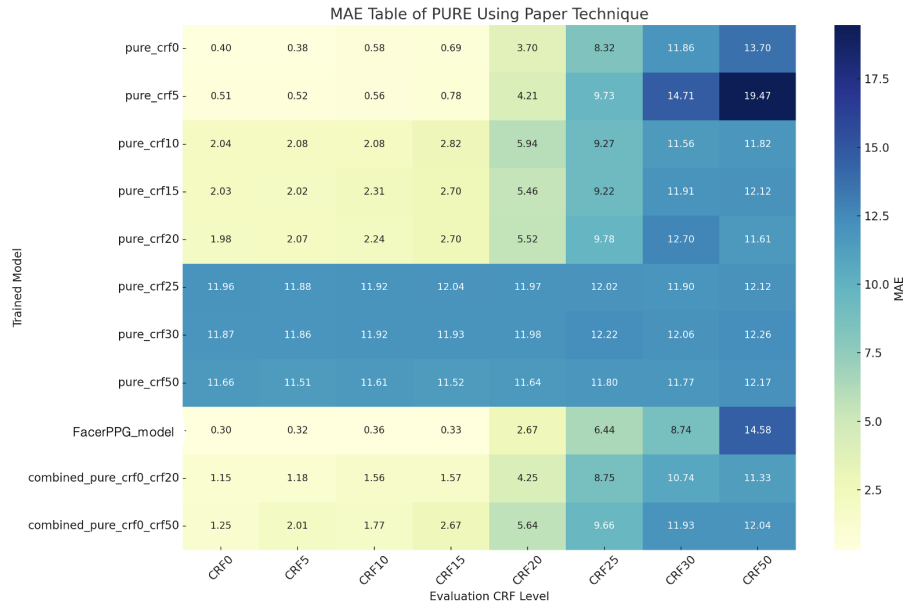


Figure 3.4: MAE results from training on PURE, compared to FacerPPG model.

The MAE table shows how heart rate estimation error grows as video compression (CRF) increases. At low CRF values (CRF0 to CRF10), all models, including the FacerPPG baseline, perform very well, with errors below 2. As compression becomes stronger (CRF20 and higher), the error rises quickly for all

single-CRF models. In contrast, the combined-CRF models are much more stable and manage to keep the MAE lower even at high compression, showing that they learn to handle different compression levels better. While FacerPPG still gives the lowest error for uncompressed videos, the combined models come very close and remain more reliable as CRF increases, making them the best overall choice for real-world compressed video.



Figure 3.5: Correlation coefficient (r) results from training on PURE, compared to FacerPPG model.

The R table shows a pattern similar to the MAE table, all models achieve excellent correlation with the reference signal when compression is low (CRF0 to CRF10), but their performance drops as the compression becomes more aggressive. What stands out here is that the combined-CRF models consistently maintain higher correlation across all CRF levels compared to single-CRF models, especially at higher compression settings. While FacerPPG still has the highest correlation at low compression, the combined models show strong resilience and remain closely aligned with the ground truth even as CRF increases. This highlights the ability of combined-CRF training to preserve signal quality and track physiological changes, even under challenging compression conditions.

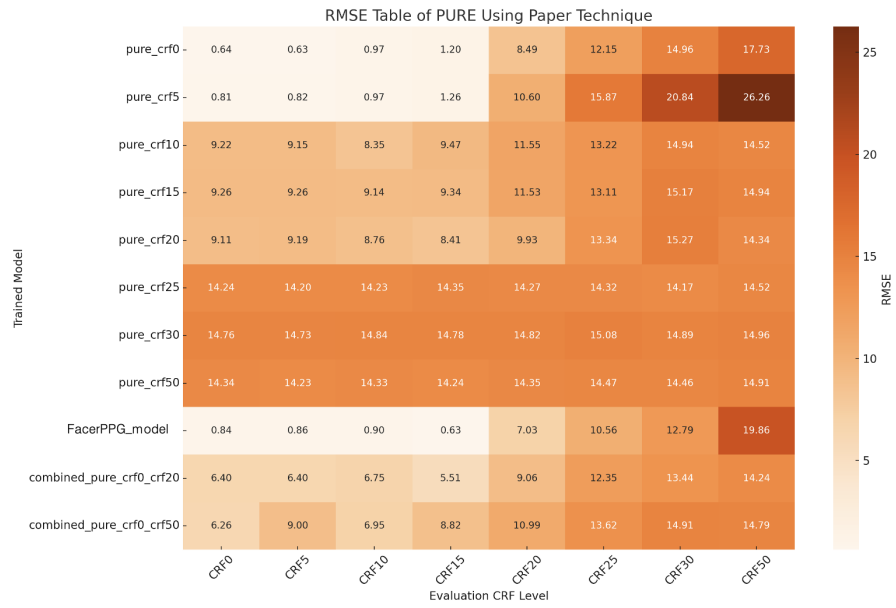


Figure 3.6: RMSE results from training on PURE, compared to FacerPPG model.

The RMSE table shows the same trend as the MAE and R tables, with errors rising sharply at higher CRF levels, especially for single-CRF models. Notably, the combined-CRF models achieve lower RMSE across a broader range of compression levels, suggesting better generalization. FacerPPG's RMSE is very low at low CRF but increases quickly as compression gets stronger. One distinct observation in this table is that, for some models, RMSE grows even faster than MAE as compression intensifies, underlining how outliers and large errors become more pronounced at high CRF.

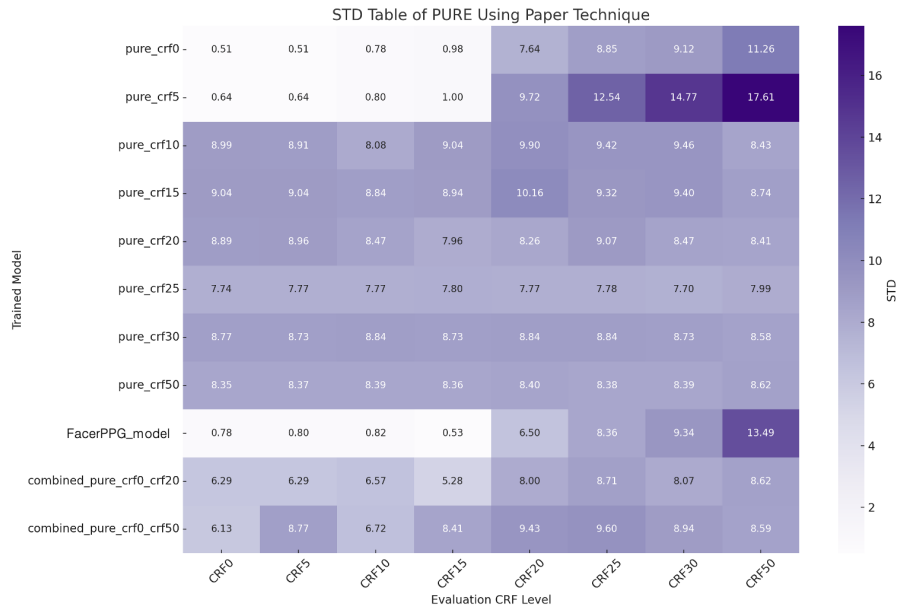


Figure 3.7: Standard deviation (STD) results from training on PURE, compared to FacerPPG.

The STD table follows the same overall pattern as the other metrics, but focuses on the variability of errors rather than their average size. What stands out here is how the standard deviation jumps

much more rapidly for high CRF levels, especially for single-CRF models and FacerPPG. This means predictions become much less stable as compression increases, with more frequent large outliers. The combined-CRF models keep STD lower and more consistent across CRF levels, which suggests they are less sensitive to heavy compression and provide more reliable predictions even as video quality drops.

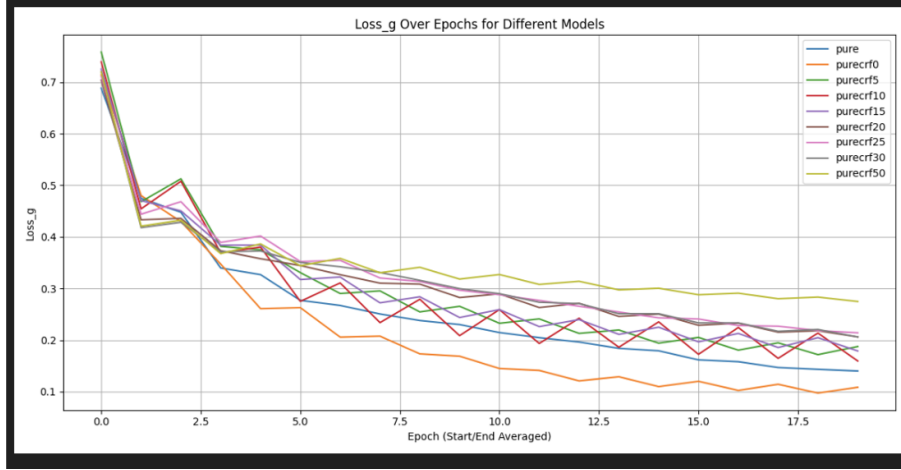


Figure 3.8: Loss curve during baseline training.

The loss curves for all single-CRF models show a clear downward trend over training epochs, meaning each model successfully learned from the data. Models trained on less compressed data (like purecrf0) reach a lower final loss, while those trained on highly compressed videos (such as purecrf50) show higher loss at the end. This highlights how strong compression makes learning more challenging, but all models show steady progress during training.

Conclusion. Looking at these results, it's clear that CRF, the video compression level, plays a major role in heart rate signal quality when working with the PURE dataset. Models trained on lower CRF values generally perform best, with higher accuracy and more stable results across all metrics. As CRF increases, the performance drops off, but interestingly, the RS-rPPG model trained on a combination of different CRF levels can better handle these changes and still find the heart rate signal even when the compression is strong. While the FacerPPG model remains a solid reference point, the combined RS-rPPG model comes surprisingly close, showing that with the right training approach, deep learning models can adapt to a range of compression settings on real-world video data like PURE. This makes CRF a key factor to consider for anyone aiming to extract physiological information from compressed video.

Pretraining and Training with Tmap0 Supervision

Next, we evaluated the effect of using the uncompressed Tmap0 as the supervision target throughout both the pretraining and main training phases. In this setup, regardless of the compression level of the input, the model always learns to reconstruct the original, high-quality physiological map (Tmap0). The idea behind this strategy is to expose the network to compressed inputs while consistently guiding

it towards the best possible physiological signal, encouraging it to develop robustness against various compression artifacts. This setup was applied to both single-CRF and combined-CRF models, allowing for a direct comparison of how Tmap0 supervision influences the model's ability to recover physiological information across different compression scenarios.

Using uncompressed Tmap0 as the supervision target shows noticeable changes compared to the baseline, especially in how well the model handles compressed inputs. The following figures make it easy to spot these differences and show how this method helps recover the rPPG signal under different compression settings.

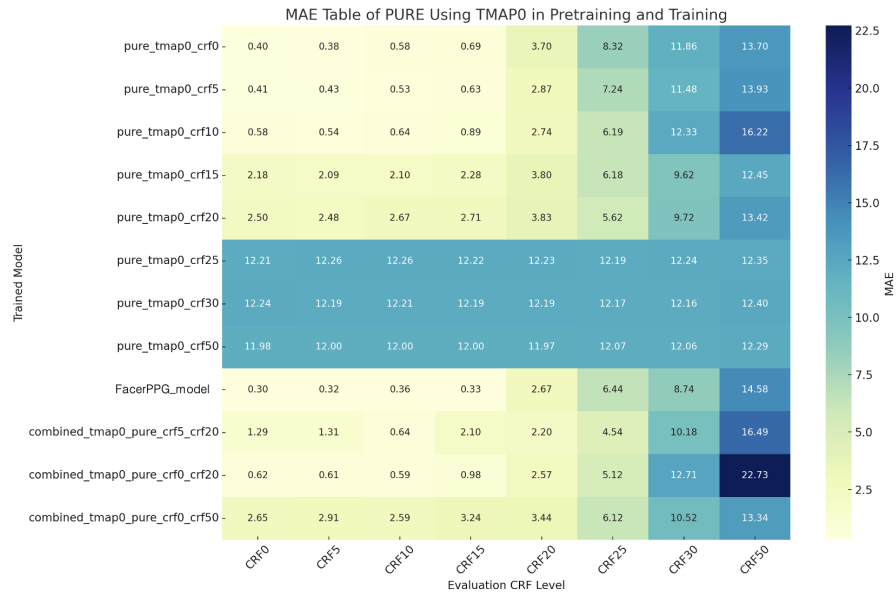
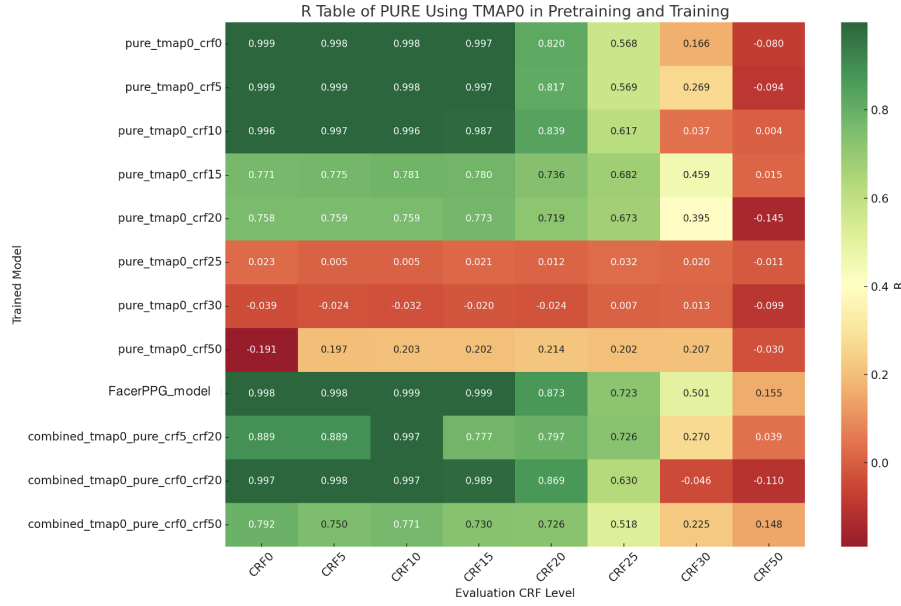


Figure 3.9: MAE results from training with tmap0 targets, across CRF levels, showing improvement over FacerPPG and the single-CRF baseline.

The MAE table above highlights a clear improvement when using Tmap0 as the ground truth in both pretraining and training, compared to the baseline results from the previous part. Across almost all single-CRF models, the errors are consistently lower at each CRF level, showing that this approach helps the network generalize better to different compression strengths. Most notably, the combined models using Tmap0 supervision adapt even more effectively: for high CRF levels, like CRF20 and CRF25, their performance even surpasses that of the FacerPPG model, which is impressive given that FacerPPG is a commercial solution. This suggests that the combined Tmap0 models are not just robust to compression, but also learn to handle the distortion patterns introduced by different CRF levels exceptionally well.

Figure 3.10: Correlation coefficient (r) results from training with Tmap0 targets

The r table for Tmap0 supervision tells a similar story as the MAE, there is clear improvement for almost all models, especially at lower and moderate CRF levels. What is nice to see with the correlation metric is that the combined models using Tmap0 not only keep their accuracy but also do a better job of following the true heart rate pattern, even as compression increases. This means they do not just get close on average, but they also track the signal shape well. Correlation does drop at the highest CRFs for all models, but the combined models hold on a bit longer, showing they can better handle signal distortion from heavy compression.

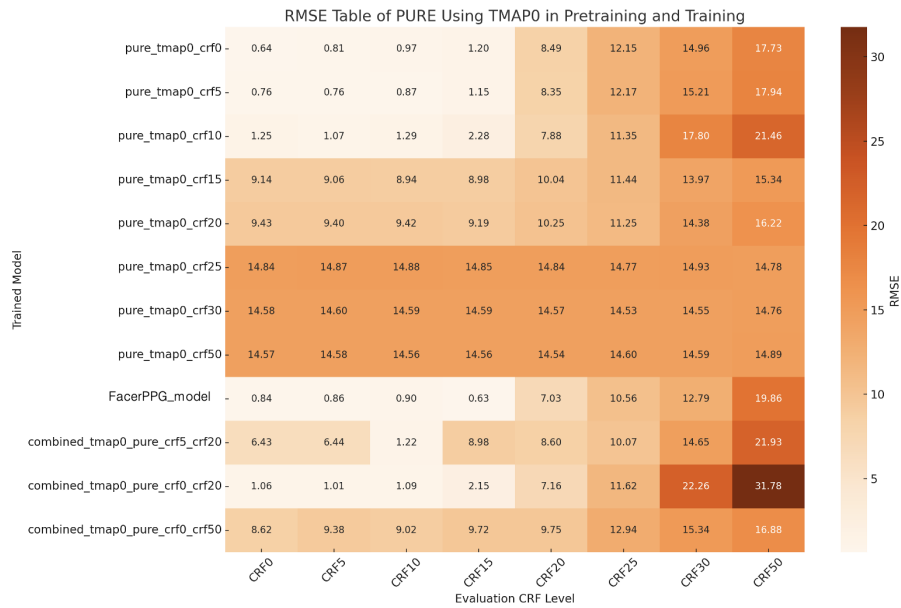


Figure 3.11: RMSE results from training with tmap0 targets

The RMSE table for tmap0 supervision mirrors the improvement trends seen in the other metrics.

Almost all single-CRF models see their errors drop, especially at low and moderate CRF levels, showing that the model makes fewer big mistakes. The real highlight is the combined model, which keeps RMSE low across compressions up to CRF20, outperforming the FacerPPG benchmark. Since RMSE highlights large prediction errors, these results make it clear that using tmap0 as the target in training helps the model handle compression artifacts more gracefully. However, as with MAE, performance drops for all models at the highest compression levels, confirming that extreme compression remains difficult to overcome.



Figure 3.12: Standard deviation (STD) results from training with Tmap0 targets

The STD table for Tmap0 supervision follows the same general pattern as the other metrics, showing improved results over the baseline for most models and CRF levels. As a metric, STD highlights how consistent the model's predictions are, lower values mean the errors are less spread out and more tightly clustered around the average. Here, using Tmap0 as the training target keeps the prediction variability lower for most models, especially at low and moderate compression. This consistency means the model is not only accurate on average but also less likely to produce outlier results, which is crucial for real-world reliability. As with the other metrics, very high compression still poses a challenge, but overall, STD confirms that Tmap0-based training helps models stay both accurate and stable across different compression levels.

Overall, these experiments on the PURE dataset show how much the choice of training strategy and supervision target matters when dealing with compressed video for remote heart rate estimation. Training with the uncompressed Tmap0 as the target made every model, both single-CRF and combined, more robust to compression and better at tracking the real heart rate, especially when video quality started to drop. The combined models, in particular, stood out for their ability to handle a range of compression levels and even matched or outperformed the FacerPPG benchmark in some cases. While strong compression still presents a challenge, these results make it clear that deep learning models, when guided by a clean physiological reference like Tmap0, can learn to extract useful signals from

compressed data. This has important implications for using rPPG in practical, bandwidth-limited settings, and provides a strong foundation for future work on model robustness and generalization in video-based physiological monitoring.

3.6 VicarrPPG Experiments and Results

In the previous section, a thorough evaluation of the PURE dataset showed how the proposed method copes with varying levels of video compression. The next step is to test whether these findings generalize to a more complex scenario, namely, the VicarrPPG dataset. Compared to PURE, VicarrPPG presents greater real-world challenges, including more diverse subjects, lighting conditions, and motion artifacts.

As an initial analysis, a heatmap was generated to visualize the mean absolute error (MAE) and correlation for each person and each video type. This revealed a clear outlier, Person 15 exhibited much higher errors across all video types than other subjects. To ensure a balanced and fair assessment, Person 15 was excluded from the validation set in all subsequent experiments.

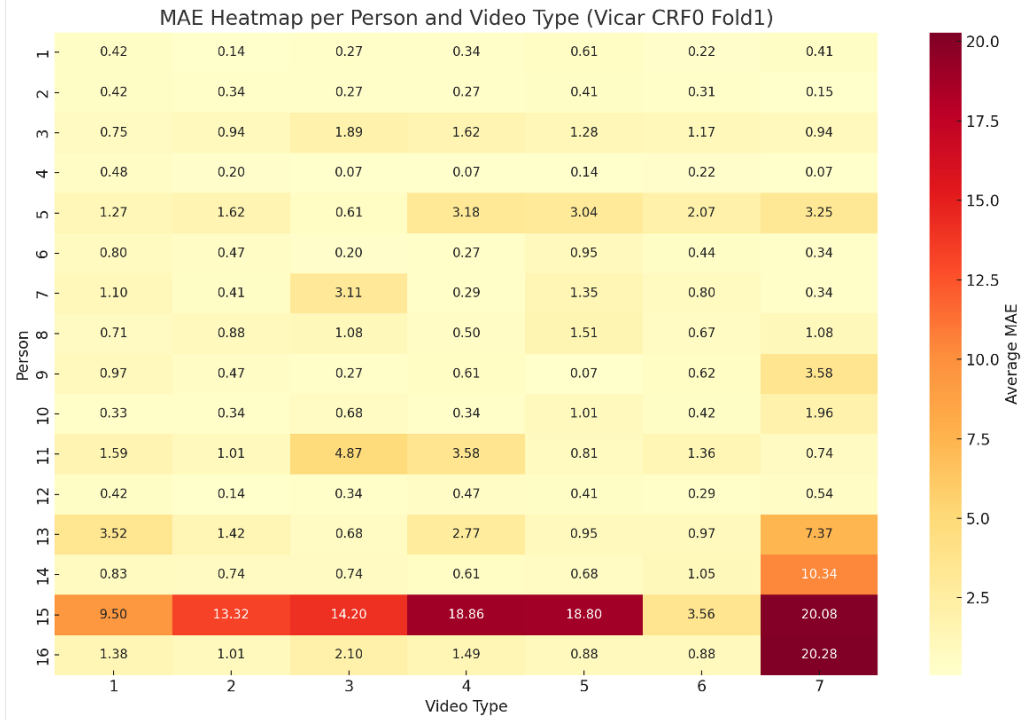


Figure 3.13: MAE heatmap for each person and video type in the VicarrPPG dataset (CRF0). Person 15 shows exceptionally high error, indicating a strong outlier effect.

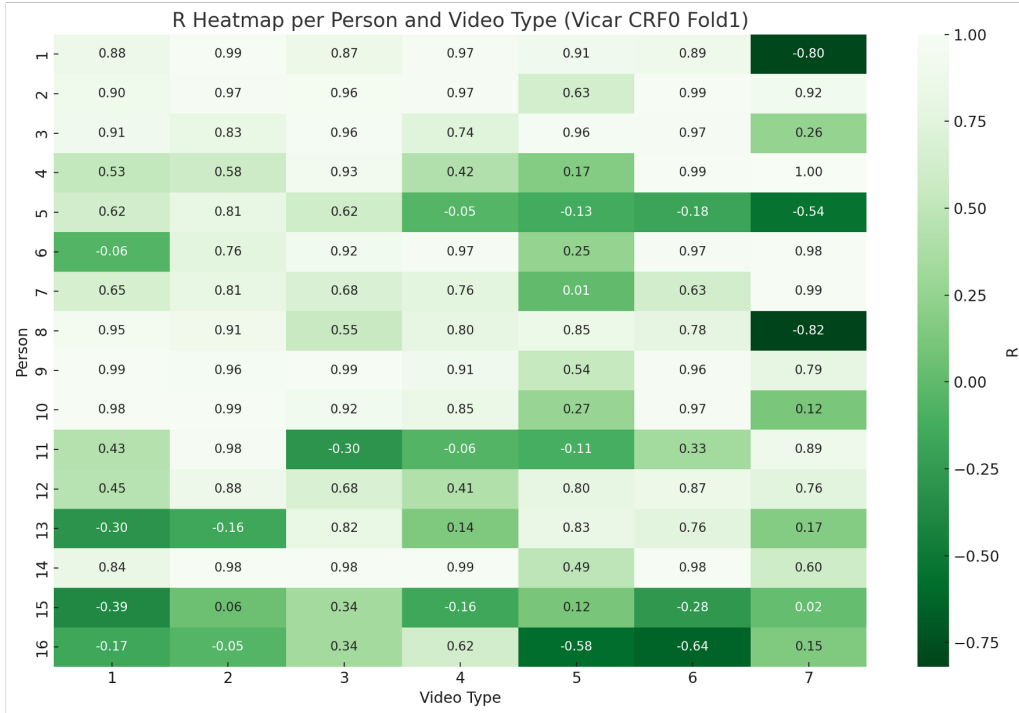


Figure 3.14: R heatmap for each person and video type in the Vicar rPPG dataset (CRF0). Person 15 consistently shows low correlation values, highlighting them as a potential outlier with unreliable signal prediction.

Drawing on the comprehensive analysis from the PURE dataset, where the combined model with Tmap0 supervision consistently outperformed other approaches, the same strategy was adopted here. In this section, both the single-CRF models and the combined model, pretrained and trained using the Tmap0 supervision. This choice was motivated by the previous section’s results, which demonstrated that using uncompressed Tmap0 as ground truth enabled more robust learning across different compression levels.

The following four figures present the results for MAE, r , RMSE, and STD across all models, illustrating each model’s ability to estimate heart rate under varying levels of CRF compression. For each metric, the performance of our approach is also compared to the commercial FacerPPG baseline, providing a clear benchmark for evaluation.

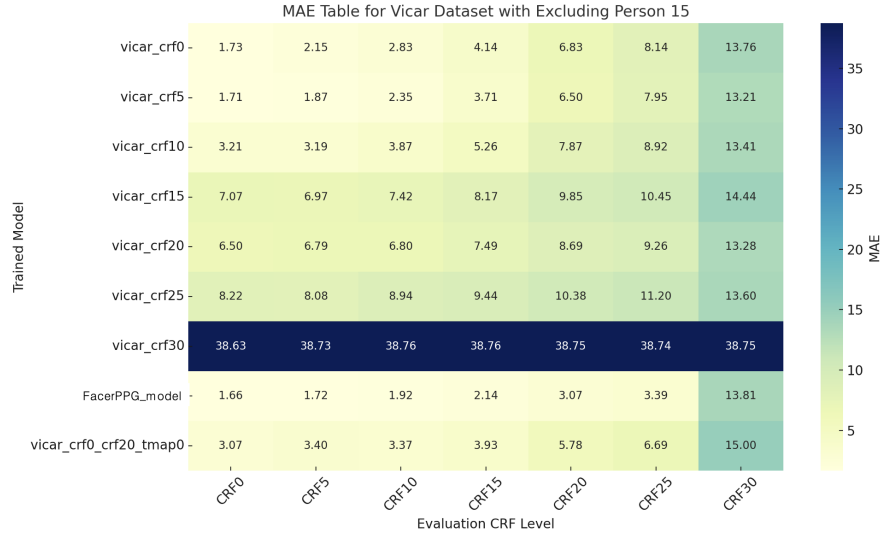


Figure 3.15: MAE table for VicarrPPG dataset with Person 15 excluded.

In the MAE table above, we observe that the VicarrPPG dataset is significantly more challenging than PURE, containing greater variability in subjects, lighting, and motion. As a result, the MAE increases sharply as compression becomes stronger. MAE is not always the most reliable metric in this context, since, similar to the BPM error discussed in Chapter 2, it is based on absolute peak differences. When compression distorts the physiological peaks, the MAE can rise disproportionately, sometimes exaggerating the true estimation error.

From the results, it is clear that the single-CRF models, especially the one trained only on CRF 30, struggle under higher compression levels, their errors increase rapidly, and they fail to follow the signal accurately. In contrast, the combined model trained with Tmap0 as ground truth achieves consistently better and more stable performance, particularly at moderate and high compression levels. Although it does not outperform the FacerPPG model overall, it demonstrates reasonable and improved results compared to the signal-only versions, showing stronger robustness against compression artifacts.

Figure 3.16: r (correlation) table for VicarrPPG dataset, Person 15 excluded.

Correlation serves as a more reliable metric in this context, as it evaluates how well the predicted heart-rate signal follows the true signal's overall pattern, rather than focusing solely on absolute peak differences. As shown in the table above, the combined model trained with Tmap0 as the baseline consistently outperforms all single-CRF models, particularly at higher compression levels. Notably, its correlation values remain close to those achieved by the FacerPPG model, indicating that the combined approach can effectively capture the overall heart-rate dynamics even under challenging compression conditions.

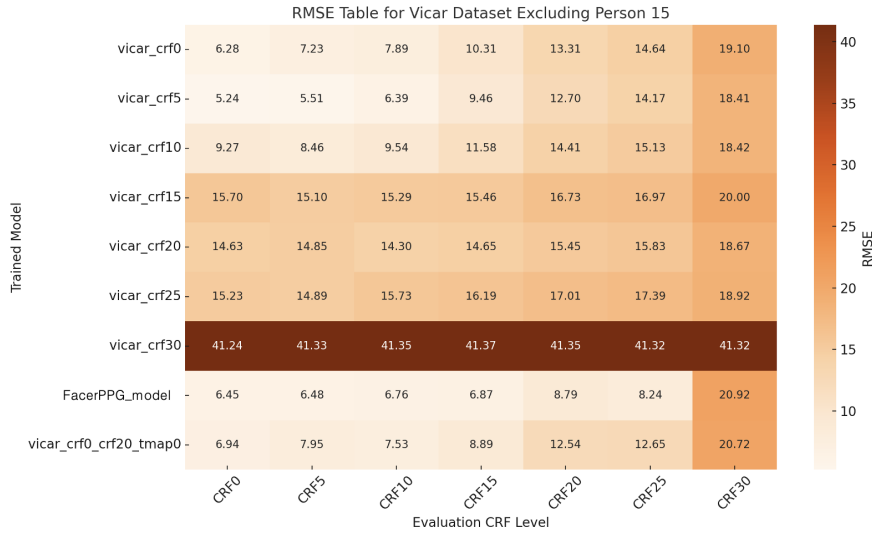


Figure 3.17: RMSE table for VicarrPPG dataset, Person 15 excluded.

The RMSE results follow a pattern similar to the correlation metrics but emphasize larger prediction errors. Unlike MAE, RMSE penalizes large deviations more heavily, making it particularly sensitive to occasional but severe inaccuracies in heart-rate estimation. As shown in the table, the combined model trained with Tmap0 as ground truth once again outperforms all single-CRF models, especially at

moderate compression levels, although it still does not reach the lower RMSE achieved by FacerPPG. At high compression levels like CRF 30, all models exhibit a sharp increase in RMSE, indicating that extreme compression not only raises average errors but also introduces more severe outliers in the predictions. This trend underscores both the difficulty of the VicarrPPG dataset and the advantage of using robust, combined-training strategies.

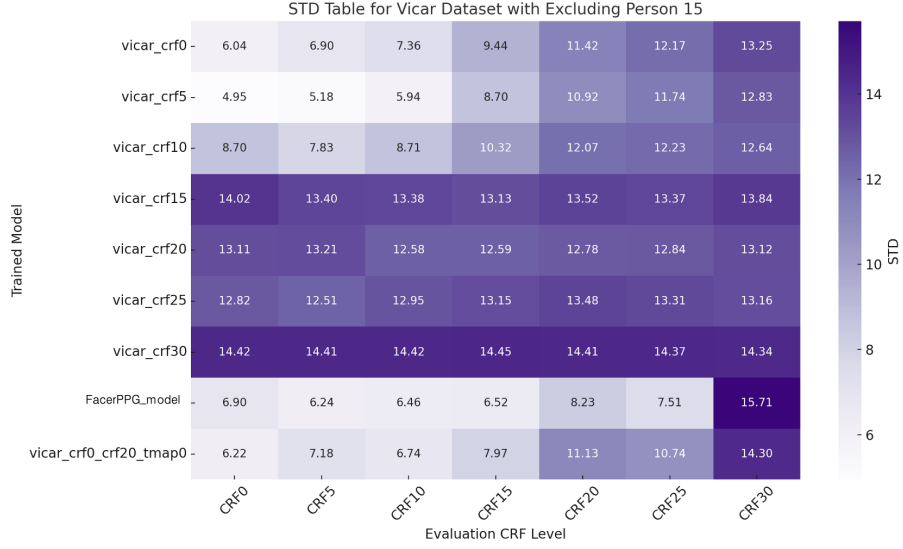


Figure 3.18: STD table for VicarrPPG dataset, Person 15 excluded.

The STD table above measures how spread out the model’s errors are across different CRF levels, giving us a sense of the prediction stability. In the VicarrPPG dataset, the single-CRF models show increasing variability as compression rises, especially from CRF10 upwards. The combined model using Tmap0 as ground truth keeps the standard deviation lower and more stable across most compression levels, which means its predictions are more consistent and less prone to outliers. However, similar to the other metrics, all models, including the combined one, struggle at the highest compression (CRF30), where STD jumps significantly. Notably, the FacerPPG model keeps the lowest STD at low and moderate compression, but its variability also increases at CRF30. This reinforces that while the combined model improves overall reliability, very strong compression remains a difficult challenge for every approach.

Overall, the VicarrPPG experiments confirm that video compression introduces a substantial challenge for remote heart-rate estimation, particularly in complex, real-world conditions. Across all four metrics, the combined model trained with Tmap0 as ground truth consistently outperforms the single-CRF models, especially at moderate to high compression levels. Nevertheless, none of the models, including the combined approach, surpass the FacerPPG baseline under extreme compression (CRF 30), reflecting the remaining limitations of current deep learning methods.

The MAE heatmap further revealed that Persons 13, 15, and 16 generated unusually high errors in certain video types, indicating potential outliers in the dataset. To assess robustness, additional experiments were conducted both with and without these subjects; the results of these extended evaluations are provided in Appendix B. Finally, Appendix C presents a theoretical explanation of the RS-rPPG model’s robustness under compression, formalizing the stability properties observed in these experiments.

3.7 Cross-Dataset Validation with PURE Folds

In this section, we investigated how well RS-rPPG models generalize to new data by performing cross-dataset validation using the PURE dataset folds as a test set. We compared two models: the first is our best-performing VicarrPPG model, trained with a combination of CRF0 to CRF20 data and using Tmap0 as supervision during training; the second is a baseline model trained only on no-compression (CRF0) VicarrPPG data. Both models were trained on the VicarrPPG dataset and then evaluated on the PURE dataset across different levels of video compression.

The heatmaps below present the results for four key metrics, as in the previous experiments in this chapter: mean absolute error (MAE), Pearson correlation coefficient (R), root mean square error (RMSE), and standard deviation (STD). All metrics are evaluated across different CRF levels and for both trained models.

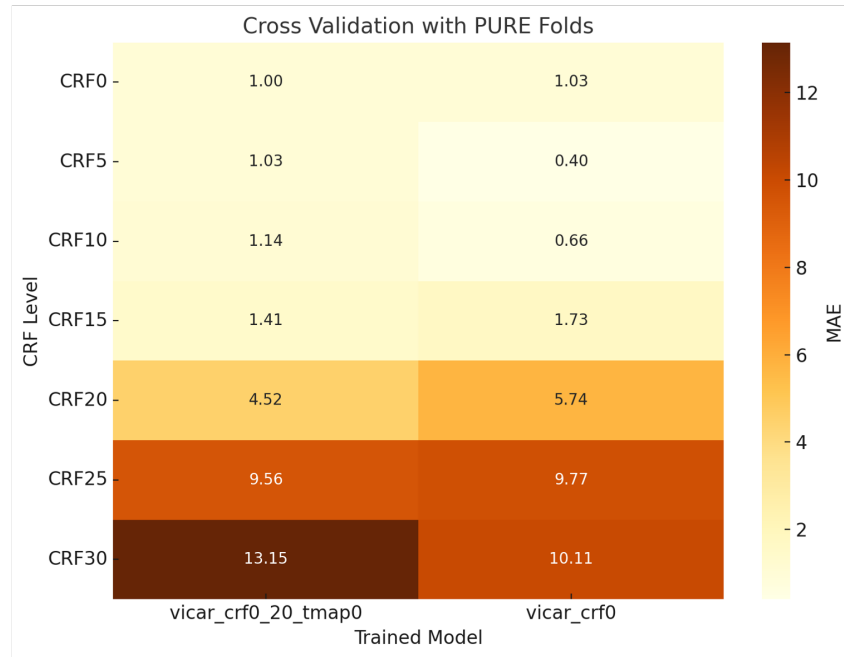


Figure 3.19: Mean Absolute Error (MAE) for cross-validation with PURE folds at different CRF levels. Lower MAE values indicate better accuracy.

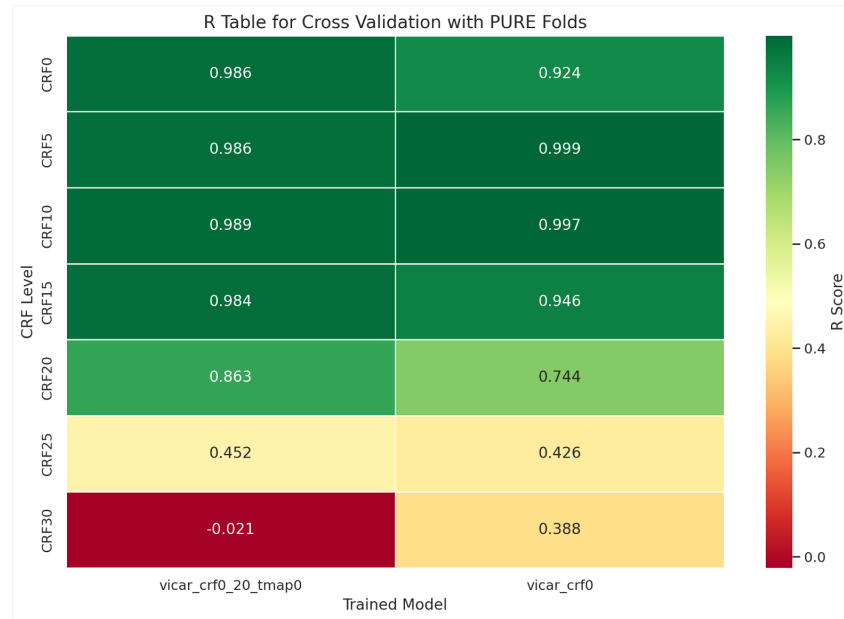


Figure 3.20: Pearson correlation (R) for cross-validation with PURE folds. Higher values indicate stronger agreement between predictions and ground truth.

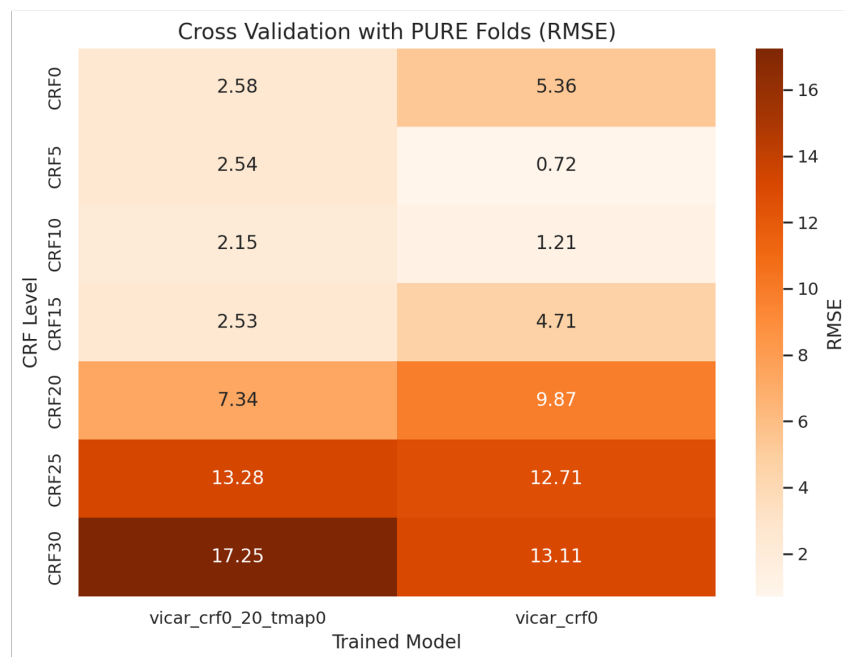


Figure 3.21: Root Mean Squared Error (RMSE) for cross-validation with PURE folds. Lower RMSE values are preferable.

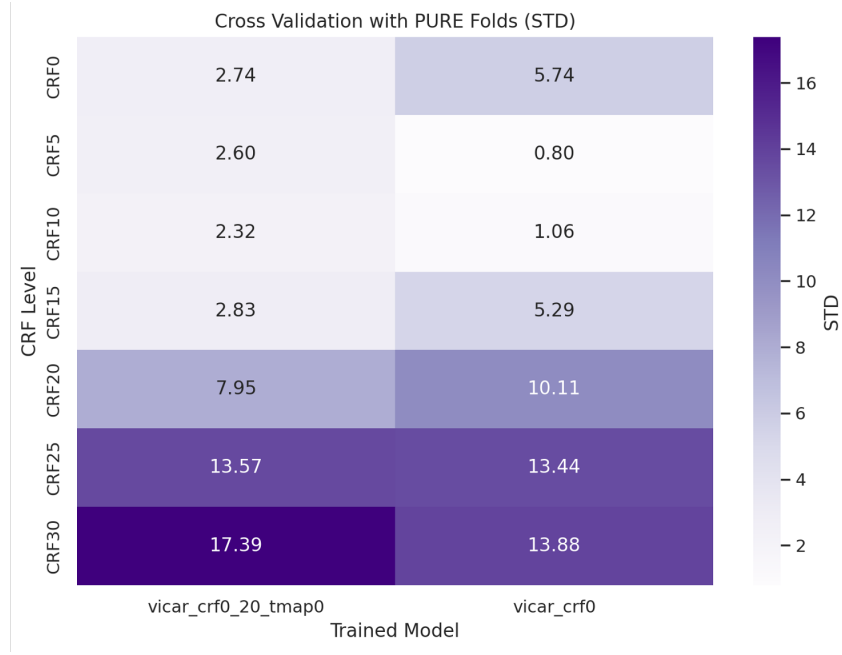


Figure 3.22: Standard Deviation (STD) of the error for cross-validation with PURE folds. Lower STD suggests more consistent predictions.

These cross-validation experiments provide valuable insight into how well the two RS-rPPG models generalize to new data and different video compression levels. As shown in Figures 3.19--3.22, both models perform strongly at lower compression settings (CRF0 to CRF15), with low MAE and RMSE, high R scores, and stable predictions. However, as the compression increases, clear differences emerge between the two approaches.

The combined model, `vicar_crf0_20_tmap0`, which was trained using multiple compression levels and supervised by the uncompressed Tmap0, performs noticeably better than the single-CRF model trained only on low-compression data. As shown in the MAE and RMSE heatmaps, the combined model maintains lower errors at moderate compression levels, for example, MAE increases modestly from 1.00 at CRF0 to 4.52 at CRF20, whereas the single-CRF model's MAE rises from 1.03 to 5.74. Similarly, its correlation values remain higher across the same range, showing a stronger ability to follow the true heart-rate pattern. Even in RMSE and STD metrics, the combined model demonstrates more stability, with slower growth in error as compression intensifies. However, when compression becomes extremely high, as at CRF30, both models experience a sharp performance drop, showing that very strong compression still degrades signal quality beyond recovery.

These findings suggest that training with a more diverse set of compression settings helps the model learn features that generalize better across different datasets and real-world video qualities. The combined approach makes the model less sensitive to changes in compression and allows it to handle a wider variety of input conditions with greater reliability.

Overall, this comparison highlights the benefit of combined supervision in developing robust rPPG models that work well not just on the training data, but also on new and unseen data with varying video quality. This generalization is important for practical use, where conditions can change and perfect video quality cannot always be guaranteed.

Chapter 4

Conclusion and Future Work

In this thesis, we addressed the problem of estimating heart rate from compressed facial videos, which is a relevant challenge for practical applications in remote health monitoring and affective computing. Our approach combined a thorough investigation of video compression effects with the implementation and enhancement of a state-of-the-art deep learning model for rPPG estimation.

We started by systematically examining the main parameters of H.264 video compression, including CRF, GOP size, B-frames, reference frames and motion estimation. Through the experiments described in Chapter 2, we explored how each parameter can influence the preservation or loss of subtle physiological signals required for accurate rPPG extraction. This analysis provided valuable guidance for the design and evaluation of rPPG models in realistic, bandwidth-limited scenarios.

Building on this foundation, we implemented the RS-rPPG model and adapted its training process by introducing uncompressed physiological maps (Tmap0) as supervision during both pretraining and training phases. We tested the model using two datasets, PURE and VicarrPPG, and evaluated its robustness across various compression settings. Our results showed that the proposed training strategy helped the model maintain reliable performance even as video quality decreased due to compression.

Importantly, we used cross-validation in our experiments to ensure that the improvements were not limited to specific data splits, but generalized across different subjects and video types. This supports the idea that the model is able to learn features that work well for a range of individuals and recording conditions. Furthermore, our analysis highlighted that the presence of certain outlier subjects could strongly influence overall performance metrics; by examining results with and without these outliers, we provided a clearer and more nuanced understanding of the model's strengths and limitations.

A particularly promising outcome was that, in some cases, our RS-rPPG-based pipeline performed as well as or even better than the commercial FacerPPG system. This suggests that integrating deep learning models like RS-rPPG into established signal processing frameworks could bring further improvements in accuracy and robustness for real-world use.

In conclusion, our work demonstrates that careful analysis of video compression effects, combined with thoughtful training strategies such as using uncompressed supervision, can substantially improve the reliability of remote heart rate estimation from facial videos. The proposed approach not only maintained strong performance under a variety of compression settings, but also showed the ability to generalize well across subjects and video types, as evidenced by our cross-validation experiments. While there remain challenges, such as handling outlier cases and adapting to diverse real-world video sources, our results suggest that deep learning-based pipelines like RS-rPPG can offer a valuable improvement over existing commercial solutions, especially when integrated with established signal processing

frameworks. At the end, this thesis provides a practical contribution towards more robust, accurate, and adaptable rPPG estimation in real-world scenarios, and lays the groundwork for future innovations in video-based physiological monitoring.

Future Work

Building on the findings and progress described in this thesis, there are several directions in which this work can be extended and improved. In the following, we outline some of the most promising avenues for future research and development related to robust rPPG estimation from compressed video.

Pulse-Signal Magnification Integration

Recently, Comas et al. [23] proposed a two-stage deep learning framework to address the challenges of video compression in remote heart rate estimation (rPPG). In their approach, the first stage involves training a heart rate estimator on high-quality, uncompressed video data. This allows the model to learn how to extract physiological signals accurately when all relevant visual information is present. The second stage introduces a pulse-signal magnification network, which is trained to enhance the subtle variations in compressed videos that correspond to the heartbeat signal. Importantly, the pulse-signal magnifier is guided by the pre-trained estimator, which is kept fixed during this phase and serves as a reference for what features are important for heart rate recovery. By focusing specifically on amplifying those features critical for rPPG estimation, rather than reconstructing the entire visual content, the method achieves strong results in both intra- and cross-dataset experiments, even under high levels of video compression.

Building on the pulse-signal magnification framework, we propose to combine the strengths of both approaches by directly replacing the generic rPPG estimator in the original pipeline (as shown in Figure 4.1) with our RS-rPPG model. In this integration, RS-rPPG serves as the heart rate estimator within the two-stage pulse-signal magnification process. The motivation for this design is to leverage the signal-enhancing capabilities of the magnifier network together with the proven robustness of RS-rPPG, especially for videos that have undergone significant compression.

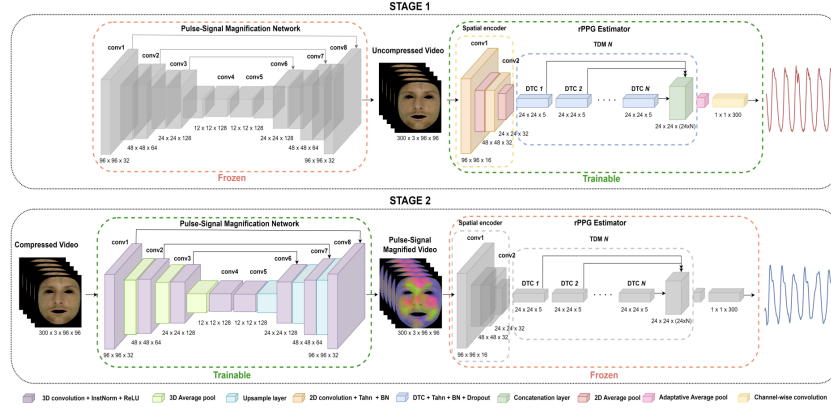


Fig. 1: Overall structure of our proposed model, which has two stages for rPPG recovery under compression. Firstly, the TDM model is trained on uncompressed data. Then, the Pulse-Signal Magnification network is trained on compressed data with fixed TDM model parameters.

Figure 4.1: Two-stage pulse-signal magnification pipeline for robust rPPG recovery under compression, adapted from Comas et al. [23]. In our proposed approach, the RS-rPPG model replaces the standard rPPG estimator component.

With this modification, the overall workflow consists of two main stages, as illustrated in Figure 4.1:

- **Stage 1 – Training RS-rPPG on Uncompressed Data:**

First, the RS-rPPG model is trained using high-quality, uncompressed facial videos. This enables the model to learn how to extract pulse signals as accurately as possible when all physiological details are preserved. The resulting RS-rPPG model serves as a reliable estimator of heart rate signals under ideal conditions.

- **Stage 2 – Training the Pulse-Signal Magnifier on Compressed Data:**

Next, the pulse-signal magnification network is trained to process compressed videos, transforming them into an intermediate form in which the rPPG signal is amplified. During this stage, the RS-rPPG estimator from Stage 1 remains fixed and is used as a reference to guide the training of the magnifier network, ensuring that the amplified features are those most relevant for heart rate estimation. Alternatively, both networks may be fine-tuned jointly, allowing for mutual adaptation and potentially better performance.

By explicitly replacing the rPPG estimator with RS-rPPG in the pulse-signal magnification pipeline, this hybrid approach brings together the ability to enhance physiological signals in compressed video and the strong extraction capabilities of RS-rPPG. The anticipated result is a more robust and accurate system for remote heart rate estimation, particularly in real-world scenarios where video quality is degraded due to compression.

Furthermore, this design makes it possible to experiment with various training strategies, such as fixing the estimator or training both components together, offering a flexible framework to optimize performance across a range of conditions. Ultimately, this integration is expected to deliver a reliable solution for heart rate estimation, even when the input video quality is far from optimal.

Upsampling and Downsampling for Data Augmentation

Another useful direction for future work is to use temporal upsampling and downsampling on the heart rate signals during training. By artificially speeding up (upsampling) or slowing down (downsampling) the heart rate signals in the training data, we can create more examples with a wider range of heart rates. This helps the RS-rPPG model learn to recognize and estimate heart rate peaks more accurately, even for people with unusually fast or slow heartbeats. Using this strategy as a form of data augmentation has been shown to improve the precision and robustness of rPPG models in previous studies, for example in [24]. Incorporating such augmentation into our training pipeline could further strengthen the model's ability to generalize to new and challenging video data.

Compression Parameter Exploration

In this thesis, we primarily validated model robustness against different CRF (Constant Rate Factor) levels, focusing on how this parameter influences rPPG signal recovery. However, as explored in detail in Chapter 2, H.264 video compression is controlled by several additional parameters, including GOP size, number of B-frames, reference frames and motion estimation settings, that can each affect the preservation of physiological signals in compressed videos. As a next step, rather than evaluating model performance solely based on CRF levels, we propose to extend validation and analysis to include these other key compression parameters, as examined in our earlier experiments.

Additionally, future research could involve experimenting with alternative video compression standards such as H.265 (HEVC) or other modern compression tools. Investigating how the model performs under different compression methods would provide an even more comprehensive understanding of its robustness and adaptability. By systematically exploring a wider variety of compression settings and standards, we can further improve and generalize rPPG models for use in diverse, real-world scenarios.

Combining RS-rPPG and FaceReader Signal Processing for Improved Results

One of the most impactful next steps is to combine the RS-rPPG model with the established FaceReader signal processing pipeline to achieve more robust and accurate physiological measurement. FaceReader, developed at VicarVision, is widely recognized for its strong performance in real-time heart rate and heart rate variability (HRV) estimation across a range of real-world scenarios [25]. During our experiments, we observed that while RS-rPPG and FaceReader can produce very similar outputs under certain conditions, each method also has unique strengths. For example, the RS-rPPG model, based on deep learning, is especially resilient to challenges such as video compression and subtle motion, whereas the FaceReader pipeline, grounded in well-established signal processing and physiological principles, excels in handling motion suppression, filtering, and reliable peak detection.

Combining these two approaches offers a logical path forward: the refined rPPG signal output by the RS-rPPG model can be directly used as input for the FaceReader signal processing modules. This means that instead of relying solely on traditional pixel-averaged and POS-extracted signals, the pipeline can start from the higher-quality, deep learning-based RS-rPPG signal. The remainder of the FaceReader pipeline, motion noise suppression, frequency filtering, and peak/inter-beat interval analysis, can then be applied without modification. This hybrid approach is expected to improve the overall accuracy and robustness of heart rate and HRV estimation, allowing each component to contribute its particular

strengths.

Overall, this thesis brings together practical experiments, domain knowledge, and deep learning approaches to improve rPPG estimation from compressed video. While the results achieved here are encouraging, there are still challenges to overcome before such systems can be widely used in everyday settings. The future work outlined above points to several promising directions for continued research and development. At the end, it is hoped that the findings and ideas presented in this thesis will support and inspire future progress toward reliable and accessible video-based heart rate monitoring.

Chapter A

Mathematical Modeling of Signal Stability under Compression

Overview

This appendix presents a mathematical framework for analyzing how video compression influences physiological signals. The compression process, denoted by $\mathcal{C}$, is modeled as a nonlinear operator acting on a Hilbert space of temporal signals. We study how small, bounded perturbations introduced by compression affect the stability of frequency estimation. By applying operator-norm analysis together with the Parseval-Plancherel identity and perturbation theory for spectral extrema, it is shown that variations in the dominant frequency, corresponding to the heart-rate component, are proportional to the relative L^2 -energy of the compression disturbance. In summary, the frequency estimation process behaves as a Lipschitz-continuous mapping with respect to compression-induced perturbations, providing a formal mathematical justification for the observed robustness of heart-rate estimation under moderate compression.

A.1 Mathematical Preliminaries

Before introducing the Hilbert-space formulation, we recall one of the key mathematical tools underlying signal-energy analysis, the *Parseval-Plancherel identity*. For a square-integrable function $f : [0, T] \rightarrow \mathbb{R}$, its Fourier transform is defined by

$$\mathcal{F}(f)(\omega) = \hat{f}(\omega) = \int_0^T f(t) e^{-i\omega t} dt, \quad \omega \in \mathbb{R}.$$

The Parseval-Plancherel identity establishes an isometry between the time and frequency domains:

$$\int_0^T |f(t)|^2 dt = \frac{1}{2\pi} \int_{\mathbb{R}} |\hat{f}(\omega)|^2 d\omega.$$

This equality shows that the total signal energy is preserved under the Fourier transform—no energy is lost when passing from the temporal representation to its spectral representation. This property makes the L^2 -space of finite-energy signals a natural mathematical setting for analyzing the effects of compression on both time and frequency content.

A.1.1 Hilbert Spaces

Let $\mathcal{H} = L^2([0, T]; \mathbb{R})$ denote the Hilbert space of square-integrable real-valued signals on $[0, T]$ with inner product

$$\langle f, g \rangle = \int_0^T f(t) g(t) dt, \quad \|f\|_2 = (\langle f, f \rangle)^{1/2}.$$

Completeness of $\mathcal{H}$ ensures that limits of Cauchy sequences of finite-energy signals remain in $\mathcal{H}$; hence every physically meaningful signal can be represented as a point of this space.

The Fourier transform $\mathcal{F} : \mathcal{H} \rightarrow L^2(\mathbb{R})$ defined by

$$(\mathcal{F}f)(\omega) = \int_0^T f(t) e^{-i\omega t} dt$$

is a linear isometry satisfying the Plancherel identity

$$\|f\|_2^2 = \frac{1}{2\pi} \int_{\mathbb{R}} |\hat{f}(\omega)|^2 d\omega.$$

Thus, L^2 shows a natural Hilbert-space framework for analyzing energy-preserving transformations and for quantifying how compression perturbs the spectral structure of physiological signals.

A.1.2 Perturbation Operators

Let $A : \mathcal{H} \rightarrow \mathcal{H}$ be a (possibly nonlinear) operator and $E : \mathcal{H} \rightarrow \mathcal{H}$ denote a perturbation. We say that $A + E$ is a *bounded perturbation* of A if the perturbation operator E itself is bounded; that is, there exists a constant $\delta > 0$ such that

$$\|E(x)\|_2 \leq \delta \|x\|_2, \quad \forall x \in \mathcal{H}.$$

Equivalently, E has a finite operator norm

$$\|E\|_{\text{op}} = \sup_{x \neq 0} \frac{\|E(x)\|_2}{\|x\|_2} \leq \delta.$$

In this case, the perturbed operator $A + E$ differs from A by a bounded mapping whose effect on any element $x \in \mathcal{H}$ is at most proportional to the norm of x .

The parameter δ measures the strength of the perturbation. Perturbation theory investigates how structural properties of A (spectrum, fixed points, norms, or functionals of its outputs) change when E is added [26]. A canonical special case is the perturbation of the identity:

$$\mathcal{C} = I + E, \quad \mathcal{C}(x) = x + E(x), \tag{A.1}$$

where E is small in operator norm. In this setting

$$\|E\|_{\text{op}} = \sup_{x \neq 0} \frac{\|E(x)\|_2}{\|x\|_2} \leq \delta \implies (1 - \delta)\|x\|_2 \leq \|\mathcal{C}(x)\|_2 \leq (1 + \delta)\|x\|_2.$$

Hence $\mathcal{C}$ is an isomorphism on $\mathcal{H}$ whenever $\delta < 1$. This bounded-perturbation model (A.1) precisely

describes a compression operator whose distortions are controlled relative to the signal's energy.

A.2 Operator Model of Compression

We model the temporal intensity signal extracted from a facial video frame sequence as $x \in \mathcal{H}$. The H.264 compression process can be regarded as the action of a (nonlinear) operator

$$\mathcal{C} : \mathcal{H} \rightarrow \mathcal{H}, \quad \mathcal{C}(x) = x + \varepsilon(x),$$

where the perturbation term $\varepsilon(x)$ comprises quantization and motion-prediction residuals. We assume a uniform energy bound

$$\|\varepsilon(x)\|_2 \leq \delta \|x\|_2, \quad 0 < \delta < 1. \quad (\text{A.2})$$

The constant δ plays the role of a normalized compression intensity; larger CRF values in H.264 correspond to larger δ .

A.3 Spectral Representation and Energy Perturbation

Define the (non-negative) spectral energy density of x as

$$S_x(\omega) = |\hat{x}(\omega)|^2.$$

By Parseval's theorem,

$$\|x\|_2^2 = \frac{1}{2\pi} \int_{\mathbb{R}} S_x(\omega) d\omega.$$

Let $x_c := \mathcal{C}(x)$, where $\mathcal{C}(x) = x + \varepsilon(x)$. For the compressed signal x_c ,

$$S_{x_c}(\omega) = |\hat{x}(\omega) + \hat{\varepsilon}(\omega)|^2 = S_x(\omega) + 2\Re[\hat{x}(\omega)\hat{\varepsilon}^*(\omega)] + |\hat{\varepsilon}(\omega)|^2.$$

Integrating over the physiological frequency band $\Omega = [\omega_{\min}, \omega_{\max}]$, we get the band-energy ratio as following:

$$R_{\Omega}(x, x_c) = \frac{\int_{\Omega} S_{x_c}(\omega) d\omega}{\int_{\Omega} S_x(\omega) d\omega}.$$

Lemma 4 (Energy Perturbation Bound). . *Let $x_c := \mathcal{C}(x)$, and the perturbation $\varepsilon(x)$ satisfies the boundedness condition*

$$\|\varepsilon(x)\|_2 \leq \delta \|x\|_2, \quad 0 < \delta < 1.$$

Then, for any measurable subset $\Omega \subseteq [0, T]$, the relative energy deviation between x and its compressed version x_c satisfies

$$|R_{\Omega}(x, x_c) - 1| \leq 2\delta + \delta^2.$$

Proof. By expanding the squared norm of x_c , we have

$$\|x_c\|_2^2 - \|x\|_2^2 = 2\Re\langle x, \varepsilon(x) \rangle + \|\varepsilon(x)\|_2^2,$$

then by applying the Cauchy–Schwarz inequality and using the bound (A.2), we get

$$\begin{aligned} |\|x_c\|_2^2 - \|x\|_2^2| &\leq 2\|x\|_2\|\varepsilon(x)\|_2 + \|\varepsilon(x)\|_2^2 \\ &\leq (2\delta + \delta^2)\|x\|_2^2. \end{aligned}$$

Since the same inequality holds when integration is restricted to any measurable subset Ω , the stated bound for $R_\Omega(x, x_c)$ follows immediately. $\square$

A.4 Functional Stability of Frequency Estimation

Let $F : \mathcal{H} \rightarrow \mathbb{R}$ denote the functional that maps each signal $x \in \mathcal{H}$ to the frequency of its dominant spectral component,

$$F(x) = \arg \max_{\omega \in \Omega} S_x(\omega),$$

where $S_x(\omega)$ is the spectral energy density of x . We investigate the stability of F with respect to bounded perturbations of the form $x \mapsto x + \varepsilon(x)$, i.e., how the dominant frequency changes under compression-induced distortions.

Theorem 5 (Lipschitz stability of dominant frequency). *Assume S_x is twice continuously differentiable near $F(x)$ and $S_x''(F(x)) < 0$. If (A.2) holds, then there exists $K > 0$ such that,*

$$|F(x_c) - F(x)| \leq K\delta.$$

Proof. Let $\Delta S(\omega) = S_{x_c}(\omega) - S_x(\omega)$. By the previous lemma, $|\Delta S(\omega)| \leq (2\delta + \delta^2)S_x(\omega)$. At the maximum point $\omega_0 = F(x)$, $S'_x(\omega_0) = 0$ and $S''_x(\omega_0) < 0$. Expanding S'_{x_c} about ω_0 ,

$$0 = S'_{x_c}(F(x_c)) = S'_x(\omega_0) + \Delta S'(\omega_0) + S''_x(\omega_0)(F(x_c) - \omega_0) + o(F(x_c) - \omega_0),$$

By solving for the frequency shift and neglecting higher-order terms, we get

$$|F(x_c) - \omega_0| \leq \frac{|\Delta S'(\omega_0)|}{|S''_x(\omega_0)|},$$

and then by differentiating the Fourier representation, we have

$$|\Delta S'(\omega)| \leq (2\delta + \delta^2)M_1,$$

where $M_1 = \max_{\Omega} |S'_x(\omega)|$. Therefore,

$$|F(x_c) - F(x)| \leq K(2\delta + \delta^2), \quad K = \frac{M_1}{|S''_x(F(x))|}.$$

For small δ , the first-order term dominates and the bound is $O(\delta)$. $\square$

Corollary. The functional $F : \mathcal{H} \rightarrow \mathbb{R}$, which assigns to each signal x the frequency of its dominant spectral component, is Fréchet differentiable in a neighbourhood of x and locally Lipschitz-continuous

with constant $K > 0$. In particular,

$$|F(x + \varepsilon) - F(x)| \leq K \frac{\|\varepsilon\|_2}{\|x\|_2},$$

where K depends on the local curvature of the spectral density around its maximum.

This result means that the dominant frequency extracted from a signal varies smoothly with respect to small perturbations of the signal itself. Fréchet differentiability ensures that F can be linearly approximated near x , while the Lipschitz bound quantifies its stability: the shift in the dominant frequency is proportional to the relative energy of the perturbation $\|\varepsilon\|_2/\|x\|_2$. Hence, if the compression distortion is bounded ($\|\varepsilon\|_2 \leq \delta\|x\|_2$), the estimated frequency changes by at most $K\delta$, showing that frequency estimation is robust to moderate compression.

The previous subsections described compression as a bounded perturbation acting directly on the temporal signal and analyzed its effect on energy and dominant-frequency estimation. To complete the theoretical picture, we now explain these results within an operator-theoretic framework. This viewpoint connects the empirical stability of frequency estimation with classical spectral perturbation theory: the dominant frequency of a signal can be regarded as the principal eigenvalue of an operator derived from its spectrum, while compression introduces a small perturbation to this operator. Consequently, the observed robustness of heart-rate estimation under moderate compression can be formally understood as a first-order spectral stability.

A.5 Operator–Spectral Interpretation

For each temporal signal $x \in \mathcal{H}$, define the integral operator

$$(\mathcal{T}_x f)(t) = \frac{1}{2\pi} \int_{\Omega} \hat{x}(\omega) e^{i\omega t} \hat{f}(\omega) d\omega,$$

with kernel

$$k_x(t, s) = \frac{1}{2\pi} \int_{\Omega} \hat{x}(\omega) e^{i\omega(t-s)} d\omega.$$

The operator $\mathcal{T}_x$ acts as a spectral filter that weights the frequency components of any function f according to the spectrum of x . The associated self-adjoint operator $\mathcal{T}_x^* \mathcal{T}_x$ has non-negative eigenvalues $\{\lambda_k\}$, which represent the energy distribution of the signal across its spectral modes. The principal eigenvalue $\lambda_1(\mathcal{T}_x^* \mathcal{T}_x)$ corresponds to the dominant energy component, hence determining the dominant frequency of x .

Compression alters the signal to $x_c = x + \varepsilon(x)$, inducing a perturbed operator

$$\mathcal{T}_{x_c} = \mathcal{T}_x + \Delta\mathcal{T}, \quad \|\Delta\mathcal{T}\|_{\text{op}} \leq C\delta,$$

where the perturbation magnitude is proportional to the normalized compression strength δ . According to classical spectral perturbation theory (Kato [26], Thm. IV.3.18),

$$|\lambda_1(\mathcal{T}_{x_c}) - \lambda_1(\mathcal{T}_x)| \leq \|\Delta\mathcal{T}\|_{\text{op}} \leq C\delta.$$

Thus, the principal eigenvalue, and consequently the dominant frequency, varies at most linearly with the perturbation amplitude. This establishes that the dominant-frequency estimator $F(x)$ exhibits first-order spectral stability with respect to compression, providing a rigorous operator-theoretic explanation for its empirical robustness.

Conclusion. The mathematical model developed above treats compression as a bounded perturbation in a Hilbert-space setting and derives quantitative frequency-stability bounds via operator and spectral perturbation theory. This formulation connects applied signal-processing observations to rigorous functional analysis, demonstrating that the rPPG frequency estimator is Lipschitz-stable under compression distortions. Future theoretical work could extend these results by relaxing the linearity assumption on $\mathcal{F}$ or by introducing stochastic perturbations to model temporally correlated quantization noise.

Chapter B

Extended experiments on VicarrPPG dataset

In addition to the main results, we conducted two extra analyses on the ViCAR dataset to further understand the impact of outliers and data complexity on remote heart rate estimation performance. The first case includes **all available data** (even Person 15, who was identified as a strong outlier), while the second applies a more stringent filter, **excluding Persons 13, 15, and 16** from the validation due to their high errors on specific video types.

B.1 All Data Included (No Outlier Exclusion)

Figures B.1, B.2, B.3, and B.4 show the results for MAE, correlation (r), RMSE, and STD when all subjects are included in the evaluation.

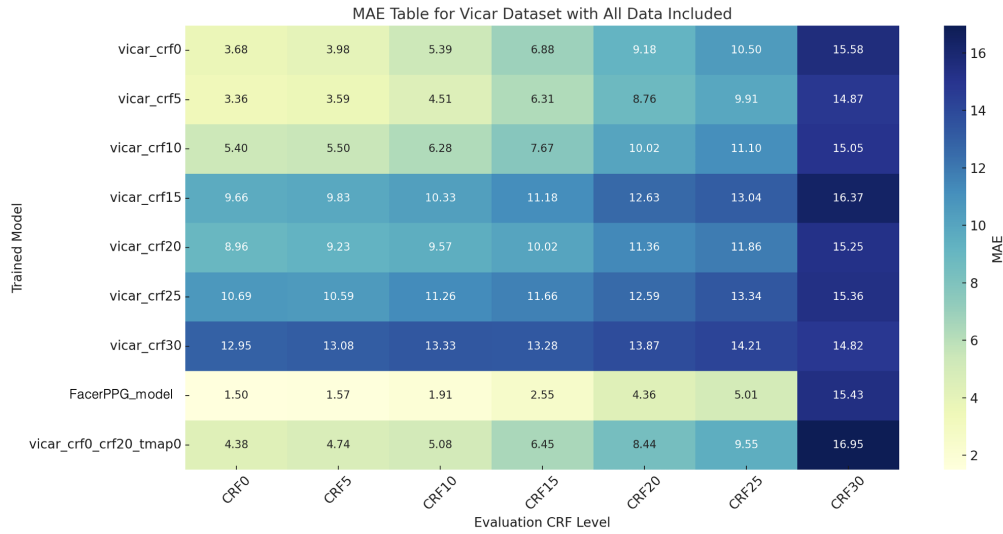


Figure B.1: MAE table for ViCAR dataset with all data included.

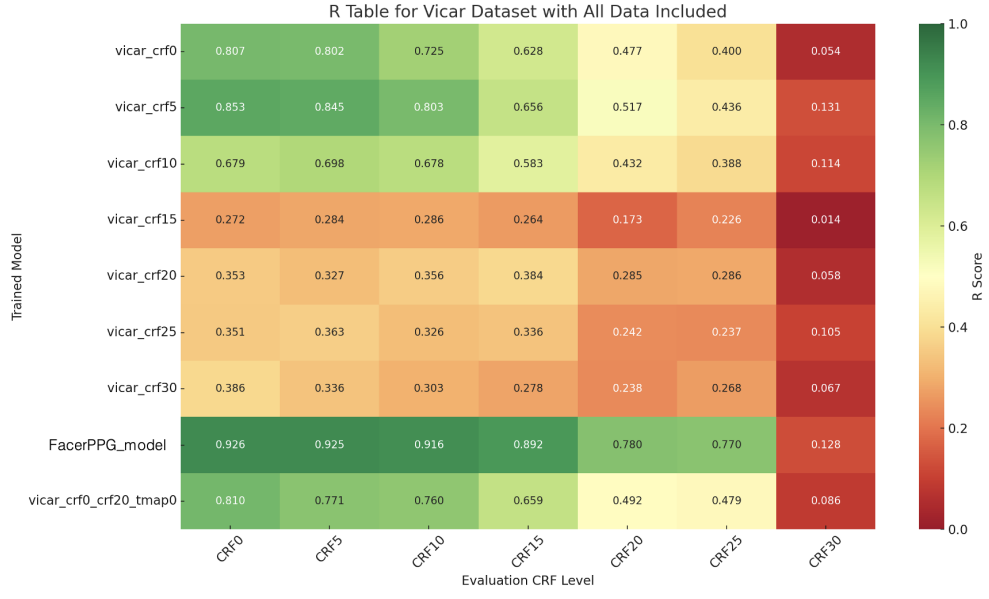


Figure B.2: Correlation (r) table for ViCAR dataset with all data included.

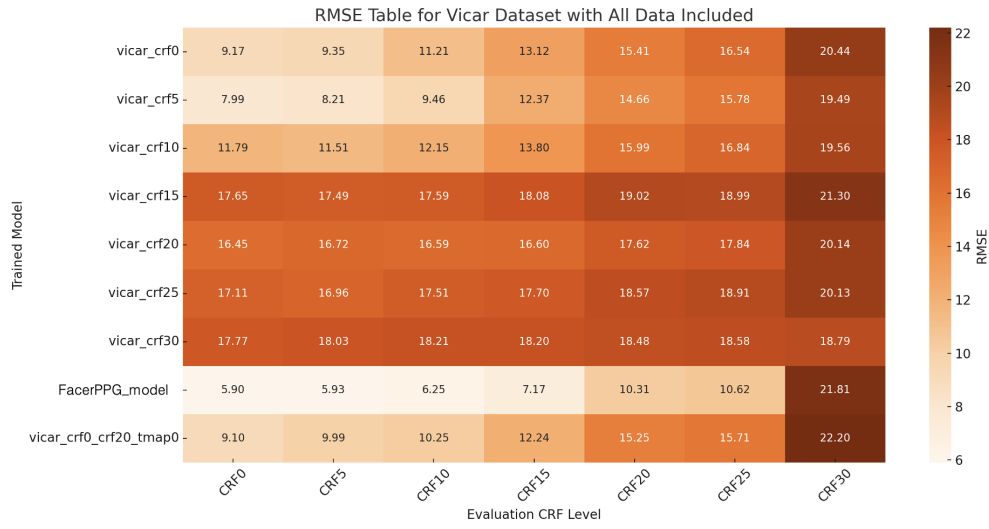


Figure B.3: RMSE table for ViCAR dataset with all data included.

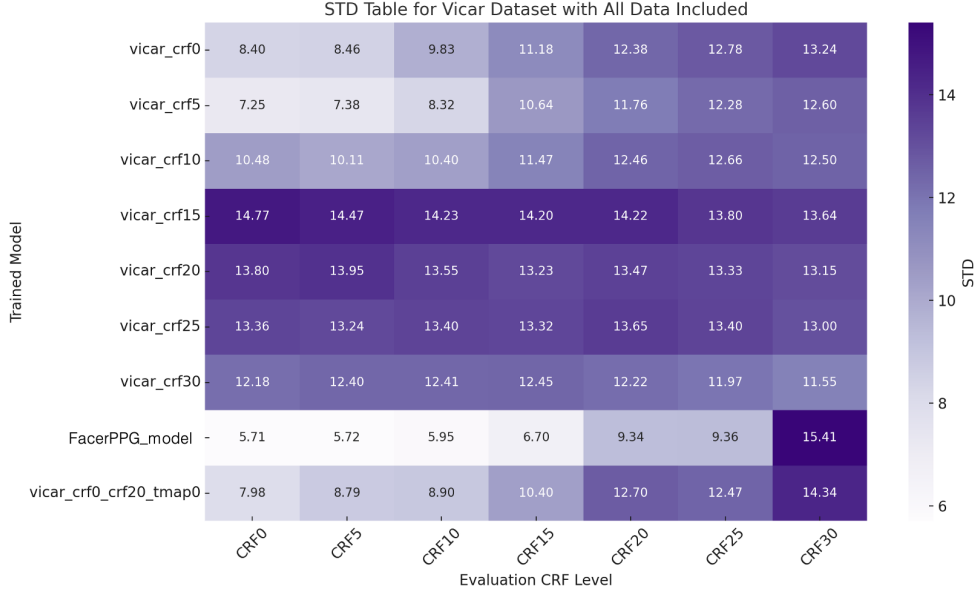


Figure B.4: STD table for ViCAR dataset with all data included.

When all data is included in the evaluation—meaning no outlier subjects are excluded—the results across all four metrics (MAE, r , RMSE, and STD) are noticeably worse than in the main section where Person 15 was omitted. The MAE values are higher for almost every CRF level and model, with even the combined model showing substantial errors at high compression, and none of the models coming close to FacerPPG performance. The r scores also drop significantly, especially for models trained on higher compression levels, confirming that the predictions lose much of their alignment with the true heart rate signal when all subjects are included. Similarly, both RMSE and STD are elevated throughout, indicating not just larger average errors, but also more frequent outliers and less stable predictions overall.

Even though the combined model still does a little better than the single-CRF models, its results are far from satisfactory, especially as compression increases. This analysis highlights just how much a single difficult or inconsistent subject, like Person 15, can distort the evaluation of deep learning models on challenging, real-world datasets like vicarrPPG. It also justifies the common practice of identifying and excluding such extreme outliers when benchmarking model performance, as their inclusion can make meaningful comparison or progress almost impossible.

B.2 Stricter Outlier Removal (Excluding Persons 13, 15, and 16)

Figures B.5, B.6, B.7, and B.8 show the results after removing Persons 13, 15, and 16 from the validation set. As discussed in Section 3.6, Person 15 was identified as an outlier, showing much higher MAE and lower correlation values than other subjects across all video types. Persons 13 and 16 also displayed similar irregularities in some cases, producing inconsistent MAE and correlation values. Therefore, this analysis investigates how removing all three subjects influences the overall model performance and stability.

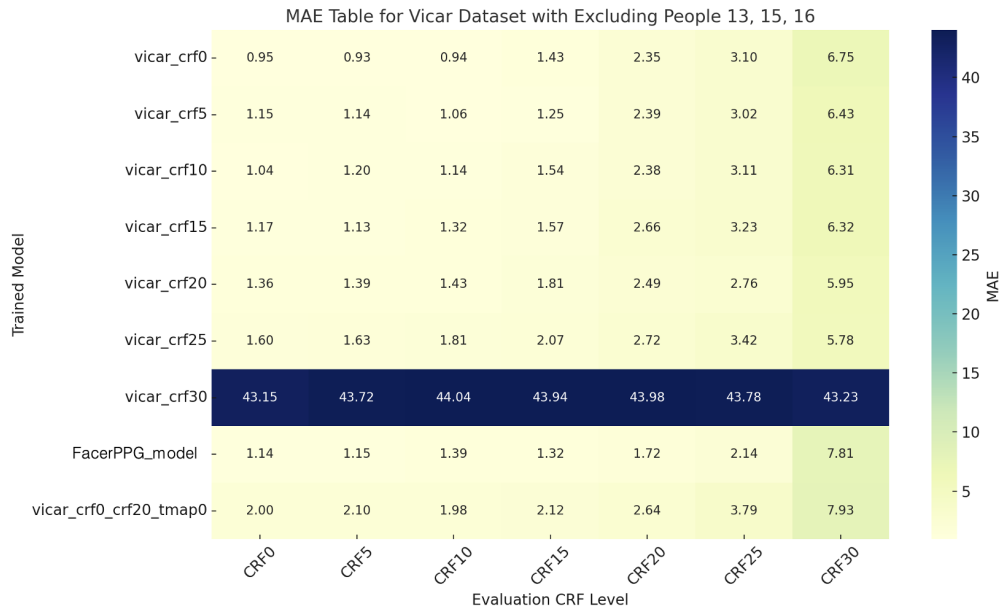


Figure B.5: MAE table for ViCAR dataset, excluding Persons 13, 15, and 16.

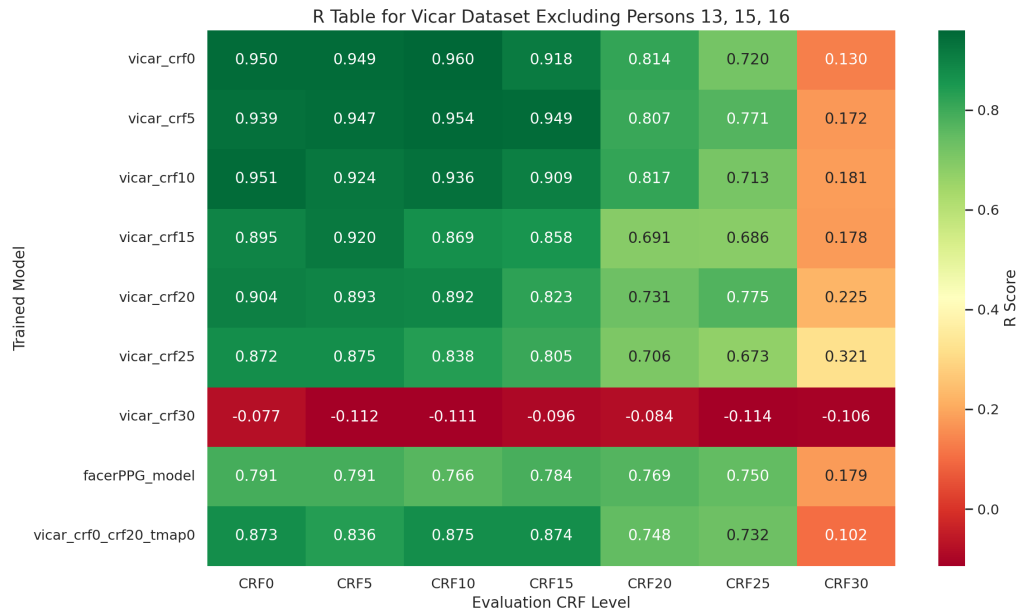


Figure B.6: Correlation (r) table for ViCAR dataset, excluding Persons 13, 15, and 16.

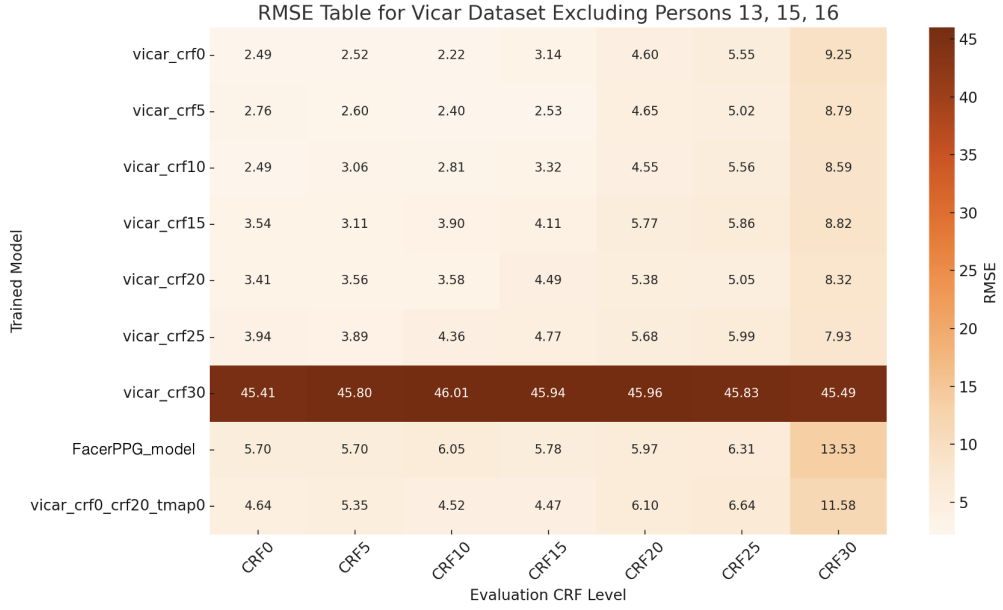


Figure B.7: RMSE table for ViCAR dataset, excluding Persons 13, 15, and 16.



Figure B.8: STD table for ViCAR dataset, excluding Persons 13, 15, and 16.

When applying the strictest filtering, excluding Persons 13, 15, and 16 from the ViCAR dataset, we see a dramatic drop in errors across nearly all metrics and models. In the MAE table, both single-CRF and combined models reach much lower absolute errors, and the commercial FaceReader baseline also appears to benefit from this cleaner data. However, as discussed before, MAE is not the most reliable indicator for rPPG performance, especially when outliers are omitted, since it only reflects average peak errors and can be sensitive to remaining outlier peaks.

Looking at the other metrics, correlation (r), RMSE, and STD, the combined model trained with tmap0 as ground truth now outperforms single-CRF models in almost every setting and even surpasses FaceReader at higher compression levels. Correlation scores are especially strong for the combined model, which

shows its ability to track the underlying heart rate signal more closely than the other models when the data is less noisy. Similarly, RMSE and STD values for the combined model are consistently lower, meaning the predictions are both more accurate and more stable.

However, this strict case is mostly a theoretical exercise. In real-world settings, data will almost always include more challenging and noisy samples. Simply omitting all “difficult” subjects is not a practical solution and can lead to overestimating model performance. This analysis mainly helps to illustrate the impact that a few hard cases can have on overall results and highlights the importance of robust modeling. In practice, models need to be evaluated and improved on datasets that reflect real-world variation, including those subjects and videos that are hardest to predict.

Chapter C

Theoretical Explanation of RS-rPPG Robustness under Compression

Overview

Building upon the perturbation framework established in Appendix A, this appendix provides a theoretical explanation of why the RS-rPPG method maintains accurate heart-rate estimation even when the input video is compressed. Using the same Hilbert-space and operator-theoretic setup, we explain the deep learning model as a sequence of Lipschitz-continuous mappings acting on the compressed temporal signal. We then show that, under the bounded perturbation assumption of the compression operator, both the learned feature representations and the final heart-rate predictions remain stable in the sense of Lipschitz continuity.

C.1 Model as a Composition of Lipschitz Operators

Let $\mathcal{H} = L^2([0, T])$ denote the Hilbert space of temporal signals as defined in Appendix A. The RS-rPPG model can be written as a composition of two mappings:

$$h(x) = g_\phi(f_\theta(x)),$$

where $f_\theta : \mathcal{H} \rightarrow \mathbb{R}^d$ is the deep encoder (e.g., Swin-UNet backbone) that extracts latent features, and $g_\phi : \mathbb{R}^d \rightarrow \mathbb{R}$ is the regression head that predicts the heart-rate-related quantity.

If both f_θ and g_ϕ are Lipschitz-continuous with constants L_f and L_g , respectively, then the overall mapping $h = g_\phi \circ f_\theta$ satisfies

$$|h(x_1) - h(x_2)| \leq L_g L_f \|x_1 - x_2\|_2, \quad \forall x_1, x_2 \in \mathcal{H}.$$

This property means that small perturbations in the input signal lead to proportionally small changes in the output prediction.

Recall from Appendix A that the compression process is modeled as,

$$\mathcal{C}(x) = x + \varepsilon(x), \quad \|\varepsilon(x)\|_2 \leq \delta \|x\|_2,$$

By applying the Lipschitz property of h , we get

$$|h(x_c) - h(x)| \leq L_g L_f \|x_c - x\|_2 \leq L_g L_f \delta \|x\|_2.$$

Thus, the model output, which represents the estimated heart rate, is bounded linearly with respect to the compression distortion strength δ . This shows that the deep model inherits the same Lipschitz stability proven for the frequency functional in Appendix A.

C.2 Self-Supervised Learning and Theoretical Stability

The RS-rPPG model employs a self-supervised pretraining phase based on contrastive learning to enforce invariance to nuisance variations such as illumination, motion, and compression. Let $f_\theta : \mathcal{H} \rightarrow \mathbb{R}^d$ denote the encoder with parameters θ , mapping each signal $x \in \mathcal{H}$ to a latent representation.

For each sample x_i , an augmented counterpart $\tilde{x}_i$ preserving physiological content is generated, and the encoder is trained via the contrastive loss

$$\mathcal{L}_{\text{contrast}} = -\log \frac{\exp(\text{sim}(f_\theta(x_i), f_\theta(\tilde{x}_i))/\tau)}{\sum_k \exp(\text{sim}(f_\theta(x_i), f_\theta(\tilde{x}_k))/\tau)},$$

where $\text{sim}(\cdot, \cdot)$ is the cosine similarity and $\tau > 0$ is the temperature. When compression acts as a perturbation

$$\tilde{x}_i = \mathcal{C}(x_i) = x_i + \varepsilon(x_i), \quad \|\varepsilon(x_i)\|_2 \leq \delta \|x_i\|_2,$$

the contrastive objective explicitly encourages $f_\theta(\mathcal{C}(x_i)) \approx f_\theta(x_i)$, forcing the encoder to learn representations invariant to such bounded distortions.

Viewing f_θ as a nonlinear operator on $\mathcal{H}$, assume it is L_f -Lipschitz:

$$\|f_\theta(x + \varepsilon(x)) - f_\theta(x)\|_2 \leq L_f \|\varepsilon(x)\|_2.$$

Contrastive pretraining minimizes the deviation between embeddings of perturbation pairs $(x_i, \tilde{x}_i)$, effectively reducing the Lipschitz constant along compression directions and yielding an *effective* constant $L_f^{\text{eff}} \ll L_f$ such that

$$\|f_\theta(x_c) - f_\theta(x)\|_2 = O(\delta L_f^{\text{eff}}).$$

Hence the learned mapping becomes non-expansive on the compression manifold, contracting perturbations in the representation space.

Let $h : \mathbb{R}^d \rightarrow \mathbb{R}$ denote the final regression head estimating the heart-rate frequency. If h is L_g -Lipschitz, then for any bounded compression disturbance $\|\varepsilon(x)\|_2 \leq \delta \|x\|_2$,

$$\begin{aligned} \|x_c - x\|_2 &\leq \delta \|x\|_2, \\ \|f_\theta(x_c) - f_\theta(x)\|_2 &\leq L_f \delta \|x\|_2, \\ |h(x_c) - h(x)| &\leq L_g L_f \delta \|x\|_2. \end{aligned}$$

After self-supervised pretraining, replacing L_f by its effective value gives

$$|h(x_c) - h(x)| = O(\delta L_f^{\text{eff}}),$$

showing that the end-to-end frequency estimator remains stable under compression. The combined Hilbert-space perturbation model and Lipschitz continuity of deep mappings thus provide a compact mathematical justification for the empirical robustness of RS-rPPG.

C.3 Generalization and Robustness Bounds in Representation Learning

The previous subsections demonstrated that RS-rPPG maintains stable predictions under compression through the Lipschitz continuity of its mappings and the invariance learned during self-supervised pretraining. In this section, we extend that reasoning to a general learning-theoretic framework, showing that the same mathematical properties ensuring stability also enhance generalization across unseen compression levels and subjects. This connects the operator-theoretic model of Appendix A with the representation-learning view developed in this chapter.

Setup. Let $\mathcal{D}$ denote the distribution of clean temporal signals and ℓ a ρ -Lipschitz loss function (e.g., spectral MAE or contrastive loss). For a model $h_\theta = g_\phi \circ f_\theta$, the expected risk is:

$$\mathcal{R}(\theta) = \mathbb{E}_{x \sim \mathcal{D}} \ell(h_\theta(x), y).$$

When compression introduces bounded perturbations,

$$x_c = \mathcal{C}(x) = x + \varepsilon(x), \quad \|\varepsilon(x)\|_2 \leq \delta \|x\|_2,$$

the *robust risk* is defined as:

$$\mathcal{R}_{\text{rob}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}} \left[\sup_{\|\varepsilon(x)\|_2 \leq \delta \|x\|_2} \ell(h_\theta(x + \varepsilon(x)), y) \right].$$

Minimizing $\mathcal{R}_{\text{rob}}(\theta)$ instead of $\mathcal{R}(\theta)$ encourages the model to remain invariant to compression, which aligns with the self-supervised objectives used in RS-rPPG.

Theorem 6 (Robust risk bound under Lipschitz continuity). *If h_θ is L_h -Lipschitz and ℓ is ρ -Lipschitz in its first argument, then*

$$\mathcal{R}_{\text{rob}}(\theta) \leq \mathcal{R}(\theta) + \rho L_h \delta \mathbb{E}_{x \sim \mathcal{D}} \|x\|_2.$$

Proof. We start from the definition of the robust risk, which measures the expected loss under the worst admissible perturbation:

$$\mathcal{R}_{\text{rob}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}} \left[\sup_{\|\varepsilon(x)\|_2 \leq \delta \|x\|_2} \ell(h_\theta(x + \varepsilon(x)), y) \right].$$

To relate this to the clean risk $\mathcal{R}(\theta) = \mathbb{E}_{x \sim \mathcal{D}} [\ell(h_\theta(x), y)]$, we need to bound the deviation in loss caused

by the perturbation $\varepsilon(x)$. Assume that the loss function $\ell(a, y)$ is ρ -Lipschitz in its first argument, i.e.

$$|\ell(a, y) - \ell(b, y)| \leq \rho|a - b|, \quad \forall a, b, y.$$

Applying this property to the perturbed and clean predictions of the model gives

$$|\ell(h_\theta(x + \varepsilon(x)), y) - \ell(h_\theta(x), y)| \leq \rho |h_\theta(x + \varepsilon(x)) - h_\theta(x)|.$$

If the model h_θ is L_h -Lipschitz, then for any perturbation $\varepsilon(x)$ we have

$$|h_\theta(x + \varepsilon(x)) - h_\theta(x)| \leq L_h \|\varepsilon(x)\|_2.$$

Combining the two inequalities yields

$$|\ell(h_\theta(x + \varepsilon(x)), y) - \ell(h_\theta(x), y)| \leq \rho L_h \|\varepsilon(x)\|_2.$$

From the compression operator model introduced in Appendix A, the perturbation satisfies $\|\varepsilon(x)\|_2 \leq \delta \|x\|_2$. Substituting this into the previous inequality gives

$$|\ell(h_\theta(x + \varepsilon(x)), y) - \ell(h_\theta(x), y)| \leq \rho L_h \delta \|x\|_2.$$

Taking expectations over $x \sim \mathcal{D}$ and using the definition of $\mathcal{R}_{\text{rob}}(\theta)$ then yields

$$\mathcal{R}_{\text{rob}}(\theta) \leq \mathcal{R}(\theta) + \rho L_h \delta \mathbb{E}_{x \sim \mathcal{D}} \|x\|_2,$$

which completes the proof. $\square$

Remark. This result follows standard Lipschitz-based robustness theory [27--30]. It shows that when compression introduces a bounded perturbation of size δ , the resulting change in the model's expected loss grows at most linearly with δ , scaled by the smoothness of the loss (ρ) and the network (L_h). In simple terms, as long as the RS-rPPG mapping remains Lipschitz-continuous, compression artifacts cause only limited changes in the predicted heart rate, providing a clear theoretical guarantee of stability.

Effect of Tmap0 Supervision.

During both pretraining and fine-tuning, RS-rPPG uses additional supervision from the uncompressed Tmap₀ targets. This means that for each video, the model sees both a clean reference signal t_0 and a compressed version $\mathcal{C}(x)$ coming from the same subject. The goal of this setup is to make sure that the encoder learns to produce almost identical feature representations for clean and compressed inputs, reducing its sensitivity to compression noise.

Formally, let $f_\theta : \mathcal{H} \rightarrow \mathbb{R}^d$ be the encoder network, and define the *compression sensitivity function*

$$\Delta_f(\delta) = \sup_{\|x' - x\|_2 \leq \delta \|x\|_2} \|f_\theta(x') - f_\theta(x)\|_2,$$

which measures how much the encoded features can change when the input is perturbed by compression

of strength δ . Without any extra supervision, this value $\Delta_f(\delta)$ reflects how much compression can distort the learned representation. Adding Tmap0 supervision introduces an extra loss term that explicitly aligns compressed and clean representations:

$$\mathcal{L}_{t0} = \mathbb{E}_x \left[\|f_\theta(\mathcal{C}(x)) - f_\theta(t_0(x))\|_2^2 \right], \quad t_0(x) = \text{Tmap}_0(x).$$

Minimizing this loss forces the encoder to map both versions—compressed and uncompressed—to nearly the same feature space, so that,

$$\|f_\theta(\mathcal{C}(x)) - f_\theta(x)\|_2 \rightarrow 0 \quad \text{for all } x \in \mathcal{H}.$$

As a result, the encoder becomes less sensitive to compression distortions, which we can express as:

$$\Delta_f^{(t0)}(\delta) \ll \Delta_f(\delta),$$

meaning that after Tmap0 supervision, the feature variations caused by compression become much smaller.

This directly improves the model’s theoretical robustness. Since the overall predictor is $h_\theta = g_\phi \circ f_\theta$ with Lipschitz constant $L_h = L_g L_f$, the robust risk after Tmap0 supervision becomes

$$\mathcal{R}_{\text{rob}}^{(t0)}(\theta) \leq \mathcal{R}(\theta) + \rho L_g \Delta_f^{(t0)}(\delta).$$

Because $\Delta_f^{(t0)}(\delta)$ is much smaller than $\Delta_f(\delta)$, the additional loss caused by compression is also smaller. In other words, Tmap0 supervision narrows the effective “perturbation radius” of the representation space, leading to more stable and consistent heart-rate predictions, even when videos are heavily compressed.

From a theoretical perspective, this can be seen as a form of *representation regularization* [30, 31]. By forcing the network to align clean and compressed features, it learns a smoother and more invariant latent space, similar to other robust learning methods [29, 32]. In RS-rPPG, this means that compression artifacts collapse toward the same manifold as the clean physiological signals, while the meaningful heart-rate information remains preserved.

Overall, adding Tmap0 supervision not only makes the model theoretically more stable under compression but also explains the improved performance observed in Chapter 3: the network learns to ignore distortions caused by compression while keeping physiologically relevant patterns intact.

Conclusion. Mathematically, the end-to-end mapping from the input signal to the estimated heart rate behaves as a Lipschitz-continuous operator, ensuring that compression-induced perturbations cause only bounded changes in output. Through contrastive and Tmap0-supervised training, the encoder further reduces its sensitivity to compression, effectively contracting feature variations along those directions. This alignment between clean and compressed representations tightens the robust risk bound and minimizes the effect of distortion. Together, these properties explain both theoretically and empirically why RS-rPPG maintains stable heart-rate estimation even under moderate to high video compression.

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